

**IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION
MONITORING, IMAGE PROCESSING AND PRELIMINARY
SCREENING USING FUZZY LOGIC-ENHANCED KNN**

by

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ABSTRACT

This study designed and developed the IoT-Based Smart Urinalysis System, a portable, offline, multi-modal urinalysis device integrating pH, Specific Gravity (SG), and Hue-Saturation-Value (HSV) color sensing within a custom opaque enclosure, paired with a cross-platform Bluetooth Low Energy (BLE) mobile application implementing the Enhanced Fuzzy k-Nearest Neighbors (EF-KNN) classification algorithm for automated hydration level classification and preliminary physiological abnormality screening. The system was evaluated across 150 independent trials using chemically modified synthetic urine formulations spanning eight hydration levels and four abnormality classifications. The pH sensor achieved a MAPE of 7.72% ($p = 0.0937$) and the Specific Gravity sensor achieved a MAPE of 0.13% ($p = 0.8843$), both passing the 10% acceptance threshold with no statistically significant difference from their respective reference instruments. The HSV optical channel recorded a MAPE of 40.03% ($p = 0.0021$) against the iPhone 16 Pro ImageJ ground truth; however, a secondary evaluation confirmed no statistically significant difference between the Arduino HSV readings and the embedded predataset centroid targets ($p = 0.5837$), establishing that the sensor and classifier share the same systematic bias and preserving classification integrity. The BLE communication pipeline achieved a Transmission Success Rate (TSR) of 99.33% with a mean end-to-end system latency of 8,576.51 ms. The EF-KNN classifier at $k = 3$ achieved 100.00% hydration classification accuracy across all eight levels with a macro-averaged AUC of 1.0000, while abnormality screening recorded an overall accuracy of 88.00% with perfect detection of Flag B (High pH) and Flag C (Low pH). Software quality evaluation by 130 respondents anchored on the ISO 25010 model yielded an overall mean of 4.42 (Agree) from subject matter experts and 4.37 (Agree) from general end-users, confirming the system's functional suitability, performance efficiency, compatibility, and reliability. These results demonstrate that reliable, automated, multi-modal urinalysis is achievable within a portable, offline embedded architecture, providing an objective alternative to subjective visual urinalysis for point-of-care hydration monitoring and preliminary physiological screening.

Keywords: *IoT, urinalysis, hydration monitoring, EF-KNN, fuzzy logic, BLE, HSV colorimetry, pH, specific gravity, synthetic urine, mobile application, ISO 25010*

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LIST OF ABBREVIATIONS

ADC – Analog-to-Digital Converter
BHW – Barangay Health Worker
CSV – Comma-Separated Values
DFD – Data Flow Diagram
HSV – Hue, Saturation, Value
IDE – Integrated Development Environment
IoT – Internet of Things
k-NN – k-Nearest Neighbors
LED – Light Emitting Diode
pH – Potential of Hydrogen
RGB – Red, Green, Blue
SDLC – Software Development Life Cycle
SG – Specific Gravity
TDS – Total Dissolved Solids
UI – User Interface
USB – Universal Serial Bus
UTI – Urinary Tract Infection
UX – User Experience

Chapter 1

INTRODUCTION

1.1 Background of the Study

Urinalysis is widely recognized as a fundamental clinical screening tool used globally to detect metabolic disturbances, kidney disease, and urinary tract infections (UTIs) (Queremel Milani & Jialal, 2023). However, traditional screening still relies on either manual interpretation of reagent strips or expensive centralized laboratory testing. Manual dipstick reading is susceptible to significant human error due to environmental factors such as lighting intensity and subjective color perception (Inagaki et al., 2025), while automated systems, though accurate, remain immobile and prohibitively expensive for resource-constrained settings (Kim et al., 2025).

These diagnostic limitations are particularly concerning given the immediate physiological and cognitive impacts of undetected fluid imbalances. It was established by Sumitra (2025) that even mild dehydration (1–2% body mass loss) precipitates significant neurocognitive deficits, including a 12–18% delay in reaction time and impaired sustained attention. Beyond immediate cognitive effects, the lack of accessible screening poses a severe long-term medical threat. Early-stage renal and metabolic disorders are often asymptomatic, leading to missed diagnoses until irreversible damage occurs. This diagnostic gap is underscored by the rising prevalence of renal diseases in the Philippines. Recent data from the National Kidney and Transplant Institute reveals that one Filipino develops chronic renal failure every hour, a crisis driven largely by undiagnosed lifestyle diseases (Naguna, 2025). Hitzeman et al. (2022) establish that physical urine parameters, specifically Specific Gravity (SG) and pH, serve as the earliest clinical indicators of these

silent pathologies. However, a critical barrier to addressing this crisis is the widespread lack of access to diagnostic equipment. In many resource-constrained settings, automated urinalysis analyzers are unavailable due to their high cost and maintenance requirements (Alberto et al., 2022).

Current smartphone-based urinalysis tools face a major bottleneck in diagnostic accuracy, particularly regarding pH and Specific Gravity (SG) levels (Kim et al. 2025). Recent evaluations by Kim et al. (2025) highlight that standard applications often struggle with these parameters due to the limitations of the classic k-Nearest Neighbor (k-NN) algorithm. Smartphone based urinalysis uses mobile cameras that are naturally sensitive to shadows and shifting light, which can completely distort the raw Red, Green, and Blue (RGB) values acquired (Chew et al., 2020). Even when using high-end devices, "visual discrepancy" remains a persistent barrier because different smartphone cameras interpret RGB data inconsistently across various brands (Azhar et al., 2023).

To address these environmental issues, Wu et al. (2023) noted that custom-built opaque boxes are necessary to isolate the testing area from external lighting interference. Complementing this hardware solution, suggest the use of Hue-Saturation-Value (HSV) color space since its superior robustness against shadows compared to standard RGB models by separating the actual color (Hue) from the lighting intensity (Value).

However, even with better hardware and color models, the underlying classification logic must be refined. Because classic k-NN relies on rigid "hard" boundaries, it often fails to distinguish the subtle transitions and overlapping color gradients found in biological samples (Ozturk Kiyak et al., 2023). To bridge this gap, research by Uddin et al. (2022) supports the implementation of Fuzzy k-NN (F-KNN). By utilizing "membership grades"

instead of fixed categories, EF-KNN allows the system to handle biological gradients and sensor noise much more accurately.

Despite these technological advancements, a significant research gap remains; there is currently no integrated system in the Philippines that combines the accuracy of hardware-based sensors with offline, algorithmic smartphone analysis. To address this gap, this study proposes the IoT-Based Smart Urinalysis System. Unlike prior attempts, this system uses IoT-controlled sensors for precise pH and Specific Gravity measurement with a camera and RGB sensor for accurate urine color analysis. The device features a custom-integrated opaque enclosure to house the sample, effectively neutralizing the ambient light variability that plagues open-air diagnostics. By combining this hardware-level light control with data-level Sensor Fusion, HSV conversion, and an Enhanced Fuzzy k-NN Algorithm, the proposed system aims to provide a reliable, automated screening tool that empowers individuals and remote communities to identify potential health risks early, facilitating timely medical intervention.

1.2 Objectives of the Study

General Objective

The general objective of this study is to design and develop an IoT-Based Smart Urinalysis System housed in a custom-integrated opaque enclosure. The system featured real-time data acquisition via an offline mobile application, utilizing multi-modal sensing, specifically pH and Specific Gravity sensors combined with image processing for urine color analysis using Hue-Saturation-Value (HSV) conversion. Furthermore, the study aims to implement a Fuzzy Logic–Enhanced k-Nearest Neighbors (k-NN) algorithm to cross-

reference these three parameters for the automated classification of hydration status and the preliminary screening of out-of-bound physiological markers.

Specific Objectives

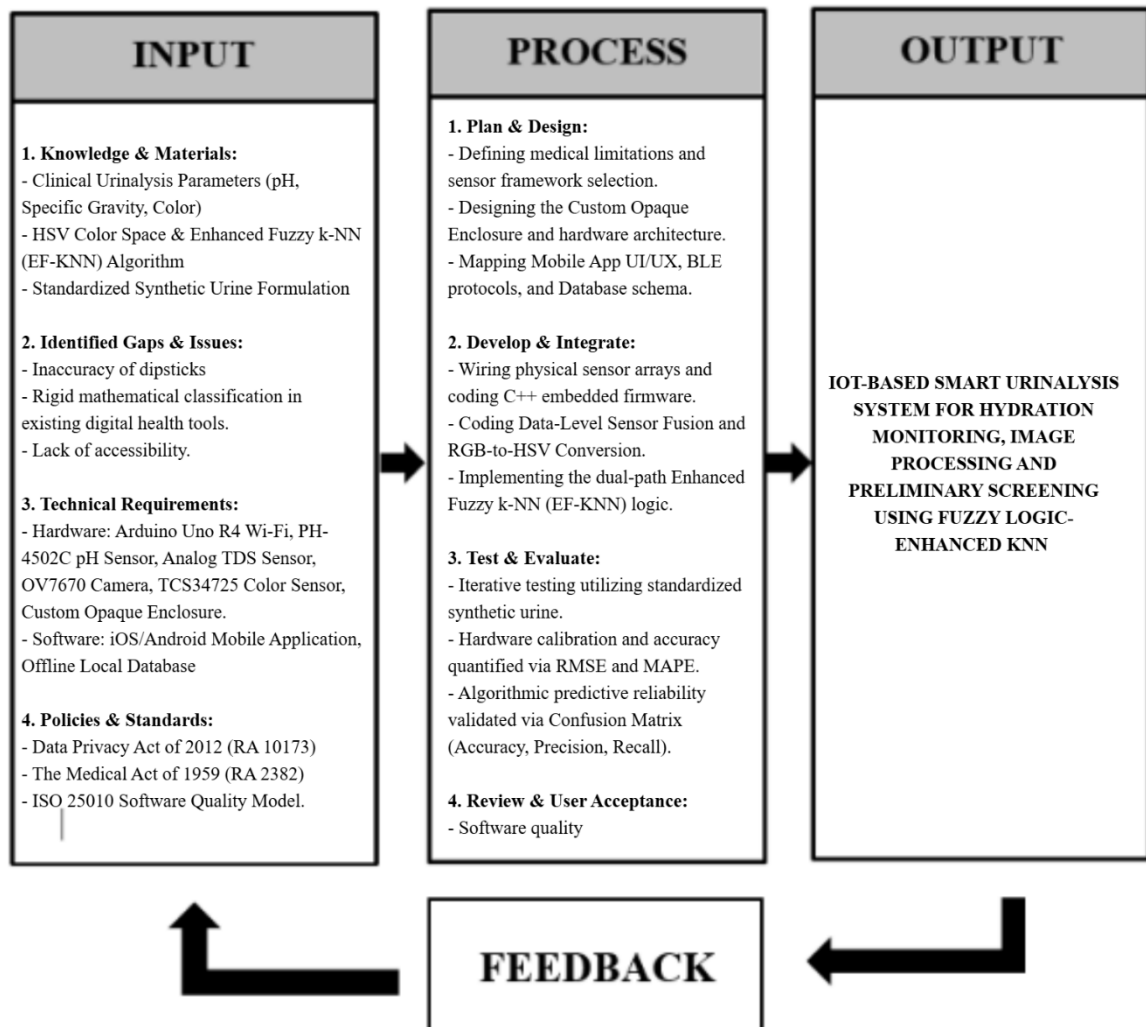
1. To design and calibrate an IoT-Based Smart Urinalysis prototype integrating a digital pH sensor, a TDS sensor (for Specific Gravity), and a dual-sensor optical array within a light-controlled enclosure, evaluating its raw sensor accuracy by calculating the RMSE and MAPE against reference tools (Digital pH Pen, Optical Refractometer, and an iPhone 16 Pro Digital Ground Truth).
2. To develop and validate a Bluetooth Low Energy (BLE) mobile application for the offline visualization of hydration status and preliminary screening, measuring system performance through Transmission Success Rate (packet delivery) and System Latency (response time) during real-time data synchronization.
3. To implement and test the Enhanced Fuzzy k-Nearest Neighbors (E-FKNN) algorithm to process fused sensor data for the categorization of hydration levels and the preliminary screening of physiological abnormalities, evaluating its predictive reliability using a Confusion Matrix to determine Accuracy, Precision, and Recall across standardized synthetic urine samples.
4. To assess the software quality of the developed mobile application based on the ISO 25010 model.

1.3 Conceptual Framework of the Study

By adopting the Input-Process-Output (IPO) model, this study visually delineates how foundational clinical requirements, specific hardware constraints, and strict legal boundaries are systematically transformed.

Figure 1.1

Input - Process - Output Diagram



As shown in figure 1.1, the input phase establishes the foundational elements required to initiate the study. This encompasses the knowledge & materials, which synthesize clinical urinalysis parameters (pH, Specific Gravity, and color) with advanced computational frameworks such as the HSV Color Space and the Enhanced Fuzzy k-

Nearest Neighbors (EF-KNN) algorithm, alongside the standardized synthetic urine formulation. It addresses the identified gaps & issues, directly targeting the inaccuracy of manual dipsticks, the rigid mathematical classification in existing digital health tools, and the overall lack of accessibility to diagnostic screening. Furthermore, the inputs define the broad technical requirements, specifying the necessary hardware (such as the Arduino Uno R4 Wi-Fi, various sensors, and the custom opaque enclosure) and software (iOS/Android mobile applications with an offline local database). Finally, it outlines the policies & standards, ensuring the system strictly adheres to the Data Privacy Act of 2012 (RA 10173), the Medical Act of 1959 (RA 2382), and the ISO 25010 Software Quality Model.

The process phase outlines the systematic workflow utilized to convert these inputs into the final system, heavily adapting an Agile development methodology. It begins with plan & design, which focuses on defining medical limitations, selecting sensor frameworks, designing the custom opaque enclosure, and mapping the mobile app UI/UX, BLE protocols, and database schema. The workflow then transitions into develop & integrate, where the physical sensor arrays are wired, C++ embedded firmware is coded, and the data-level sensor fusion, RGB-to-HSV conversion, and dual-path EF-KNN logic are strictly implemented. This is followed by test & evaluation, which involves iterative testing utilizing standardized synthetic urine. During this stage, hardware calibration and accuracy are quantified via Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE), while algorithmic predictive reliability is validated via a Confusion Matrix to determine accuracy, precision, and recall. The phase concludes with review & user acceptance, evaluating the overall software quality.

Finally, the output phase presents the culmination of the research. As indicated in the final block of the framework, the direct result is the fully developed project itself: the IoT-Based Smart Urinalysis System for hydration monitoring, image processing and preliminary screening using fuzzy logic-enhanced knn. Additionally, a feedback loop illustrates that iterative testing and user evaluation from the output phase are continuously used to refine the initial inputs and processing parameters until optimal system accuracy and reliability are achieved.

1.4 Scope and Delimitations

The scope of this study is centered on the design, development, and technical validation of a portable, IoT-Based Smart Urinalysis System utilizing multi-modal sensor fusion architecture. The system integrates hardware-level data acquisition powered by an Arduino Uno R4 Wi-Fi microcontroller, utilizing a pH Sensor, an Analog TDS Sensor, and a dual-optical array (Camera Module and Color Sensor) with a dedicated image processing module for RGB-to-HSV color space conversion. To eliminate ambient interference, all optical data capture and system validation occurred strictly within a custom-integrated opaque enclosure under uniformly controlled lighting conditions. The study implements an Enhanced Fuzzy k-Nearest Neighbors (EF-KNN) algorithm within an offline mobile application to process sensor data via Bluetooth Low Energy (BLE). This enables the automated classification of hydration status and the preliminary screening of physiological markers while utilizing a local database for secure, offline data persistence. Furthermore, the scope includes the physical calibration of the Arduino sensors and the in-house formulation of standardized synthetic urine samples. These samples were systematically

controlled using synthetic urine, sodium chloride, food-grade dyes, diluted acetic acid, and sodium bicarbonate, and were utilized exclusively to test the system's analytical capacity without human biological exposure. The evaluation of the system is strictly limited to technical reliability. Hardware precision and becomes "calibration accuracy was quantified using Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) against standard reference tools (digital pH pen, optical refractometer, and an iPhone 16 pro via ImageJ). Algorithmic predictive reliability was evaluated using a Confusion Matrix (Accuracy, Precision, Recall). Finally, the software's functional suitability, performance efficiency, compatibility, and reliability was assessed based on the ISO 25010 model by 100 general end-users and 30 subject matter experts. The mobile application supported iOS and Android devices running firmware and operating-system versions that meet the study's predefined minimum technical requirements, as older firmware versions may not reliably support Bluetooth Low Energy communication, image processing functions, and secure local data storage.

This study is strictly limited to the architectural design, algorithmic development, and technical validation of an IoT-Based Smart Urinalysis System. The system's capability is restricted to hydration monitoring and the preliminary screening of physiological abnormalities based entirely on pH, specific gravity (SG), and color. It is not legally intended to diagnose medical diseases; therefore, any condition that does not manifest measurable deviations in these three variables cannot be detected. Furthermore, biochemical parameters such as glucose, protein, ketones, nitrites, leukocytes and temperature are excluded, as their detection requires chemical reagent strips or specialized lab equipment incompatible with the system's non-invasive electronic sensors. The

algorithm evaluates real-time sensor values and cannot account for urine color alterations caused by recent diet, oral medication, or supplements. This study does not involve clinical human trials, longitudinal health monitoring, or patient pre and post surveys.

Due to the academic timeframe and the ethical constraints of primary clinical data collection, the testing phase relies exclusively on standardized synthetic urine samples. While this approach effectively proves the technical concept, the system is not validated against a diverse human demographic and does not account for complex biological variances found in a wider population. Additionally, no original medical dataset was formulated from human subjects. Instead, the predataset for the Enhanced Fuzzy k-Nearest Neighbors (EF-KNN) algorithm is aggregated from established clinical literature and peer-reviewed studies. Thus, the system's baseline reference relies heavily on the accuracy and representativeness of these secondary medical sources rather than empirical measurements generated by the proprietary hardware.

Finally, several hardware and technical constraints define the limits of the prototype. Measurement accuracy is inherently bound by the capabilities of consumer-grade Arduino modules rather than industrial or medical-grade instrumentation, and long-term calibration drift over extended use is not experimentally quantified. For software validation, mobile application testing is confined to the specific smartphone available to the researchers, meaning other device models and hardware variations are not evaluated. In the absence of an affordable clinical colorimeter, this study establishes a digital ground truth for color analysis using a high-resolution mobile CMOS sensor (iPhone 16 pro). To ensure methodological rigor, reference color coordinates are extracted using ImageJ, an open-source scientific image processing software widely utilized for multidimensional

analysis. While this methodology provides a highly practical and cost-effective baseline for validating the prototype's optical sensor array, it serves strictly as a benchmark and does not substitute for laboratory-grade spectrophotometric validation.

1.5 Significance of the Study

This study bridges the critical gap between cost-prohibitive clinical laboratory analyzers and the subjective inaccuracies of manual visual screening. By developing a multi-modal sensor architecture within a custom opaque enclosure and integrating it with an offline Enhanced Fuzzy k-Nearest Neighbors (EF-KNN) mobile application, the system delivers objective, preliminary screening of hydration status and urinary abnormalities. While the IoT-Based Smart Urinalysis System is a one-time unit cost which makes it a significant investment for a single household, the same cost becomes highly economical in shared settings, where one reusable device serves many individuals at minimal per-test cost. Its successful implementation specifically benefits the following:

Healthcare Facilities. Hospitals, diagnostic centers, and rural clinics manage continuous patient flow while relying on costly reagents and equipment for every test. This system serves as a first-line screening station that delivers objective hydration and acidity readings in seconds, allowing staff to identify dehydrated or at-risk patients before formal laboratory processing begins. By filtering routine cases early, the device reserves expensive laboratory resources strictly for patients it flags as high-risk. This reduces congestion during high-volume periods while lowering operational costs. As a shared, reusable unit, it remains economical across the large patient volumes these facilities handle.

Community and Public Health Programs. Local Government Units and barangay health workers deliver public health services across many communities on constrained budgets and often without laboratory infrastructure or stable internet. Because the device operates fully offline via Bluetooth Low Energy and interprets automatically, frontline workers can conduct rapid point-of-care screening directly in the field without specialized training. A single shared unit can support recurring health drives, senior check-ups, wellness campaigns, and emergency relief operations in evacuation centers at minimal per-test cost. This allows one device to serve thousands of residents over its service life, extending preventive care to underserved, geographically isolated, and disaster-affected areas.

Specialized Care and At-Risk Group Monitoring. Certain populations face elevated dehydration risk that often goes unnoticed until it becomes severe, including the elderly in geriatric facilities, and students, athletes, and laborers in physically demanding environments. This device enables caregivers and clinic staff to monitor these groups routinely and non-invasively, whether as ongoing care for vulnerable residents or as an on-site safety check before high-exertion activity. Concentrated urine can be flagged early, prompting timely correction of fluid intake. This proactive approach helps prevent complications such as urinary tract infections, heat-related illness, and avoidable hospitalization. It supports both individual welfare and institutional safety compliance.

Physicians and Medical Practitioners. Physicians require objective data to make informed decisions during initial patient consultations. This system provides immediate baseline metrics for pH, Specific Gravity, and HSV color data. Strictly adhering to the boundaries of the Medical Act of 1959, the device does not diagnose diseases. Instead, it

supplies physicians with reliable physiological indicators to guide their clinical judgment. This helps them quickly determine which patients require urgent and extensive laboratory workups.

Medical Technologists and Allied Health Workers. Medical technologists often interpret traditional chemical dipsticks manually, a process vulnerable to human error. Visual fatigue and inconsistent ambient lighting can distort color-based readings, especially during long shifts. This system automates urine assessment within a sealed, light-controlled environment. By removing subjective visual interpretation, it produces consistent and repeatable results. This improves both the speed and accuracy of the initial triage process, particularly during high-volume medical missions.

Patients and At-Risk Populations. Many patients face long waiting periods when relying solely on conventional laboratory routing. This system offers immediate, transparent, and non-invasive feedback through a real-time mobile application. It is particularly relevant to individuals whose conditions are reflected in hydration, urine concentration, and acidity, such as those with recurrent urinary tract infections, a tendency toward kidney stones, or chronic kidney concerns. These patients gain a clearer understanding of their hydration status and potential abnormalities. This empowers them to seek appropriate follow-up care sooner.

Individual and Home-Based Users. Individuals may choose to invest in the device for personal daily monitoring. Rather than relying on thirst or subjective guesses about urine color, they obtain exact digital readings of fluid concentration and acidity at home. This enables immediate lifestyle adjustments, such as increasing water intake before mild dehydration worsens. Because the application uses localized storage, users maintain a

private, offline log of their daily metrics. This historical record can later be shared with their physicians to support more informed consultations.

Future Researchers and Developers. This study contributes to the academic fields of Computer Science and Biomedical Engineering. It serves as a foundational reference for integrating the Enhanced Fuzzy k-Nearest Neighbors (EF-KNN) algorithm within low-cost microcontroller ecosystems. The research documents reusable methodologies, including the formulation of synthetic urine for non-clinical validation. It also details hardware error quantification using RMSE and MAPE alongside localized offline software architecture. Future scholars can adapt these methods to engineer accessible, legally compliant smart health innovations.

1.6 Definition of Terms

To ensure clarity and precision, the following terms are defined according to their usage in this study. Operational definitions describe how specific concepts are measured and applied within the context of the developed system, while technical definitions provide the standard scientific or theoretical meaning of the algorithms and biological parameters employed.

Operational Terms

IoT-Based Smart Urinalysis System. This refers to the IoT-Based Smart Urinalysis System in this study, which utilizes an Arduino microcontroller combined with sensors, camera and a mobile application to assess urine pH, specific gravity, and color of urine for the purpose of monitoring hydration levels and identifying possible abnormalities.

Hydration Status. This refers to the body's fluid balance level as identified by the system, which categorizes individuals as hydrated, mildly dehydrated, or dehydrated based on urine pH, specific gravity, and color measurements.

Mobile Application. This refers to the smartphone application developed for the study, which receives data from the Arduino based hardware, processes and classifies images, and presents hydration status along with preliminary health information to the user.

Preliminary Disease Detection. This refers to a non-diagnostic indication of potential urine abnormalities derived only from variations in urine pH, specific gravity, and color, and which must be confirmed by a licensed healthcare professional.

Synthetic Urine. This refers to a non-biological, laboratory-prepared liquid solution utilized in this study for sensor calibration and performance testing.

Technical Terms

Colorimetry. This is a quantitative technique that is used to analyze color information by measuring color intensity and variation from images. Applied in this study to assess urine color for hydration monitoring using smartphone-based image processing (Azhar et al., 2020).

Fuzzy Logic. It is a computational decision-making approach that allows partial truth values between 0 and 1, used in the study to combine urine pH, specific gravity, and color parameters for flexible hydration classification. (Utomo et al., 2024).

k-Nearest Neighbors (k-NN). This is a supervised machine learning algorithm that classifies data based on the closest data points in the feature space, utilized to classify

hydration levels by comparing urine parameters with reference datasets (Uddin et al., 2022).

Specific Gravity (SG). A physical urinary parameter that measures the concentration of solutes in the urine compared to pure water. In this study, it acts as a primary indicator of the kidney's concentrating ability and hydration status, measured via the Arduino sensor module (Queremel Milani & Jialal, 2023).

Urine pH. A measure of the acidity or alkalinity of urine. In this system, it is measured electronically to assess tubular function and screen conditions such as potential bacterial infections, which often result in elevated (alkaline) pH levels (National Library of Medicine, 2025).

HSV Color Space. This is a color representation model that consists of hue, saturation, and value that is used to improve robustness in urine color analysis by separating color information from lighting intensity. (Wu et al., 2023).

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of related literature and studies, both local and foreign, that are significant to the current research. It covers the medical significance of urinalysis parameters, the current state of renal health in the Philippines, and the technological advancements in portable health monitoring. This review provides the theoretical framework and justification for the development of an IoT-Based Smart Urinalysis System using Fuzzy Logic.

2.1 Foreign Literature

Clinical Baselines and the Medical Significance of Urinalysis

The foundational parameters of clinical urinalysis provide critical insights into a patient's systemic hydration and metabolic state. According to a comprehensive review by Hitzeman et al. (2022), the preliminary visual inspection and chemical analysis of urine are standard diagnostic procedures in primary care. The color of urine, driven by the concentration of urochrome, serves as a primary indicator of hydration. The importance of tracking this hydration extends far beyond renal health. A 2025 study by Sumitra et al. evaluated the impact of mild hydration deficits on the cognitive performance of young adults. The researchers found that participants with highly concentrated urine exhibited significantly degraded executive function, impaired working memory, and a 12% to 18% delay in neuromotor reaction times. Because the human brain is highly sensitive to fluid imbalances, the authors advocated for routine hydration monitoring to prevent cognitive

fatigue and diminished alertness commonly experienced in academic and occupational settings.

Figure 2.1

Urine Color Chart for Hydration Level Assessment (Adapted from Healthdirect Australia (2026))

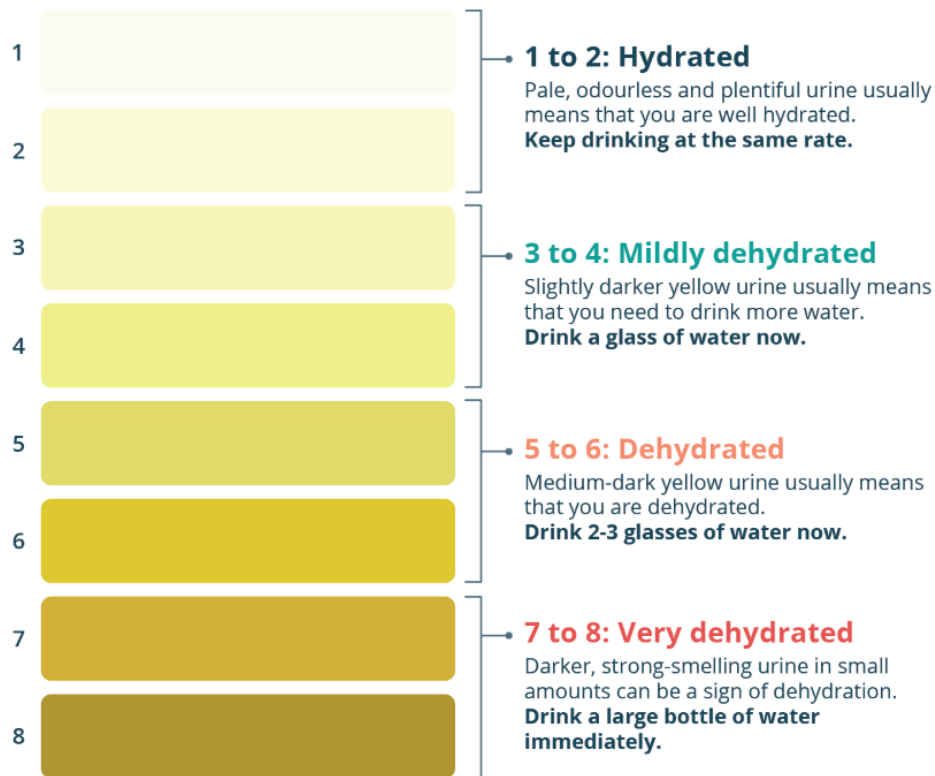


Figure 2.1 presents the standard eight-level urine color chart used to visually assess hydration status. Color levels 1 to 2 indicate a well-hydrated state characterized by pale and odorless urine, signifying that fluid intake is adequate and should be maintained at the same rate. Levels 3 to 4 represent mild dehydration, indicated by a slightly darker yellow urine color, suggesting that additional fluid intake is needed. Levels 5 to 6 denote a dehydrated state, where medium to dark yellow urine signals that the individual must consume two to three glasses of water immediately. Levels 7 to 8 represent severe dehydration, characterized by dark, strong-smelling urine produced in small amounts,

requiring urgent and immediate fluid replenishment. Notably, the chart also acknowledges that certain foods, medications, and vitamin supplements may alter urine color independent of hydration status, reinforcing the limitation of relying solely on visual color observation. This eight-level classification scale serves as the foundational reference for the hydration level framework adopted by the system, which replaces subjective visual color matching with objective sensor-based HSV colorimetry through the AS7341 spectral sensor, thereby eliminating human interpretation error and providing a more accurate and consistent assessment of hydration status.

To accurately quantify this fluid concentration, clinicians rely on Urine Specific Gravity (SG) as the definitive metric for evaluating urinary osmolality. Hitzeman et al. (2022) note that an SG near 1.010 indicates normal euvoemia, whereas values exceeding 1.020 signify systemic dehydration. Building upon this baseline, clinical guidelines from the Cleveland Clinic (2025) specify that an SG falling below 1.005 indicates excessive dilution often associated with impaired kidney function, while an SG exceeding 1.030 signals severe dehydration or restricted renal blood flow. Traditionally, these density parameters are assessed manually in clinical settings using optical refractometers or chemical dipsticks, both of which require trained medical personnel and are inherently prone to human interpretation errors.

Equally critical to fluid density is the evaluation of urinary acidity, which serves as a primary diagnostic metric for determining systemic acid-base balance. According to clinical standards outlined by the National Library of Medicine (2025), the physiological baseline for human urine pH ranges from 4.6 to 8.0. Deviations outside this established spectrum serve as critical indicators for severe physiological distress; highly acidic urine

frequently signals diabetic ketoacidosis, whereas highly alkaline urine is a primary marker for kidney failure or urinary tract infections (UTIs). However, as highlighted in a 2025 clinical review by Kim et al, conventional chemical dipsticks only measure pH in discrete 0.5 intervals. This lack of granularity severely restricts diagnostic precision, failing to capture the minute but clinically relevant pH fluctuations necessary for predicting the onset of Chronic Kidney Disease (CKD) and Renal Tubular Acidosis (RTA). To achieve the precision necessary for accurate medical assessment, the authors note that direct-contact electronic pH meters utilizing glass electrodes are required.

The continuous monitoring of both specific gravity and pH is clinically indispensable, as the intersection of these two metrics directly dictates the risk of nephrolithiasis (kidney stone disease). A 2024 review by Tamborino et al. established that the formation and recurrence of urinary calculi are heavily dictated by urine supersaturation and abnormal pH levels. High fluid concentration (indicated by elevated specific gravity) leads to mineral crystallization, while extreme pH deviations dictate the specific morphology of the stones: highly acidic urine frequently precipitates uric acid and calcium oxalate stones, whereas highly alkaline urine promotes the formation of calcium phosphate and infectious struvite stones. Thus, routine urinalysis of both variables is critical for identifying metabolic risk factors before physical stones can form.

Limitations of Traditional Diagnostics and Alternative Sensors

Urinalysis remains one of the most fundamental and widely utilized diagnostic tools for screening, diagnosing, and monitoring systemic and renal conditions. According to clinical guidelines provided by the Mayo Clinic (2023), a standard laboratory urinalysis is

divided into three primary evaluations: a visual examination of the fluid's color and clarity, a chemical dipstick test to measure critical biomarkers such as acidity (pH) and particle concentration (Specific Gravity), and a microscopic analysis. While these parameters which are color, pH, and concentration are clinically indispensable, the traditional testing methodology relies heavily on manual human observation and consumable chemical strips, both of which introduce significant potential for subjective interpretation errors.

This vulnerability to human error is particularly evident when relying on the visual observation of urine color (U_c) to estimate hydration status. A 2021 study by Wardenaar et al. evaluated the accuracy of self-reported urine color assessments among athletes utilizing standardized multi-shade color charts. The researchers found that human visual accuracy in classifying hydration status based solely on color matching was relatively poor, ranging only from 63.5% to 77.2%. The study concluded that subjective color perception is heavily distorted by ambient lighting conditions and container variables, governed by the Beer-Lambert law of light absorbance. Furthermore, the researchers established that Urine Specific Gravity (USG) is a vastly more sensitive and accurate clinical marker for dehydration than optical color alone, highlighting the inadequacy of relying purely on visual metrics.

To bypass the subjective limitations of manual urinalysis, significant engineering efforts have focused on developing non-invasive wearable sensors; however, recent comprehensive reviews indicate these technologies remain clinically unreliable outside of controlled laboratory environments. Gray et al. (2023) evaluated the diagnostic accuracy of emerging hydration-tracking modalities, including sweat composition analysis, skin bioimpedance, and saliva osmolality. The researchers concluded that sweat sensors suffer

from severe physiological inconsistencies, as electrolyte concentrations vary drastically based on diet and environment rather than true hydration status. Similarly, skin bioimpedance sensors are heavily skewed by ambient temperature, postural changes, and external moisture, rendering them ineffective for daily at-home monitoring. Because these external, non-invasive metrics are heavily confounded by environmental noise and high inter-subject variability, direct analysis of internal biological exudates (urine) remains the most viable approach for accurate screening.

Environmental Constraints and Pre-Analytical Sample Integrity

According to a clinical review by Queremel Milani and Jialal (2023) published via the National Institutes of Health, urinalysis remains one of the most cost-effective and critical diagnostic screening tools in modern medicine, providing a direct window into the body's metabolic and renal functions. The study defines Urine Specific Gravity (USG) as a primary measure of the kidney's ability to concentrate or dilute urine, with normal euvoletic values typically ranging between 1.002 and 1.035. Furthermore, urinary pH is identified as a vital indicator of renal tubular function, where deviations from the normal physiological range of 4.5 to 8.0 can signal systemic acid-base imbalances, dehydration, or predispositions to calculi formation. Importantly, the researchers note that urine is a highly unstable fluid; external interfering factors such as prolonged light exposure can cause the rapid decomposition of key chemical components, leading to false or inaccurate diagnostic readings.

Beyond the immediate threat of photodegradation, the diagnostic validity of any automated urinalysis system is inherently dependent on strict adherence to time-sensitive

sample handling protocols. A 2024 review by Skrajnowska and Bobrowska-Korczak emphasized that the physical and chemical characteristics of urine are highly volatile and susceptible to rapid environmental degradation over time. The researchers established that urine samples must be analyzed within a strict two-hour window when kept at room temperature. If processing is delayed beyond 2.5 hours, accelerated bacterial proliferation degrades urea into ammonia, artificially inducing highly alkaline pH readings and significantly increasing the fluid's turbidity. Such degradation inherently corrupts both physical pH probes and optical colorimeter readings, rendering delayed clinical analysis completely invalid.

Hardware Architecture for Point-of-Care Diagnostics

The democratization of medical diagnostics relies heavily on the integration of accessible, open-source hardware capable of processing real-time biological data. A 2025 review by Tsebesebe et al. analyzed the application of microcontroller-based devices in healthcare, concluding that the Arduino platform is the premier open-source hardware for developing Point-of-Care (PoC) biosensing equipment. The researchers identified that while microcomputers like the Raspberry Pi offer broader operating system capabilities, they suffer from higher costs, greater power consumption, and thermal management issues that compromise device dependability in remote settings. As a result, Arduino microcontrollers excel in rapid prototyping and real-time analog sensor interfacing, providing a highly stable, cost-effective foundation for embedded medical systems. Furthermore, the study emphasized that coupling Arduino-based sensor arrays with

machine learning algorithms enables the creation of highly sensitive, automated predictive models for disease diagnosis.

Beyond the initial platform selection, the transition of diagnostic tools from traditional laboratory environments to portable, point-of-care systems requires highly energy-efficient hardware architectures, particularly regarding signal acquisition. According to Arafa et al. (2023), the critical bottleneck in wireless biomedical sensors is the conversion of raw, continuous analog signals into processable digital data without exhausting battery life. Their comprehensive review establishes that the Successive Approximation Register Analog-to-Digital Converter (SAR ADC) is the optimal micro-architecture for medical applications. This specific architecture offers an unparalleled balance between high-resolution data conversion and ultra-low power consumption. While industrial clinical equipment relies on power-intensive processing, portable diagnostic systems must utilize these efficient analog-to-digital pathways to remain viable in the field.

Data Privacy and Ethical Considerations in mHealth

As point-of-care diagnostics increasingly rely on mobile health (mHealth) applications, ensuring the privacy and security of sensitive medical data has become a critical ethical imperative. A 2021 study by Bartlett Ellis et al. highlighted the severe regulatory and ethical vulnerabilities inherent in modern digital health technologies. The researchers emphasized that when health monitoring systems rely on cloud-based processing or third-party servers, both developers and users lose control over data management, exposing highly sensitive physiological information to severe privacy risks. To address these vulnerabilities, the study advocated for the strict application of the Digital

Health Checklist for Researchers (DHC-R), prioritizing secure data management and localized privacy control as foundational requirements for the development of any new medical technology.

2.2 Foreign Studies

Clinical Significance of Hydration and Renal Biomarkers

The physiological impact of inadequate fluid intake remains a prevalent, yet often undetected, public health issue. A 2022 study by Rauf and Lestaluhu evaluated the hydration status of primary school students and its direct correlation to cognitive and fine motor decline. Utilizing Urine Specific Gravity (USG) and Urine Color (UC) as their primary clinical biomarkers, the researchers identified that over 65% of the observed students were in a state of dehydration, with a significant portion exhibiting severe dehydration levels indicated by a USG of > 1.030 . Furthermore, the study established a strong, positive statistical correlation between the darkening of urine color and the elevation of specific gravity. The findings demonstrated that this severe dehydration directly impaired the subjects' visual attention, information processing, and fine motor functions, which were only restored upon rehydration. Compounding this physiological threat is a documented behavioral gap between hydration awareness and actual fluid intake. In a study of secondary students, Ahmeda et al. (2021) found that despite possessing a strong understanding of the clinical importance of water, most subjects consistently failed to meet daily hydration requirements, leading to chronic, undetected mild dehydration that negatively correlated with physical health metrics and academic performance.

A 2021 study by Nainggolan et al. investigated the diagnostic utility of Urine Specific Gravity (SG) in detecting subclinical kidney damage among heat-exposed factory workers. The researchers observed that chronic occupational heat stress led to repetitive dehydration, which subsequently caused elevated urinary nephrin which a primary biomarker for glomerular damage. Crucially, the study determined that a simple Urine Specific Gravity threshold of 1.018 served as a highly sensitive and specific predictor for this early-stage kidney injury. Because advanced laboratory assays for nephrin are cost-prohibitive and inaccessible for routine screening, the authors concluded that continuous SG monitoring is the most viable, cost-effective method for the early detection of renal impairment in vulnerable populations.

Limitations of Traditional and Wearable Diagnostics

While automated point-of-care urinalysis machines were historically developed to eliminate human visual bias, recent studies indicate that they inherit the fundamental flaws of the chemical dipsticks they analyze. A 2025 study by Cox et al. evaluated the efficacy of commercial automated dipstick readers in general practice settings. The researchers found that automated optical interpretation yielded a poor diagnostic specificity of approximately 41%, performing no better than subjective visual analysis. Furthermore, the study exposed significant hardware vulnerabilities: automated photometric readers frequently suffered from measurement errors caused by reagent smudging, where distinct chemical colors bled across adjacent test fields, disrupting the machine's fixed wavelength calibrations. These findings demonstrate that utilizing chemically reactive paper as an intermediary step inherently compromises digital accuracy.

To bypass these chemical diagnostics altogether, recent advancements in point-of-care diagnostics have explored wearable electronics for hydration monitoring; however, these methods struggle with severe environmental interference. A study by AlDisi et al. (2022) investigated the use of Bio-electrical Impedance Analysis (BIA) via wearable skin sensors to measure human hydration. Although theoretical simulations suggested that skin conductivity correlates with water retention, in-vivo human testing revealed highly inconsistent data. The BIA sensors failed to accurately classify mild dehydration, as the electrical capacitance readings were easily distorted by external variables such as sensor placement, physical movement, and fluctuating body temperatures, proving them too volatile for reliable, standalone clinical screening.

Optical Engineering and the HSV Color Space

The accurate optical evaluation of urine color is a critical component of automated diagnostics, though it presents significant engineering challenges regarding ambient light interference and hardware bias. A 2023 study by Wu et al. developed a cloud-based image processing system to automate the reading of multi-parameter urine test strips. The researchers identified that relying on the standard RGB (Red, Green, Blue) color model frequently resulted in misinterpretations, as RGB values are highly susceptible to distortions from uneven illumination and shadows. To resolve this, the study utilized the HSV (Hue, Saturation, Value) color space, demonstrating that separating the chromaticity (Hue) from the intensity (Value) provided a much more stable and accurate color comparison that closely mimics human vision.

Compounding these ambient light distortions, research conducted by Azhar et al. (2023) established that digital image-based colorimetry is severely bottlenecked by hardware inconsistencies. Their evaluation of urine colorimetry across five different smartphone models revealed that raw RGB values exhibited poor to moderate inter-device agreement, proving that uncalibrated mobile cameras are highly susceptible to optical distortion and hardware bias. However, once optical standardization is achieved through controlled environments, the diagnostic viability of urine colorimetry becomes highly accurate. In a clinical evaluation using a controlled imaging setup, Bustam et al. (2023) demonstrated that standardized, light-invariant color values yield "good" to "excellent" diagnostic accuracy, correlating strongly with clinical markers such as Specific Gravity (SG) and urine osmolality to achieve over 90% sensitivity in detecting fluid imbalances. Further emphasizing the necessity of controlled optical environments, Feng et al. (2022) developed a hydration monitoring system that isolates urine samples within a custom enclosure featuring a standard LED light source and a digital camera. Their study determined that converting raw image data into the CIE L*a*b* color space yields a strong correlation with Urine Specific Gravity, allowing them to map specific hydration levels to fluid density.

Table 2.1*Hydration Levels Based on Urine Specific Gravity (Adapted from Feng et al. (2022))*

Hydration level	Urine specific gravity	Hydration status
Level 1	1.000 – 1.005	Well hydrated
Level 2	1.005 – 1.010	Well hydrated
Level 3	1.010 – 1.015	Minimal dehydration
Level 4	1.015 – 1.020	Minimal dehydration
Level 5	1.020 – 1.025	Significant dehydration
Level 6	1.025 – 1.030	Significant dehydration
Level 7	1.030 – 1.035	Serious dehydration
Level 8	> 1.035	Serious dehydration

The transition from subjective visual urinalysis to objective digital colorimetry also reveals complex correlations with other dry chemical parameters. A 2025 study by Liu et al. demonstrated that quantifying urine color through three-dimensional digital color spaces provides a highly accurate, objective assessment of a patient's physiological state. The researchers observed that while Specific Gravity scaled positively with digitized color concentration, urinary pH exhibited a highly complex, non-linear relationship with the optical properties of the fluid, requiring advanced processing to accurately classify.

The Necessity of Multi-Sensor Fusion (Optical and Physical)

Recent medical literature emphasizes that combining multiple non-invasive urinary biomarkers yields diagnostic accuracy comparable to invasive clinical blood tests. A 2025

study by Keefe et al. evaluated the efficacy of multi-marker hydration assessments, specifically correlating urine color against clinical indices like Urine Specific Gravity (SG) and plasma osmolality. The researchers demonstrated that utilizing a combined-marker approach resulted in exceptionally high diagnostic specificity (ranging from 91% to 99%) for detecting dehydration across both male and female demographics. The study concluded that urine color and SG are robust indicators of systemic fluid balance that, when used together, reliably mirror invasive hematologic testing. However, practical clinical tools utilized in the study, such as the WUT (Weight, Urine Color, Thirst) Venn diagram, still rely heavily on the subjective human perception of color and thirst, which inherently introduces user error and limits automated tracking.

To eliminate this subjectivity, recent foreign studies have explored digitizing these visual parameters. Chew et al. (2020) demonstrated that the optical properties of urine, when quantified into digital color spaces within a controlled, light-blocking enclosure, share a highly significant correlation with laboratory-grade hydration markers such as SG and osmolality. Their research proved that digital color analysis achieves over 90% sensitivity in predicting dehydration severity, effectively eliminating the inaccuracies of subjective visual assessments. However, while their methodology successfully established this medical correlation, it relied heavily on manual image capture via generic smartphones and subsequent post-processing using desktop photo-editing software. Furthermore, relying solely on camera-based colorimetry to extract physical fluid density metrics presents severe limitations. Inagaki et al. (2025) evaluated the reliability of an automated smartphone program designed to read urinalysis test strips inside a dedicated enclosed photography box. While the optical program achieved high accuracy for standard chemical

markers such as glucose and protein, the study revealed a critical vulnerability in optical-only systems: the colorimetric method failed drastically in measuring Specific Gravity (SG), achieving a mere 14.4% exact concordance rate. The researchers concluded that determining fluid density (SG) through optical color comparison is inherently unreliable.

Hardware Enclosures, Calibration, and Accessibility

The modernization of point-of-care urinalysis is increasingly focused on eliminating the subjective vulnerabilities of traditional chemical dipsticks through enclosed, digital architectures. A 2024 study by Siu et al. detailed the development of a portable, digital urinalysis tool designed to quantify urinary biomarkers, including pH. The researchers identified that conventional dipstick analysis is inherently prone to error due to inconsistent ambient lighting, improper sample saturation, and subjective visual interpretation. To resolve these inconsistencies, they engineered a 3D-printed, enclosed colorimeter utilizing LEDs and photodetectors managed by an Arduino microcontroller.

This architecture successfully isolated the fluid samples from environmental interference, significantly outperforming commercial hospital dipstick readers by yielding highly accurate and quantifiable measurements. However, while the clinical efficacy and patient acceptability of such remote digital urinalysis tools have been well-documented, significant socioeconomic barriers to accessibility remain. A 2021 study by Stauss et al. evaluated the use of smartphone-based urinalysis for the remote management of patients with glomerular diseases. While the digital process successfully eliminated interobserver variability and facilitated at-home testing, the study highlighted critical limitations to current commercial systems: the proprietary test kits were prohibitively expensive, and the

system's reliance on cloud processing required a continuous internet connection, ultimately exacerbating the "digital divide" for vulnerable populations.

A 2025 study by Safii and Ghozali investigated the efficacy of an Arduino Uno-based pH sensor specifically designed for human urinalysis. Upon initial calibration with standard buffer solutions, the microcontroller-driven sensor demonstrated exceptional precision, achieving an accuracy rate of 99.89%. However, the researchers documented a slight increase in data deviation during the transition from pure buffer solutions to actual urine. This variance was attributed to the complex biochemical composition of urine which contains proteins, uric acid, and inorganic minerals that leave microscopic residue on the sensor's glass electrode. Furthermore, even when sensors are properly maintained, inherent hardware bias must be accounted for. A study by du Preez et al. (2020) evaluated the accuracy and interchangeability of various clinical refractometers in measuring mammalian urine Specific Gravity (SG). The researchers determined that while standard optical refractometers generally exhibit excellent diagnostic agreement, distinct hardware models can produce significant proportional bias and constant errors, leading to skewed concentration readings that require strict algorithmic benchmarking.

Algorithmic Processing: k-NN and Fuzzy Logic Classifiers

Table 2.2

Accuracy vs. Processing Time Trade-off in Sensor Classification (Adapted from Afuan & Isnanto, 2025)

Algorithm	Accuracy	Processing Time (s)
SVM	0.87	0.020
KNN	0.84	0.015
Decision Tree	0.81	0.010
Random Forest	0.89	0.030
LSTM	0.92	0.060
CNN	0.94	0.090

Table 2.2 illustrates the critical trade-off between diagnostic accuracy and computational efficiency when deploying machine learning models in sensor-based healthcare systems. According to Afuan and Isnanto (2025), evaluating six classification algorithms SVM, KNN, Decision Tree, Random Forest, LSTM, and CNN reveal distinct operational advantages depending on hardware constraints. As demonstrated in their data, CNN and LSTM achieved the highest accuracy (94% and 92%), but their longer processing times limit their suitability for real-time applications on portable edge devices. Random Forest balanced accuracy (89%) and efficiency, making it suitable for applications requiring both. SVM and KNN offered significantly faster processing times (0.020 seconds and 0.015 seconds, respectively), making them ideal for resource-limited systems.

For a portable, microcontroller-based diagnostic tool, the processing delay of deep learning models like CNN (requiring 0.090 seconds, or six times the processing time of

KNN) introduces unacceptable latency and power consumption. The exceptionally low computational footprint of the KNN algorithm establishes it as the superior foundational model for edge computing, provided its baseline accuracy (84%) can be algorithmically enhanced to approach the precision of heavier deep learning models.

Table 2.3

Comparative Studies of ML and AI Models

Author/Year	Application	Method / Model Used	Key Findings	Limitations
Afuan & Isnanto (2025)	Sensor-based Healthcare / Fall Detection	SVM, KNN, Decision Tree, Random Forest, LSTM, CNN	CNN/LSTM achieved the highest accuracy (94%). KNN achieved 84% accuracy with incredibly fast processing times (0.015s).	CNN/LSTM processing times are too slow and resource-heavy for edge devices.
Rekkas et al. (2021)	General Model Analysis	ANN, KNN, Naive Bayes, Decision Tree, Random Forest, CNN, RNN	KNN is a "lazy learner" requiring no initial training phase, making it easily implemented for multi-class categorization.	KNN is highly sensitive to environmental noise and high computational costs for massive datasets.

Kim et al. (2025)	Colorimetric Classification of Urine Strips	k-NN	Achieved over 97% accuracy for distinct chromatic shifts (glucose, nitrites).	Failed significantly when estimating physical fluid density (SG), yielding only 33.4% sensitivity.
Uddin et al. (2022)	Disease Prediction	Classic k- NN, Fuzzy k- NN	Fuzzy k-NN consistently outperformed classic k-NN (81.44% vs 80.93% accuracy) by utilizing probabilistic membership assignments.	Standard models struggle due to rigid majority voting on medical datasets.
Lu & Gweon (2025)	High- dimensional Data Classification	standard k- NN	Proved that standard distance calculations are essential for baseline accuracy.	Noisy and non- informative variables dilute calculations, increasing classification errors.

Airlangga (2024)	Diagnosing Urinary Tract Infections	SVM, XGBoost	Achieved up to 98.25% accuracy classifying urinary abnormalities using color, pH, and SG.	Relied entirely on retrospective laboratory data; not viable for real-time PoC hardware.
Nogalo et al. (2025)	Urine Crystal Detection and Classification	Logistic Regression, YOLOv8 (CNN)	Fused physical sensor data with deep learning image classification for 96.67% overall accuracy.	Deep learning models require heavy hardware (microscopes, Raspberry Pi) and high power.
Yudhana et al. (2023)	Optical Urine Glucose Identification	k-NN	Effectively classified varying urine color intensities with a 96.33% diagnostic accuracy via microcontrollers.	Relied heavily on consumable chemical reagents to induce color changes.
Ahad et al. (2025)	Stroke Prediction in	F-KNN, ANN (with	F-KNN achieved 96.3% accuracy,	F-KNN relies on distance-

Smart Healthcare	Chi-Square and Best First Search)	precision, and recall when optimized with feature selection, demonstrating highly robust medical classification.	based similarity, which is slightly less adaptable to massive, highly non-linear deep-learning patterns than ANN.
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Based on the comparative analysis presented in table 2.3, the k-Nearest Neighbors (k-NN) algorithm was selected as the foundational classification model for the IoT-Based Smart Urinalysis System. The primary justification lies in its operational efficiency; as a "lazy learner," k-NN does not require a resource-heavy initial training phase. This characteristic makes it highly suitable for deployment within a local, offline mobile application that must function with limited smartphone computational overhead. Furthermore, its documented advantage of being "easily implemented for multi-class categorization" perfectly aligns with the system's core requirement: accurately sorting physiological data into eight distinct hydration levels and screening for various health abnormalities without requiring the heavy architectural trial-and-error associated with Artificial Neural Networks (ANNs) or Convolutional Neural Networks (CNNs).

While Table 2.2 identifies critical limitations of k-NN, these drawbacks are actively mitigated by the study's design and proposed algorithmic enhancements. The limitation regarding "high computational cost for massive datasets" is rendered negligible, as the offline system processes a highly focused, controlled dataset of synthetic samples rather

than big-data streams. Most importantly, the algorithm's documented vulnerability of being "highly sensitive to environmental noise" serves as the direct rationale for this study's proposed intervention. By upgrading the baseline model to an Enhanced Fuzzy k-NN (EF-KNN), the system replaces rigid classification boundaries with continuous membership weights. This fuzzy logic integration effectively filters out the analog noise generated by the physical Arduino sensors, directly resolving the standard algorithm's primary theoretical weakness while retaining all its computational advantages.

The application of machine learning algorithms for optical urinalysis has proven highly effective for color classification, though it remains inherently limited when applied to physical fluid density. A 2025 study by Kim et al. evaluated a smartphone-based AI system that utilized the k-Nearest Neighbor (k-NN) algorithm to interpret standard urine test strips. The researchers demonstrated that the k-NN model is exceptionally proficient at colorimetric classification, achieving over 97% diagnostic accuracy for parameters with distinct chromatic shifts, such as glucose and nitrites. However, the study revealed a critical vulnerability in relying solely on optical AI: the algorithm failed significantly when attempting to measure Specific Gravity (SG), yielding a poor sensitivity of 33.4% and the lowest overall performance metrics of the entire study. Kim et al (2025) also noted that the standard k-nearest neighbor (k-NN) has limitations, increasing the dataset size did not always correspond to enhanced performance. It should be improved to perform well for all parameters with various methods.

Building upon these limitations, recent research has focused on comparing k-NN variants to identify more robust models for medical risk prediction. A 2022 study by Uddin et al. conducted a comparative performance analysis of k-NN variants using medical

benchmark datasets. Their findings highlight that specialized variants consistently outperform the standard k-NN in medical contexts. Specifically, the Fuzzy K-Nearest Neighbor (F-KNN) variant enhances diagnostic probability by utilizing membership assignments rather than rigid majority voting. This allows the algorithm to calculate probabilistic chances for a classification based on class frequency and weight, resulting in an average accuracy of 81.44% in disease-related datasets.

Table 2.4

Comparative Performance of k-NN Variants in Disease Prediction (Adapted from Uddin et al. (2022))

KNN Variant	Average Accuracy (%)	Average Precision (%)	Average Recall (%)
Classic k-NN	80.93%	77.53%	76.53%
Fuzzy k-NN	81.44%	78.54%	76.92%

This algorithmic vulnerability was further corroborated by Lu and Gweon (2025), who demonstrated that standard k-NN becomes increasingly less meaningful in complex, high-dimensional environments. Their research highlighted that as the number of features increases, noisy and non-informative variables often dilute the distance calculations essential for nearest-neighbor classification, leading to a significant rise in classification error.

Artificial Urine formulation in Experimental Studies

The use of synthetic urine has become increasingly prevalent as a substitute for authentic human urine. Gallardo Molina et al. (2026) introduced SimUrine, a fully defined artificial urinary medium designed to mimic the composition of human urine for

microbiological studies. The study emphasizes the limitations of using real urine, including variability between samples, potential pathogen exposure, and ethical concerns in human sample collection. The study introduced a chemically defined artificial urine medium specifically designed to simulate the biochemical environment of human urine. Additionally, the specific gravity values of commercially available synthetic urine were tested and found to range from 1.005 to 1.016, while the pH values ranged from 5.8 to 7.6. (Franke et al., 2023). And according to Soltani et al. (2023), the researchers used synthetic urine in their studies due to biohazard issues in real human urine. Their approach was to formulate synthetic urine recipes based on elemental urine composition. In addition to formulating synthetic urine, Noble et al. (2024) developed an artificial urine composition and adjusted its formulation according to typical urinary parameters. When used in controlled amounts, samples with a pH range of 4 to 8 represented well-hydrated urine, those within 5.5 to 6.4 indicated mildly concentrated urine, and samples with 4.5 to 5.4 represented dehydrated urine. Acidic samples were prepared by adding small amounts of vinegar or lemon juice until the pH level reached between 4.5 and 5.4. This method followed the approach described by Rajšić et al. (2020), who reported that acetic acid and citric acid effectively lower urine pH and influence its chemical properties. Alkaline samples were also prepared by adding small, measured amounts of baking soda or sodium bicarbonate to raise the pH to 6.5–7.5, consistent with the findings of Wissenbach and Steuer (2023), who noted that sodium bicarbonate reliably increases urine pH when used in controlled amounts. Sodium chloride (NaCl) was added to adjust the specific gravity, while yellow food coloring was used to control the urine color. adding 1.0 to 1.5 grams of NaCl per liter of distilled water produced a specific gravity between 1.000 and 1.010. When

the salt concentration increases to around 4.0 grams per liter, the specific gravity can rise above 1.035. The color variations were based on the findings of Kim et al (2025), who reported that using 0.05 grams of yellow food coloring per liter resulted in a pale-yellow tone, while 0.1 grams per liter produced a deeper brown shade. The pH levels were adjusted to represent three hydration conditions: well-hydrated (6.5–7.5), mildly concentrated (5.5–6.4), and dehydrated (4.5–5.4). This setup allowed the device to detect and classify hydration levels more accurately during the testing process.

Dilution as an Accepted Sample Preparation Method in Urine Testing

Tamama (2023) reviewed the dilute-and-shoot method as a recognized and validated sample preparation technique in toxicology testing, establishing that dilution is an officially accepted pre-treatment procedure for urine samples used across clinical, forensic, and doping control laboratories. The study categorizes dilution alongside other established pre-treatment techniques such as extraction, filtration, centrifugation, and solid-phase extraction, confirming that it is not an unconventional or experimental approach but rather a standard procedure practiced in busy, high-throughput laboratory settings. One of the most significant findings of the study is that the dilution process is non-selective and causes no analyte loss, meaning that every substance originally present in the urine sample is fully retained in the diluted specimen, only at a lower concentration. This is particularly important because it confirms that dilution does not alter or remove any of the key parameters being measured, preserving the diagnostic integrity of the sample. Furthermore, Tamama (2023) noted that urine is the most common specimen type processed through the dilute-and-shoot method precisely because of its complexity as a

biological matrix, containing highly variable levels of compounds influenced by hydration status, diet, medications, and environmental factors. These matrix components, if left unaddressed, can interfere with sensor or instrument readings and produce inaccurate results.

Sundström et al. (2024) developed and validated a comprehensive urine drug screening method employing a 5-fold dilution of urine samples prior to analysis, which was validated across 56 different drugs and 192 authentic clinical urine samples collected from real patients in a clinical laboratory setting. The study explicitly states that the main purpose of sample dilution is to reduce matrix interferences, allowing for better sensitivity and accuracy in measurement, and the validation results demonstrated acceptable performance across all key metrics including repeatability, mass accuracy, specificity, and stability. Notably, the study also found that the dilute-and-shoot method using simple dilution yielded comparable or better results than more complex and labor-intensive extraction methods such as solid-phase extraction, indicating that dilution is not only an accepted method but also a highly effective one that does not sacrifice accuracy for simplicity. The study further noted that since sample preparation in the dilute-and-shoot approach involves only dilution, analyte loss is minimal and recovery approaches nearly 100% for most compounds, further reinforcing the reliability of dilution as a preparatory step.

2.3 Local Literature

The Epidemiological Burden of Renal Disease in the Philippines

The escalating prevalence of renal disorders in the Philippines has created a critical public health burden, underscoring the urgent need for accessible early-detection technologies. According to a 2025 report by Naguna published via the Philippine Information Agency (PIA), the National Kidney and Transplant Institute (NKTi) estimates that approximately 2.3 million Filipinos are currently at risk of developing Chronic Kidney Disease (CKD). The severity of this crisis is further emphasized by a 2024 health advisory published by the Philippine Society of Nephrology (PSN), which highlights that the incidence of CKD has reached critical levels, with approximately one Filipino developing chronic renal failure every single hour. A primary clinical challenge in mitigating this crisis is the asymptomatic nature of early-stage CKD; because individuals typically experience no physical symptoms during the onset of renal decline, severe underdiagnosis remains a persistent issue nationwide.

To combat this silent progression, local nephrologists and health officials strongly advocate proactive preventative measures, specifically emphasizing the continuous monitoring of hydration and the necessity of routine community-level urinalysis. Identifying early warning signs of metabolic and renal decline is the most effective strategy to prevent irreversible kidney damage and prohibitive hospitalization costs. However, relying on traditional clinical laboratories or sporadic medical missions poses a significant barrier to consistent monitoring. Many individuals delay or forget diagnostic testing entirely due to the time constraints of daily life and the logistical inconveniences associated with formal facility visits, effectively preventing equitable access to early intervention.

Local Innovations in Microcontroller and Sensor-Based Diagnostics

Recently local engineering initiatives have emphasized the integration of multi-modal hardware for urinalysis to improve diagnostic accuracy. A 2025 study by Nogalo et al. from the University of Southeastern Philippines developed a microcontroller-based diagnostic system combining chemical sensors (pH and turbidity) with optical imaging to assess kidney stone risks. Utilizing a Raspberry Pi, the system processed physical sensor data via logistic regression while simultaneously running a YOLOv8 deep learning model to classify microscopic urine crystals, achieving an impressive overall diagnostic accuracy of 96.67%. This dual-modality approach successfully proves that combining physical sensor readings with optical data yields highly accurate medical diagnostics.

However, while these local iterations are highly accurate, they frequently remain computationally and physically burdensome. The architecture developed by Nogalo et al. requires a dedicated optical microscope to capture the crystalline structures, computationally intensive deep-learning processing to execute the YOLOv8 model, and a locally hosted web server to manage the data. These heavy hardware and software dependencies severely limit the device's portability, making it difficult to deploy as a rapid, decentralized tool in resource-constrained community settings.

Republic Act No. 10173 (Data Privacy Act of 2012)

Republic Act No. 10173 protects the personal information of individuals and regulates communications systems in both the government and private sectors. Section 2 of the Act recognizes the vital role of information and communications technology while mandating that the personal information of all individuals remains secure. Furthermore,

Section 3 requires explicit, documented consent from the data subject prior to the collection and processing of their personal information. Because urinalysis involves highly sensitive health-related data (Section 4), the proposed IoT-Based Smart Urinalysis System is engineered specifically for localized privacy. By housing the Enhanced Fuzzy k-NN (EF-KNN) classification algorithm entirely within an offline mobile application, the system prevents sensitive physiological data from being exposed to vulnerable cloud servers. Furthermore, to ensure absolute compliance during the system's development and validation phases, the researchers utilized only their own biological samples, thereby avoiding the processing of third-party sensitive personal information while adhering to the research exemptions provided under the Act.

Republic Act No. 2382 (The Medical Act of 1959)

In addition to data privacy, the deployment of diagnostic technology must strictly observe the legal boundaries of medical practice. Republic Act No. 2382 establishes the legal framework for medicine in the Philippines. Specifically, Section 10 dictates that only individuals possessing a valid medical license are authorized to diagnose medical conditions or prescribe treatments. To guarantee full compliance with this mandate, the proposed IoT-Based Smart Urinalysis System is strictly classified as a preliminary screening tool, not a diagnostic medical device. The system's algorithmic functions are limited to measuring physical urine parameters (pH, Specific Gravity, color). It provides objective health indicators but explicitly does not diagnose specific diseases or prescribe therapeutic interventions.

2.4 Local Studies

The Epidemiological Burden: Dehydration and Renal Diseases in the Philippines

While chronic renal diseases represent the severe end of urological health, acute dehydration remains a widespread and critically overlooked public health crisis in the Philippines. A 2022 study by Sangalang et al. assessed the health impacts of inadequate water, sanitation, and hygiene (WaSH) infrastructure among public school students in Metro Manila. Utilizing point-of-care urine test strips, the researchers measured Urine Specific Gravity (USG) as their primary clinical biomarker for hydration, defining clinical dehydration at a USG threshold of ≥ 1.020 . The findings were alarming: 68% of the 1,558 students tested exhibited highly concentrated urine indicative of acute dehydration. The researchers attributed this to "voluntary dehydration," where students purposefully restrict fluid intake to avoid using poorly maintained school restrooms. This massive local prevalence underscores the urgent need for accessible, continuous hydration monitoring, though the study's reliance on manual chemical test strips for mass screening inherently exposed the data to logistical and visual interpretation errors.

Compounding the physiological threat of dehydration is a severe local burden of infectious urological diseases that heavily rely on these same basic urinalysis parameters for early detection. A 2020 study by Cano et al. investigated the prevalence of Urinary Tract Infections (UTIs) among marginalized indigenous groups in the country. Utilizing manual chemical dipsticks to measure urinary pH and specific gravity, the researchers discovered an alarmingly high UTI prevalence rate of 51.2%. The study emphasized that because geographically isolated communities lack consistent access to formal healthcare facilities, rapid point-of-care urinalysis is critical; yet the reliance on visual dipstick

interpretation remains highly susceptible to human error and environmental degradation. Beyond common UTIs, urinalysis serves as a vital frontline defense against highly destructive, insidious infections.

A 2021 study by So and Villanueva at the Philippine General Hospital (PGH) analyzed patients with Genitourinary Tuberculosis (GUTB), noting that delayed diagnoses frequently result in high rates of mortality and structural organ failure. Crucially, the researchers found that up to 90% of patients exhibited abnormal urinary parameters specifically, a high prevalence of acidic urine (45.63%) and abnormally low specific gravity (31.07%) proving that routine pH and USG monitoring are indispensable clinical indicators for flagging asymptomatic renal infections.

Healthcare Disparities and the Need for Point-of-Care Diagnostics

Despite national mandates intended to provide universal healthcare, a severe disparity exists in the availability of essential diagnostic tools across the Philippine archipelago. According to Alberto et al. (2022), while basic screening procedures such as urinalysis are theoretically covered under the Philippine Health Insurance Corporation's Primary Care Benefit Package, rural and geographically isolated primary care facilities remain critically ill-equipped to perform them. This diagnostic bottleneck is driven by severe underfunding, a lack of trained medical technologists in rural sectors, and the logistical failure to distribute hospital-grade diagnostic machines outside of highly urbanized centers in Luzon. Due to this systemic lack of accessible infrastructure, millions of Filipinos face prohibitive out-of-pocket costs or delayed diagnoses for manageable

conditions like urinary tract infections and renal diseases, underscoring the urgent need for decentralized, point-of-care medical technology.

To bridge this diagnostic gap, recent local engineering initiatives have explored low-cost microcontroller-based healthcare devices. A 2024 study by Estoquia et al. demonstrated the viability of this approach through the development of "PeeCheck 2.0," a portable urine analyzer designed for rural health units. By utilizing a color sensor integrated with a central microcontroller, the device successfully automated the reading of standard 14-parameter chemical reagent strips, yielding a high overall accuracy of 92.47% against conventional laboratory urinalysis. However, while the system effectively addressed initial hardware costs, its architecture relied heavily on continuous Wi-Fi connectivity for cloud-based data storage, and its diagnostic capacity was entirely dependent on a constant supply of consumable chemical dipsticks. These dependencies present significant logistical barriers, as deeply isolated communities frequently lack both stable internet infrastructure and reliable medical supply chains.

Local Innovations in Microcontroller and Sensor-Based Diagnostics

The application of machine learning algorithms within microcontroller-based fluid analysis systems has been successfully demonstrated in recent regional engineering initiatives. A 2023 study by Yudhana et al. developed an Arduino-based prototype to non-invasively classify urinary glucose levels as an alternative to traditional blood-draw methods. The researchers utilized an AS7262 multi-spectral color sensor to detect the precise optical changes in the urine samples after chemical reagent processing. To analyze the sensor data, the system employed the K-Nearest Neighbor (k-NN) machine learning

algorithm, which calculated the Euclidean distance between the real-time sample data and the nearest historical training data points to determine the disease classification. The study reported that the k-NN algorithm effectively classified the varying intensities of the urine colors, achieving an overall diagnostic accuracy of 96.33%, strongly validating that proximity-based classification models are highly effective even when deployed on lightweight, low-cost microcontrollers.

Building upon the proven efficacy of algorithmic analysis, recent local studies have further validated the use of machine learning to reduce manual interpretation errors in Philippine urinalysis. A 2024 study by Airlangga demonstrated that machine learning models, specifically Support Vector Machines (SVM) and XGBoost, can achieve up to 98.25% accuracy in classifying urinary abnormalities based on parameters such as urine color, pH, and specific gravity. However, while this study validates the advanced algorithmic processing of urine biomarkers, it relies entirely on retrospective, manually gathered laboratory datasets. This approach does not resolve the logistical bottleneck of real-time data acquisition in resource-limited clinics, highlighting a critical gap between high-level algorithms and point-of-care hardware.

To address the need for real-time fluid data acquisition in local settings, researchers have successfully utilized microcontroller-based sensor arrays. In a recent study, Calajate et al. (2024) engineered a portable, Arduino-based multi-sensor kit designed to monitor the water quality of aquaculture farms in Camarines Norte. Their prototype, which integrated physical sensors for pH and Total Dissolved Solids (TDS), demonstrated high precision with minimal standard deviation during field testing, leading experts to rate the system as "Highly Acceptable" in terms of reliability and portability. However, while the study

validated the hardware's accuracy, the researchers noted that the system relied solely on raw data display and basic LED alerts, explicitly recommending the future integration of mobile applications for deeper, automated data analysis.

2.5 Synthesis

The comprehensive review of both foreign and local literature establishes a clear and urgent directive for the modernization of point-of-care (PoC) urinalysis. Clinical studies confirm that foundational parameters, specifically Urine Specific Gravity (SG) and pH, are highly sensitive biomarkers critical for detecting dehydration, chronic kidney disease, and infectious urological conditions (Hitzeman et al., 2022; Sumitra et al., 2025; Tamborino et al., 2024). Locally, epidemiological data underscores a severe public health concern in the Philippines, worsened by significant healthcare disparities that leave rural and marginalized communities without access to standard laboratory infrastructure (Alberto et al., 2022; Naguna, 2025; Sangalang et al., 2022). While recent engineering initiatives have attempted to decentralize diagnostics, existing solutions possess critical vulnerabilities. Manual dipsticks are inherently subjective and prone to interpretation errors (Kim et al., 2025; Wardenaar et al., 2021), wearable biosensors suffer from severe external interference (AlDisi et al., 2022; Gray et al., 2023), and camera-based colorimetry frequently fails to accurately estimate physical fluid density (Inagaki et al., 2025; Kim et al., 2025). Furthermore, current digital PoC alternatives heavily rely on expensive chemical consumables (Estoquia et al., 2024), computationally burdensome architectures such as deep-learning microscopes (Nogalo et al., 2025), or constant cloud connectivity that threatens data privacy (Bartlett Ellis et al., 2021; Stauss et al., 2021).

The literature reveals a distinct research gap: there is a critical need for a fully automated, multi-modal diagnostic tool that is highly accurate, computationally efficient, and completely independent of chemical reagents and internet connectivity. The proposed IoT-Based Smart Urinalysis System is engineered specifically to fill this void. By adopting an Arduino-based enclosed hardware architecture (Tsebesebe et al., 2025) that fuses controlled lighting and the HSV color space (Wu et al., 2023) with direct-contact physical probes for pH and SG (Siu et al., 2024), the system eliminates both visual subjectivity and the documented failures of optical-only density estimation. Furthermore, by processing these uncorrupted physiological metrics through an offline Enhanced Fuzzy k-NN (EF-KNN) algorithm, the device transcends the mathematical rigidity and noise susceptibility of standard machine learning classifiers (Lu & Gweon, 2025; Uddin et al., 2022). Finally, by ensuring localized, offline data processing and restricting its functionality to preliminary biomarker screening, the system strictly adheres to the regulatory boundaries of the Data Privacy Act of 2012 and the Medical Act of 1959. Ultimately, this system operationalizes the global demand for automated diagnostics into an accessible, low-cost, and legally compliant tool tailored to address the urological health concerns of resource-constrained Philippine communities.

Chapter 3

METHODOLOGY

This chapter outlines the framework and processes used in this study, focusing on the IoT-Based Smart Urinalysis System. This section includes requirements analysis, project design, system architecture, software development methodology, data gathering techniques, evaluation procedures, and statistical treatment of data.

3.1 Requirements Analysis

Operational Feasibility

The operational feasibility of the IoT-Based Smart Urinalysis System is highly viable, as end-user access requires only proprietary IoT-Based hardware and a Bluetooth-capable smartphone. The system is designed to require minimal technical or medical expertise; it was designed with a highly intuitive user interface (UI) to facilitate seamless navigation. Furthermore, because the mobile application operates entirely offline utilizing a local database, users in resource-constrained or geographically isolated areas can conduct tests and securely log their physiological data without relying on a stable internet connection. To further maximize usability, the application integrated step-by-step testing instructions and provided clear, non-diagnostic interpretations of the generated health flags. With these accessible features, a diverse demographic ranging from the public and students to Barangay Health Workers (BHWs) were able to effectively operate the system for routine hydration monitoring and preliminary health screening.

Technical Feasibility

Table 3.1

Hardware and Software Specifications for Developers

Requirements	Description
Hardware	
Processor	Apple M4 Chip (Mac) AND Intel Core i5 (Windows)
Memory	8 GB RAM
Storage / Ports	256 GB SSD / Minimum one USB Type-C Port
Test Devices	Physical iOS device (iOS 17+) AND Physical Android Phone (Android 13+)
Microcontroller	Arduino Uno R4 Wi-Fi
Optical Sensors	ESP32 Camera & AS7341 11-Channel visible light sensor
Physical Sensors	PH-4502C pH Sensor & Analog TDS Sensor
Software	
IDE / Code Editor	Xcode (v26.2), Android Studio (panda), Arduino IDE (v2.3.7+)
Programming Languages	Swift (iOS), Kotlin (Android), AND C++ (Arduino)
UI Framework	Swift UI (iOS) & XML (Android)
Database / Storage	SQLite (Swift Data for iOS / Room for Android)
Operating System	macOS Sequoia AND Windows 10/11

Table 3.1 details the hardware and software specifications required to develop the mobile applications and the embedded firmware for the IoT-Based Smart Urinalysis System. An Apple machine powered by an M4 Chip running macOS Sequoia is required

to utilize Xcode (v26.2) for iOS development, while a standard Windows 10/11 machine equipped with an Intel Core i5 processor is utilized to run Android Studio (panda) and compile the C++ firmware via the Arduino IDE (v2.3.7+). A baseline of 8 GB RAM and 256 GB SSD ensures the integrated development environments run smoothly, while a USB Type-C port is specifically required to physically interface with and flash the microcontroller.

For software architecture, the mobile applications utilize Swift and SwiftUI for iOS, alongside Kotlin and XML for Android, while securely maintaining local offline data through SQLite integrations specifically, Swift Data for iOS and Room for Android. The embedded hardware, programmed in C++, is powered by the Arduino Uno R4 Wi-Fi, which processes real-time data from an integrated suite of optical sensors (ESP32 Camera & AS7341 11-Channel visible light sens) and physical sensors (PH-4502C pH Sensor and Analog TDS Sensor). Finally, physical test devices, specifically an iOS device running iOS 17 and an Android phone running Android 13 are the requirements for on-device system testing and evaluation.

Table 3.2

Minimum Hardware and Software Specifications for Users

Requirements	Description
Hardware	
Mobile Device	Android Smartphone or Physical iOS device
Memory	8 GB RAM
Software	
Operating System	Android 13+ / iOS 17+
Connectivity	Bluetooth 4.0+

Table 3.2 outlines the minimum hardware and software specifications required for end-users to successfully install and operate the IoT-Based Smart Urinalysis System mobile application. To ensure the smooth execution of the offline classification algorithms and local database management, users must utilize an Android smartphone or a physical iOS device equipped with a minimum of 8 GB of RAM. Furthermore, the device operating systems must be updated to at least Android 13 or iOS 17 to guarantee native system support and app compatibility. Finally, a Bluetooth 4.0+ connection is a strict requirement, as the system relies exclusively on Bluetooth Low Energy (BLE) protocols to establish a stable, wireless data transmission pipeline from the Arduino hardware directly to the user's smartphone.

Economic Feasibility

Table 3.3

Project Development Cost Breakdown

Item / Description	Quantity	Total Cost (PHP)
Hardware costs		
Arduino Uno R4 Wi-Fi Kit	1	₱ 1,700.00
PH-4502C pH Sensor	1	₱ 1,100.00
Analog TDS Sensor	1	₱ 4990.00
ESP32 Camera	1	₱ 456.00
AS7341 11-Channel visible light sensor	1	₱ 1579.00
Wires (Male-to-Male)	1	₱ 60.00
Wires (Male-to-Female)	1	₱ 60.00
Estimated Cost:		₱ 9,945

Software costs		
Android Studio / Arduino IDE / Xcode	-	-
SQLite Database (Local Storage)	-	-
Figma (UI/UX Design)	-	-
Claude Pro Subscription	1	₱ 1300
Total Estimated Cost:		₱ 11,245.00

Table 3.3 presents the detailed cost breakdown for the development of the IoT-Based Smart Urinalysis System. The total one-time development cost of ₱ 11,245.00 represents the full investment required to produce the IoT-Based Smart Urinalysis System, inclusive of all hardware components and the sole recurring software expense. Software development environments specifically for Android Studio, Xcode, the Arduino IDE, and Figma for UI/UX design were utilized at no cost under their respective free and open-source licenses. Local SQLite databases were implemented on both mobile applications for offline data storage, eliminating the need for paid cloud-hosting subscriptions entirely.

Table 3.4

Cost Benefit Analysis

	IoT-Based Smart Urinalysis System	Manual Dipstick	Clinical Lab Urinalysis
Unit / One-Time Cost	₱ 11,245.00	₱ 5.00 – ₱ 15.00 per strip	₱ 150.00 – ₱ 350.00 per test
Per-Test Consumable Cost	₱ 8.00 – ₱ 25.00 (urine cup)	Included in strip price	Included in lab fee
Total Per-Test Cost (after purchase)	₱ 8.00 – ₱ 25.00	₱ 8.00 – ₱ 25.00	₱ 150.00 – ₱ 350.00
Reusable Hardware	Yes	No	No

Multi-User Capability	Yes (shareable across multiple users)	No (single use per strip)	Yes (lab serves many patients)
Requires Clinical Personnel	No	No	Yes
Result Turnaround	~8.6 seconds (BLE)	~2 minutes (manual read)	3–48 hours
Objective Classification	Yes (EF-KNN)	No (visual interpretation)	Yes
Home-Based Use	Yes	Partial	No
Multi-Parameter (pH + SG + HSV)	Yes	Partial (pH + SG only)	Yes

Table 3.4 presents a comparative cost-benefit analysis between the IoT-Based Smart Urinalysis System and the two most common traditional urinalysis modalities available in the Philippines: manual dipstick testing and clinical laboratory urinalysis. While the system requires a one-time development investment of ₱ 11,245.00, its per-test consumable cost of ₱ 8.00 to ₱ 25.00, limited to a sterile urine specimen cup is comparable to the per-strip cost of manual dipstick testing and significantly lower than the ₱ 150.00 to ₱ 350.00 charged per clinical laboratory urinalysis session. Unlike both traditional methods, the hardware is fully reusable across an unlimited number of tests, meaning the one-time device cost is amortized over every subsequent use. Assuming a monitoring frequency of twice per week, the annual consumable cost of the system ranges from approximately ₱ 832.00 to ₱ 2,600.00, compared to ₱ 15,600.00 to ₱ 36,400.00 for equivalent clinical laboratory testing, yielding a break-even point of approximately 4 to 9 months.

Beyond cost, the system demonstrates clear operational advantages over its traditional counterparts. Clinical laboratory urinalysis requires licensed medical personnel, a physical visit to a diagnostic facility, and a result of turnaround of 24 to 48 hours, rendering it impractical for frequent home-based monitoring. Manual dipstick testing, though accessible, relies on subjective visual interpretation that is inherently prone to human error, and only partially covers pH and Specific Gravity parameters without HSV color classification. The system addresses all of these limitations by delivering automated, objective, multi-parameter classification across pH, Specific Gravity, and HSV color channels through the EF-KNN algorithm, with a mean result turnaround of approximately 8.6 seconds via Bluetooth Low Energy transmission and without requiring clinical personnel or facility access, collectively establishing its economic viability as a practical and affordable alternative for home-based urinalysis.

Schedule Feasibility

Figure 3.1

Gantt Chart (phase I)

ACTIVITY/TASK	START	END	DEC	JAN					FEB				MAR		
			W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	
PHASE 1															
Adviser Selection	12/15/2025	12/15/2025													
Chapter 1	1/5/2026	1/17/2026													
Chapter 2	1/12/2026	1/24/2026													
Chapter 3	1/19/2026	1/31/2026													
Preparation	2/2/2026	2/21/2026													
Mock defense	2/23/2026	2/28/2026													
Project defense	3/2/2026	3/7/2026													
Revision and submission	3/9/2026	3/21/2026													

Figure 3.1 illustrates the first phase of the project schedule, the researchers allocated to pre-development and the formal research proposal. Spanning from December 2025 to March 2026, the researchers allocated time during this phase to draft the

preliminary chapters and secure adviser approval. This period culminated in a four-week preparation for the mock defense, ensuring the researchers were fully prepared for the official Project Proposal Defense in early March. Following the defense, the researchers dedicated a two-week period to integrating panelist recommendations, revising the manuscript, and submitting the final proposal document.

Figure 3.2

Gantt Chart (phase 2)

ACTIVITY/TASK	START	END	APR			MAY				JUN					JUL	
			W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
PHASE 2																
Hardware development and testing	4/13/2026	5/9/2026														
Application development and testing	4/13/2026	5/9/2026														
Quality Evaluation	5/11/2026	5/23/2026														
Chapter 4	5/18/2026	5/30/2026														
Chapter 5	5/25/2026	5/30/2026														
Preparation	6/1/2026	6/6/2026														
Mock defense	6/8/2026	6/13/2026														
Final defense	6/15/2026	7/11/2026														
Revision and submission	7/13/2026	7/18/2026														

Figure 3.2 details the second phase of the schedule, which focused entirely on core system construction, evaluation, and final oral defense. The researchers initiated a parallel four-week timeline to simultaneously assemble the embedded hardware and develop the mobile application. This synchronized approach ensured that the researchers fully integrate the IoT sensors with the application before conducting the two-week Quality Evaluation phase in mid-May. Finally, the researchers allocated dedicated weeks for statistical data analysis, manuscript finalization (Chapters 4 and 5), and mock defense preparations, leading up to the final oral defense and official manuscript submission in July.

3.2 Project Design

Figure 3.3

Use Case Diagram

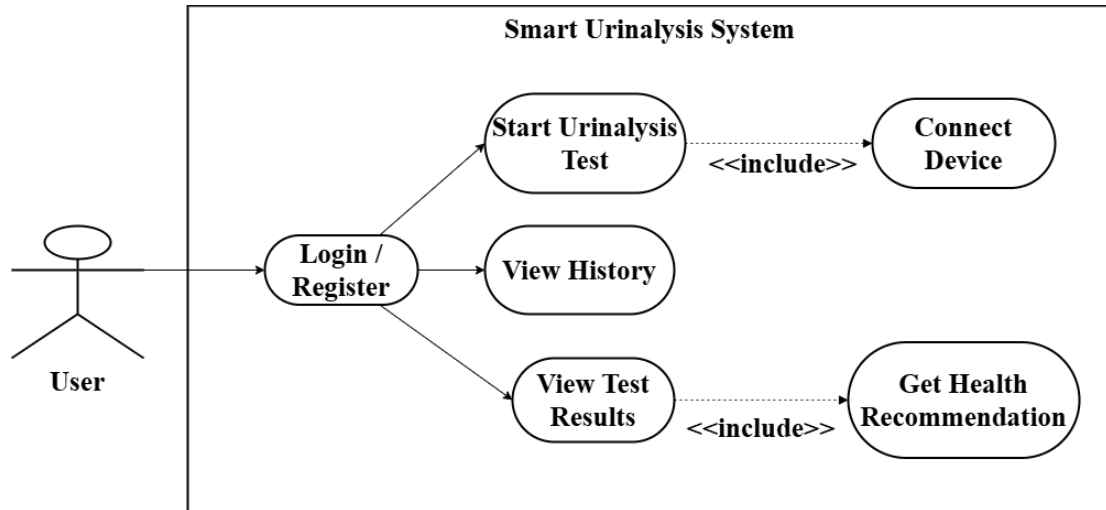


Figure 3.3 illustrates the Use Case Diagram for the IoT-Based Smart Urinalysis System, mapping the direct interactions between the User (the primary actor) and the application's core functions. The workflow mandates that the user first authenticates through the Login / Register. Upon successful access, the user can initiate three primary use cases: Start Urinalysis Test, View History, and View Test Results. The system utilizes **<<include>>** dependencies to denote mandatory background processes. Specifically, the Start Urinalysis Test use case incorporates the Connect Device function, ensuring that the mobile application successfully establishes a secure connection with the hardware before any sensor data acquisition can commence. Similarly, accessing the View Test Results triggers Get Health Recommendation, guaranteeing that the user is provided with actionable, contextual health advice alongside their calculated hydration status and screening data.

Figure 3.4

Data Flow Diagram Level 0 (Context Diagram)

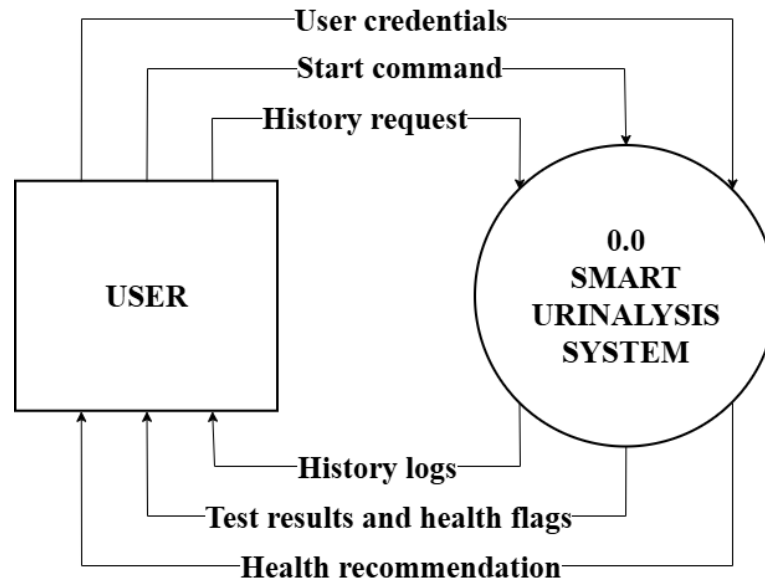


Figure 3.4 illustrates the Data Flow Diagram (DFD) Level 0, also known as the Context Diagram, for the IoT-Based Smart Urinalysis System. This diagram defines the system's scope by representing the entire diagnostic process as a single entity (Process 0.0) interacting with one external entity: the User / Patient. The data flow demonstrates the exchange of information required for operation. The user inputs three key data elements: User Credentials for system access, a Start command to trigger the sensors, and a History Request to retrieve past logs. In response, the system processes these inputs and returns three corresponding outputs: Test Results & Health Alerts derived from the immediate analysis, History logs from the database, and a Health Recommendation.

Figure 3.5

Data Flow Diagram Level 1 (Detailed System Process)

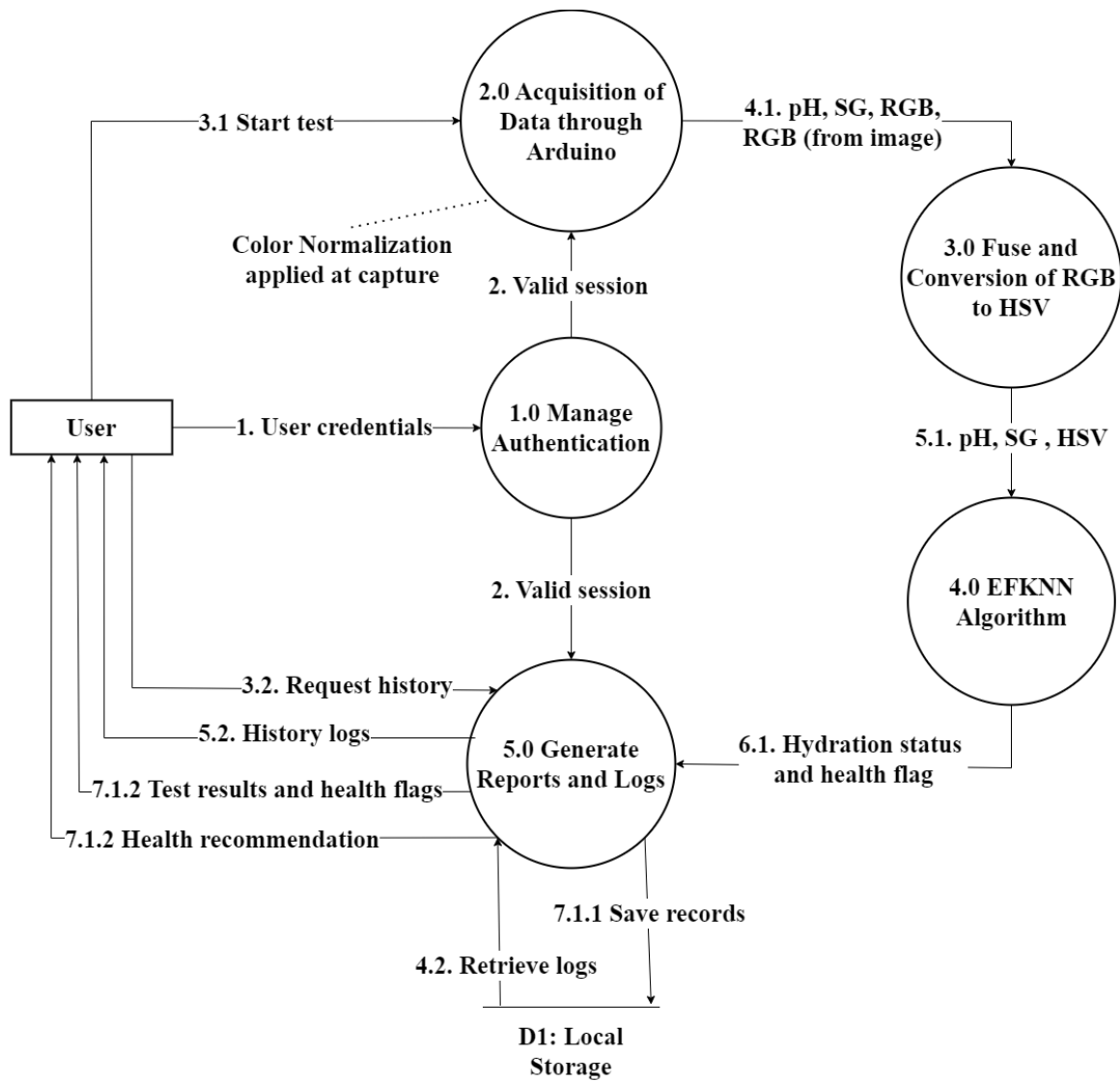


Figure 3.5 details the functional decomposition of the IoT-Based Smart Urinalysis System, breaking down the single process from the Context Diagram into five distinct sub-processes to illustrate the internal data transformation.

- **1.0 Manage Authentication:** The data flow begins with the User inputting their 1. User credentials. The system verifies these to establish a 2. Valid session, which is required and distributed to both the hardware execution and reporting processes before any physical testing or database retrieval can occur.

- **2.0 Acquisition of Data through Arduino:** Once a valid session is confirmed, this process awaits a 3.1 Start test command from the User. The system then acquires the sample's physiological readings. During this acquisition, each optical sensor applies Color Normalization where a two-point dark/white calibration of the form $\text{corrected} = (\text{sample} - \text{dark}) / (\text{white} - \text{dark})$, so that the captured color data is already calibrated against a clear-water white reference and a lights-off dark reference before it leaves the hardware. Process 2.0 then outputs the raw data (4.1. pH, SG, RGB, RGB (from image)) and push them forward into the next process.
- **3.0 Fuse and Convert RGB to HSV:** This process performs sensor fusion by calculating the arithmetic mean of the color-normalized RGB values from both the dedicated color sensor and the camera, combining two independent optical sources into a single, more reliable color reading. The fused output is then converted into the HSV (Hue, Saturation, Value) color space, which separates chromatic information (Hue and Saturation) from brightness (Value). Together, the sealed enclosure, the sensor-level color normalization, and the HSV conversion ensure that the color features reflect the true color of the sample rather than ambient lighting conditions. The process outputs a clean 5.1. pH, SG, HSV data stream.
- **4.0 EF-KNN Algorithm:** The pH, SG, and HSV data is passed to this singular diagnostic engine. The system executes the Enhanced Fuzzy k-Nearest Neighbors algorithm to simultaneously evaluate hydration status and screen for abnormalities. To process this data accurately, the algorithm relies on two computational features:
 - Feature Weighting and Normalization: Since the sensors operate on drastically different mathematical scales, the algorithm normalizes the

inputs into a uniform scale and assigns specific mathematical weights to each parameter based on its clinical priority.

- Fuzzy Logic Distance Calculation: The algorithm calculates the Euclidean distance between the user's sample and a pre-trained clinical dataset. Instead of forcing a strict majority vote (as in standard k-NN), it assigns continuous membership weights percentages of belonging to each class allowing the system to accurately evaluate biological "gray areas" that do not fall within absolute boundaries. This unified calculation outputs the definitive 6.1.

Hydration status and health flag.

- **5.0 Generate Reports and Logs:** This final process manages data persistence and UI rendering, handling two distinct data routes. When completing a new test, it compiles the algorithm's output, generates health recommendations, securely sends 7.1.1 Save records to D1: Local Storage, and delivers the 7.1.2 Test result, health flags, and health recommendation to the User. Alternatively, if the User issues a 3.2. Request history command, this process performs 4.2. Retrieve logs from the database and delivers the 5.2. History logs back to the User.

Figure 3.6

System Flowchart

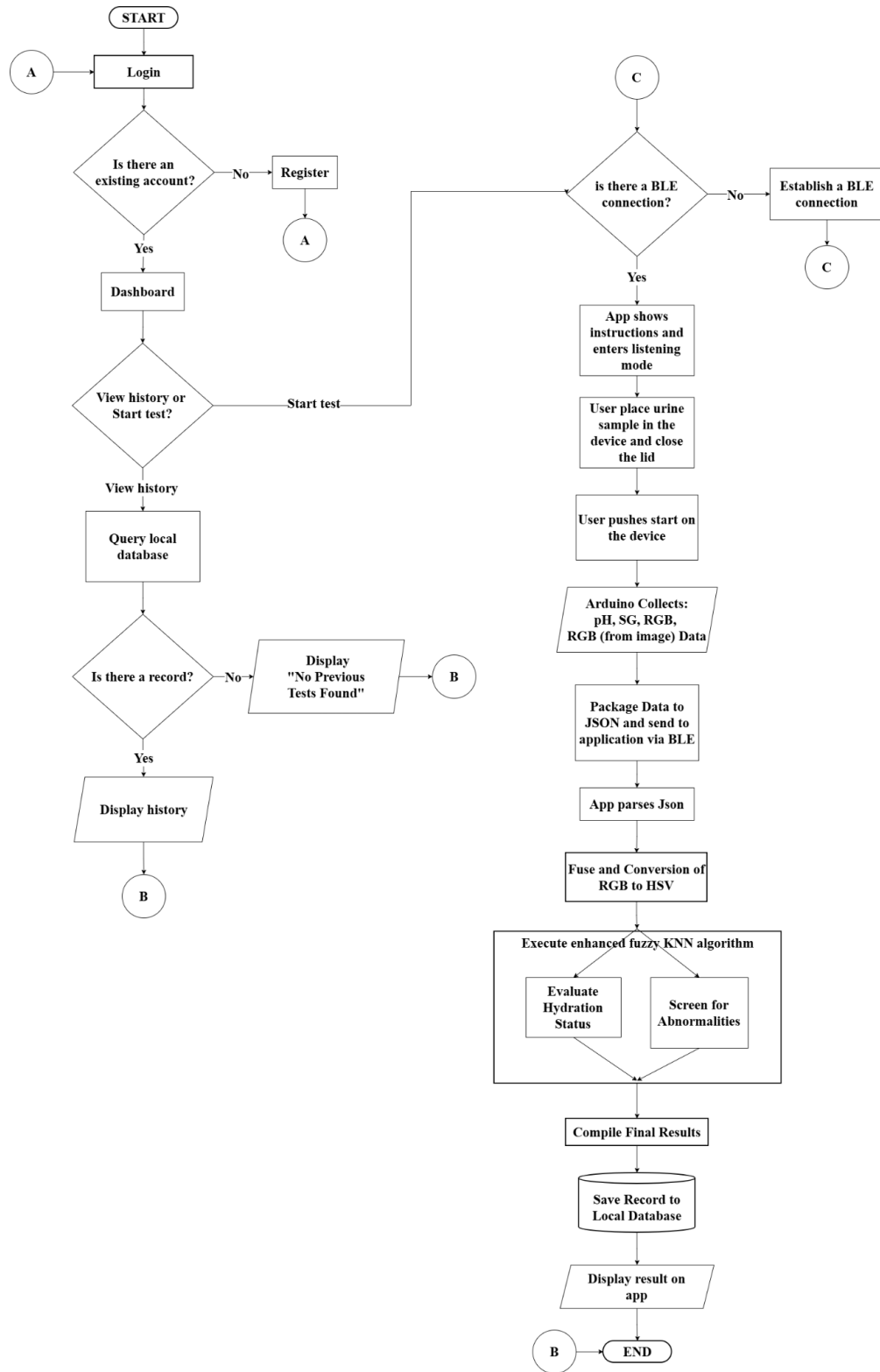


Figure 3.6 presents the System Flowchart, illustrating the sequential logic of the application from start to finish. The process begins at the "START" node, leading into the "Login" step. The system then evaluates the decision point, "Is there an existing account?". If the user does not have an account, they are directed to "Register" and routed back to the login phase via connector A. Upon a successful login, the user progresses to the "Dashboard".

At the dashboard, the system presents the primary operational decision node: "View history or Start test?". If the user chooses to view history, the application executes a "Query local database" command and evaluates "Is there a record?". If no prior tests exist, the system outputs a "Display 'No Previous Tests Found'" and routes to the "END" node via connector B. If records are successfully located, the application triggers a "Display history" output before concluding the process.

On the other hand, if the user initiates a new session by selecting "Start test", the workflow immediately transitions to check "is there a BLE connection?". If no connection is detected, the system executes "Establish a BLE connection" and loops back via connector C to re-verify. Once the BLE connection is secure, the "App shows instructions and enters listening mode". The sequence then pauses for physical user intervention, instructing that the "User place urine sample in the device and close the lid", followed directly by the "User pushes start on the device" action.

Triggered by the hardware start button, the "Arduino Collects: pH, SG, Image and RGB Data" from the physical sample. The hardware then executes the "Package Data to JSON and sends to application via BLE" process. Upon reception, the mobile software executes "App parses JSON" and inputs the RGB data to the process "Fuse and Conversion

of RGB to HSV" step to ensure an accurate color of urine. This data is then pushed into the process "Execute enhanced fuzzy KNN algorithm". Inside this algorithmic process, the system branches simultaneously to "Evaluate Hydration Status" and "Screen for Abnormalities".

Once the algorithm completes its processing, the application converges the diagnostic data to "Compile Final Results". The workflow then performs a "Save Record to Local Database" operation to ensure the test is logged for future retrieval. Finally, the system executes a "Display result on app" output, providing the user with their medical screening feedback before routing to the "END" node.

Figure 3.7

Entity Relationship Diagram

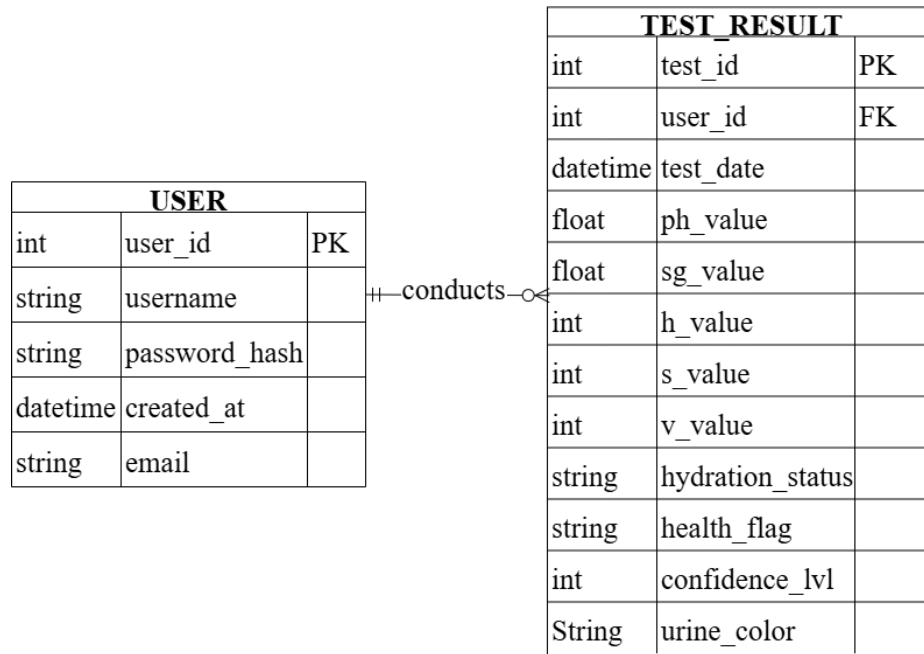


Figure 3.7 presents the Entity Relationship Diagram (ERD), which illustrates the local database schema designed to efficiently manage user records and diagnostic data. The structure consists of two primary entities: the USER table and the TEST_RESULT table, connected by a One-to-Many relationship denoted by the "conducts" connector. This

structural logic dictates that a single user can perform and store multiple urinalysis tests over time, while each specific test record is linked back to exactly one unique user. Serving as the parent entity, the USER table stores essential account credentials and user information. It utilizes an integer-based user_id as the Primary Key to uniquely identify each account. Additional attributes within this table include the username (string), a securely encrypted password_hash (string), the account created_at timestamp (datetime), and the user's email address (string).

The TEST_RESULT table acts as the child entity, storing the comprehensive diagnostic data generated from each individual screening session. It is identified by a unique test_id (integer) as the Primary Key and utilizes the user_id (integer) as a Foreign Key to securely link the diagnostic record to the respective user. The stored data points encompass the exact test_date (datetime) the session was conducted, along with the empirical physical readings captured by the analog sensors, specifically the ph_value (float) and sg_value (float). Furthermore, it records the optical data, including the converted h_value, s_value, and v_value (integers) mapping the HSV color space, alongside a descriptive urine_color (string). Finally, the table houses the definitive diagnostic outputs from the algorithm, stored as the hydration_status (string) and health_flag (string), which are paired with the fuzzy logic algorithm's certainty score represented by the confidence_lvl (integer).

3.3 System Architecture

Figure 3.8

System Architecture

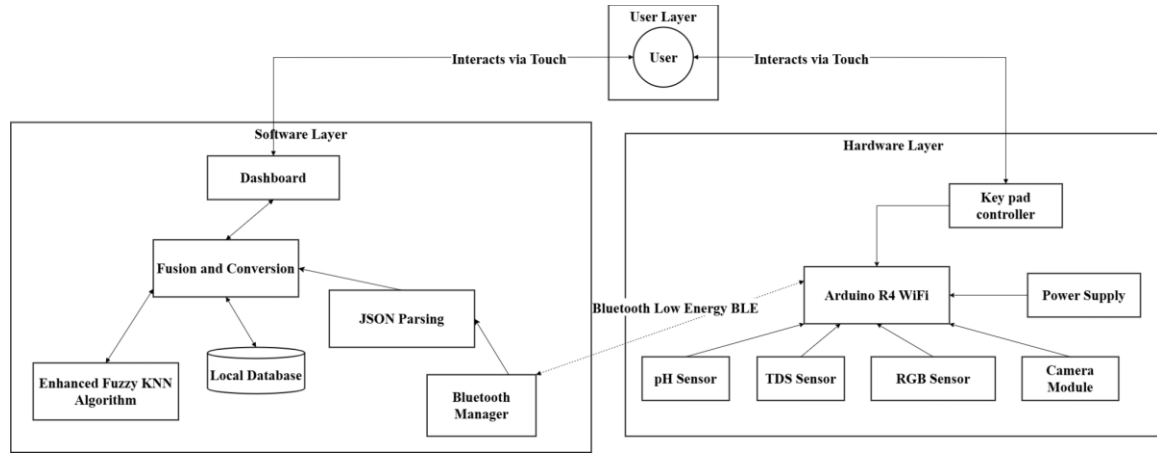


Figure 3.8 depicts the System Architecture of the IoT-Based Smart Urinalysis System, organized into three distinct layers to illustrate the clear separation of duties between the user, the software, and the hardware. The User Layer represents the end-user, who utilizes two distinct "Interacts via Touch" pathways to command the system. One pathway connects directly to the Software Layer to navigate the mobile application, while the second pathway connects to the "Keypad controller" within the Hardware Layer, allowing the user to physically interact with the device.

The Software Layer acts as the data processing and visualization hub of the mobile environment. Wireless data enters this layer through the "Bluetooth Manager," which routes the transmission to the "JSON Parsing" module to extract the raw data. This parsed data then flows into the central "Fusion and Conversion" module, which standardizes the inputs. This central parsing unit manages the flow of information, securely interfacing with the "Dashboard" for user interface visualization, the "Enhanced Fuzzy KNN Algorithm" for diagnostic classification, and the "Local Database" for offline data persistence.

The Hardware Layer is built around the central "Arduino R4 Wi-Fi" microcontroller, which is sustained by a dedicated "Power Supply" and receives manual input triggers from the "Keypad controller". This microcontroller is responsible for gathering real-time physiological data from an integrated multi-sensor array comprising a "pH Sensor," a "TDS Sensor," an "RGB Sensor," and a "Camera Module".

Bridging the digital and physical domains, the Hardware and Software layers are seamlessly linked via a "Bluetooth Low Energy BLE" connection. This wireless pathway specifically connects the Arduino R4 Wi-Fi to the mobile application's Bluetooth Manager, ensuring efficient, real-time data transmission from the physical device directly to the software.

3.4 Software Development Methodology

The study utilized the Agile Software Development Life Cycle (SDLC) methodology for the development of the IoT-Based Smart Urinalysis System.

Figure 3.9

SDLC Agile (Adapted from Laoyan, 2025)



Unlike sequential and rigid linear models, Agile is an iterative development methodology that breaks the project down into short-term cycles, or "sprints." This approach allows for continuous feedback, adaptability, and the simultaneous development of both hardware and software components. While medical and diagnostic systems require high discipline and precision, Agile ensures that every component is technically verified, calibrated, and adjusted in real-time throughout the development pipeline, rather than waiting for a final, end-stage testing phase.

Software Development Process

Plan: In this initial phase, the researchers establish the system's technical and physical specifications to ensure the device accurately monitors preliminary health indicators. The core requirements and goals are defined: measuring physical parameters via dedicated hardware sensors and ensuring accurate data fusion for preliminary diagnostic screening.

Develop: This phase involves the physical construction and software coding necessary to execute the system design. The researchers assemble a functional prototype by integrating the sensors with the microcontroller. Simultaneously, the core user interface of the cross-platform mobile application is developed. During this stage, embedded firmware is written to handle raw data acquisition, and the classification algorithm is coded directly into the application.

Test: Software and hardware testing is conducted continuously. To ensure accurate readings, the researchers iteratively calibrate the hardware components before system integration. The pH and Specific Gravity (SG) sensors are strictly calibrated using standardized pH and SG chemical buffer solutions to establish baseline accuracy. The

optical sensors are calibrated using a calibration liquids to ensure precise colorimetry and reliable RGB-to-HSV conversion. Furthermore, integration testing is performed to stress-test the wireless connectivity pipeline, verifying seamless transmission with zero data loss between the hardware device and the smartphone. The researchers systematically identify and filter out any signal noise or hardware instability within the firmware.

Deploy: Once a stable build of the system is integrated and verified internally, the prototype is prepared for execution. The fully connected Arduino enclosure and the compiled mobile application are deployed into the testing environment to be utilized with synthetic urine, ensuring the physical output strictly matches the parameters set during the planning and design phases.

Review: After the system is fully calibrated and deployed, the researchers conduct a validation and review phase with the intended end-users and subject matter experts. These evaluators assess the finalized device and application using a structured questionnaire based on the ISO 25010 software quality model. Feedback gathered during this phase is used to identify areas for refinement in subsequent development sprints.

3.5 Software Testing

This section outlines the testing methodologies the researchers utilized to assess the functionality, accuracy, and reliability of the IoT-Based Smart Urinalysis System. Aligning with the Agile framework, this is a multi-tiered, continuous testing approach performed throughout the sprints.

Unit Testing

In this initial and ongoing testing phase, the researchers isolated and evaluated individual software and hardware components to verify independent functionality as soon as they are built. For the software, testing focuses specific code modules, including the RGB-to-HSV conversion logic and the local database storage functions. Hardware unit testing focused on the individual accuracy of the physical and optical sensors. The pH and SG sensors were explicitly tested and calibrated using their respective standardized buffer solutions. Meanwhile, the optical sensor array was evaluated under controlled lighting using a calibration liquid. This ensures calibration errors are identified early, guaranteeing that each sensor independently acquires accurate raw data prior to full system integration.

Integration Testing

Following the unit tests within each sprint, the researchers performed Integration Testing to validate the continuous interaction between the embedded hardware and the mobile application. The primary objective is to ensure real-time data flow between the microcontroller and the smartphone via BLE protocols. The researchers systematically identify and eliminate any transmission latency, data loss, or parsing errors, ensuring the application correctly receives and interprets the raw JSON data file sent by the hardware module.

System Testing

Once the components are successfully integrated into a complete working build, the researchers conducted comprehensive system testing against predefined technical requirements. The researchers evaluated the diagnostic accuracy of the Enhanced Fuzzy k-NN algorithm by directly comparing the system's generated readings against standard

reference instruments, specifically a calibrated Digital pH Meter, a Clinical Refractometer, and an iPhone 16 pro (via ImageJ) for optical ground truth. This technical validation ensures that the device's algorithmic classification of hydration and preliminary screening, pH levels, and urine color strictly matches standard medical and optical interpretations before exposure to human evaluators.

Acceptance Testing

As the final validation phase of the Agile release cycle, Acceptance Testing was to ensure the IoT-Based Smart Urinalysis System meets the practical requirements and usability expectations of its target demographic. The researchers deployed the fully calibrated system to end-users and subject matter experts to assess user experience, system responsiveness, and the clarity of the mobile dashboard. Quantitative feedback was gathered utilizing a 5-point Likert Scale questionnaire anchored on the ISO 25010 Software Quality Model.

3.6 Data Gathering and Techniques

Population and Sample

In strict accordance with the study's delimitations regarding clinical human trials, the selected human population is utilized exclusively for the software quality evaluation of the developed mobile application based on the ISO 25010 model. The physical hardware sensors and the classification algorithm are tested separately using standardized synthetic urine. The user evaluation was conducted within Muntinlupa City, National Capital Region (NCR), a highly urbanized location selected to capture a diverse demographic of

smartphone users. To determine the appropriate sample size for the public evaluation, the researchers utilized Slovin's formula.

The final evaluation group consists of:

1. **100 randomly selected participants** who provided insights into the system's everyday usability, mobile application interface, and overall user experience.
2. **30 subject matter experts**, who assessed the system's diagnostic accuracy, technical architecture, and hardware-software integration.

Sampling Technique

The researchers utilized cluster random sampling to select the public participants, with the barangays of Muntinlupa City serving as geographic clusters. Four barangays were selected at random to serve as the sample clusters, with an equal allocation of 25 participants drawn from each, as shown in Table 3.5. This ensures an unbiased sample that reflects a wide range of user demographics and digital familiarity, strictly evaluating the software interface without gathering personal medical data.

Table 3.5

Cluster Distribution for General Users

Barangay (Cluster)	Participants
Sucat	25
Alabang	25
Tunasan	25
Poblacion	25
Total	100

For the subject matter experts, the researchers utilized purposive sampling, selecting professionals based on their qualifications in the medical and allied health or technology and IT fields rather than their geographic location, as shown in Table 3.6.

Table 3.6

Distribution of Subject Matter Experts

Field of Expertise	Participants
Medical / Allied Health professionals	15
Technology / IT professionals	15
Total	30

1. Users:

- Participants must own a Bluetooth-capable smartphone (Android or iOS) and possess basic digital literacy to navigate the mobile application.

2. Subject Matter Experts:

- Active professionals specializing in medical and allied health fields, or experts functioning within the Technology and IT industry.

Research Instruments

The study utilizes two distinct sets of instruments to evaluate the hardware prototype and the mobile application separately:

To validate the hardware and algorithm without conducting human clinical trials, the researchers utilized synthetic urine. The samples were formulated using a synthetic urine, modified with precise measurements of sodium chloride to manipulate Specific Gravity (SG). The optical properties (Hue, Saturation, and Value) were altered using

varying concentrations of yellow and amber food-grade dyes. The pH levels were systematically controlled, diluted acetic acid was used to decrease the pH (simulating high acidity), while sodium bicarbonate was introduced to increase the pH (simulating high alkalinity).

- **Digital pH Pen:** To measure the exact acidity/alkalinity of the sample.
- **Optical Refractometer:** To establish the precise Specific Gravity (SG) ground truth.
- **iPhone 16 Pro (via ImageJ):** To extract the digital ground truth for Hue, Saturation, and Value (HSV).

the researchers utilized a structured 5-point Likert scale questionnaire to evaluate the mobile application. The evaluation relies on the ISO 25010 Software Quality Model, focusing on four key characteristics:

1. **Functional Suitability** - Measures the degree to which the application provides complete, accurate, and appropriate presentation of hydration classification results and abnormality flags to the user.
2. **Performance Efficiency** - Evaluates the speed of data processing and the application's responsiveness without excessive battery or memory usage.
3. **Compatibility** - Assesses the application's ability to run on different mobile devices and screen sizes.
4. **Reliability** - Evaluates the consistency of sensor readings and diagnostic results across repeated tests of the exact same sample.

3.7 Evaluation Procedures

Before evaluating the classification algorithm, the raw accuracy of the IoT sensors was tested using the formulated synthetic urine samples. Sensor readings from the prototype were directly compared against the ground truth established by the calibrated reference instruments (digital pH pen, optical refractometer, and iPhone 16 Pro via ImageJ). To accurately quantify the hardware's precision, the researchers calculated the Root Mean Square Error (RMSE) and the Mean Absolute Percentage Error (MAPE).

Root Mean Square Error (RMSE) was utilized to determine the average magnitude of the sensor error in its original, physical units. It is calculated using the following formula:

Figure 3.10

RMSE Formula

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$$

Mean Absolute Percentage Error (MAPE) was utilized to normalize errors across all three distinct sensors, providing a standardized percentage that validates whether the hardware meets the acceptable operational threshold. It is calculated using the following formula:

Figure 3.11

MAPE Formula

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{Y_i - \hat{Y}_i}{Y_i} \right|$$

Where:

- n = Total number of tested synthetic samples
- Y_i = The actual ground truth value from the reference tool
- $\hat{Y}_i$ = The predicted sensor value from the Arduino prototype

To evaluate the classification performance of the standard k-NN and the proposed Enhanced Fuzzy k-Nearest Neighbors (EF-KNN) models, a k -fold cross-validation technique was employed. Specifically, a 5-fold cross-validation was utilized to partition the chemically modified synthetic dataset. In k -fold cross-validation, the entire dataset is randomly partitioned into k mutually exclusive subsets of approximately equal size. The model is trained and tested k times, ensuring that every sample is used for both training and testing. This prevents algorithmic overfitting and guarantees that the model's accuracy is robust across the entire dataset.

Once the cross-validation is complete, the system's ability to process the fused parameters (pH, SG, and HSV) to categorize hydration levels and screen for abnormalities were analyzed using a Confusion Matrix. This provides a structured summary by comparing predicted categories with actual synthetic conditions.

The confusion matrix consists of the following components:

- True Positive (TP) – Samples correctly classified under their actual category/abnormality.
- False Positive (FP) – Samples incorrectly classified into a category/abnormality they do not belong to.
- True Negative (TN) – Samples correctly identified as not belonging to a specific category/abnormality.

- False Negative (FN) – Samples belonging to a category/abnormality but missed by the system.

Using these values, the following performance metrics are computed to evaluate the classification accuracy and reliability of the system:

Accuracy measures the overall correctness of the system in classifying hydration levels/abnormality:

Figure 3.12

Accuracy Formula

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measures of how many samples are classified into a specific category are correct:

Figure 3.13

Precision Formula

$$Precision = \frac{TP}{TP + FP}$$

Recall evaluates the system's ability to correctly detect actual conditions:

Figure 3.14

Recall Formula

$$Recall = \frac{TP}{TP + FN}$$

The F1-score represents the harmonic means of precision and recall, providing a balanced metric:

Figure 3.15

F1 Formula

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Furthermore, the diagnostic capability of the algorithms was mapped using a Receiver Operating Characteristic (ROC) curve. This visualizes the models' sensitivity and

specificity by plotting the True Positive Rate (which is mathematically equivalent to Recall) against the False Positive Rate (FPR) across various threshold settings.

Figure 3.16

False Positive Rate (FPR) Formula

$$FPR = \frac{FP}{FP + TN}$$

To quantify this visual data into a definitive comparative metric, the Area Under the Curve (AUC) is calculated by computing the definite integral of the ROC curve. An AUC score closer to 1.0 statistically validated the model's high predictive accuracy and its successful ability to distinguish between targeted hydration levels and abnormalities without false positive categorizations.

Figure 3.17

Area Under the Curve (AUC) Formula

$$AUC = \int_0^1 TPR(FPR) d(FPR)$$

To quantify the user experience data gathered during the ISO 25010 evaluation, the researchers utilized a structured 5-point Likert scale.

Table 3.7

Likert Scale Interval Table

Numerical Equivalence	Ranges	Interpretations
5	4.50 - 5.00	Strongly Agree
4	3.50 - 4.49	Agree
3	2.50 - 3.49	Neutral
2	1.50 - 2.49	Disagree
1	1.00 - 1.49	Strongly Disagree

3.8 Data Processing and Statistical Treatment of Data

To statistically validate the performance of the developed urinalysis system, an independent samples t-test was conducted to compare the results obtained from synthetic urine samples analyzed by the proposed system (experimental condition) and the measurements obtained using a standard reference tool (controlled condition). This statistical test determines whether a significant difference exists between the mean measurements produced by the developed system and those obtained from the reference instrument.

The independent samples t-test evaluates the null hypothesis (H_0) stating that there is no significant difference between the mean measurement results of the proposed system and the reference tool when analyzing synthetic urine samples. The alternative hypothesis (H_1) states that a statistically significant difference exists between the two measurement methods, indicating a discrepancy between the developed system and the reference instrument.

The t-test statistics are computed using the following formula:

Figure 3.18

Independent Samples t-Test Formula

$$t = \frac{x_1 - x_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

- $\bar{X}_1$ = Mean measurement obtained from the reference tool (controlled condition)
- $\bar{X}_2$ = Mean measurement obtained from the developed urinalysis system (experimental condition)
- s_1^2 = Variance of measurements from the reference tool

- s_2^2 = Variance of measurements from the proposed system
- n_1 = Number of synthetic urine samples measured using the reference tool
- n_2 = Number of synthetic urine samples measured using the developed system

The computed p-value was compared with a significance level of $\alpha = 0.05$. If the calculated p-value is greater than 0.05, the null hypothesis was not rejected, indicating that no significant difference exists between the measurements of the proposed system and the reference tool, suggesting that the developed system produces results comparable to the standard instrument. If the p-value is less than 0.05, the null hypothesis was rejected, indicating a statistically significant difference between the two measurement methods. This statistical analysis provides additional validation of the developed system by determining whether the measurements obtained from the proposed urinalysis device are consistent with those produced by the reference tool when analyzing synthetic urine samples, thereby supporting the accuracy and reliability of the system.

To analyze the quantitative data gathered from the survey questionnaires, the researchers utilized the weighted mean as the primary statistical tool.

Figure 3.19

Weighted mean Formula

$$\underline{x} = \frac{\sum fx}{N}$$

Where:

- $\underline{x}$ = Weighted Mean
- f = Frequency of each response
- x = Weight of each point on the scale (1 to 5)
- N = Total number of respondents

Chapter 4

RESULTS AND DISCUSSION

This chapter presents the project description, structure, and its capabilities and limitations. The results obtained from the technical validation of the hardware, the algorithmic classification, and the final software quality evaluation are presented and discussed.

4.1 Project Description

The IoT-Based Smart Urinalysis System is designed to provide objective, localized physiological monitoring. The system replaces traditional, subjective visual color interpretations with a multi-modal hardware-software framework that analyzes chemical and optical parameters simultaneously. The system architecture is built around an embedded hardware prototype linked dynamically to an offline-capable cross-platform mobile application.

The physical assembly is integrated within a light-controlled, opaque testing enclosure featuring a matte-black interior and dedicated white lighting. This configuration completely seals the sample chamber from outside ambient light variations, ensuring consistent data acquisition conditions. By combining physical fluid parameters and digital optical capture, the system executes real-time data fusion to run targeted classifications. These fused outputs drive an Enhanced Fuzzy k-Nearest Neighbors (E-FKNN) model embedded directly into the application layer, automatically evaluating the user's continuous hydration tier (Levels 1 to 8) and screening for discrete chemical abnormalities (Flags A, B, and C).

4.2 Project Structure

The engineering architecture of the IoT-Based Smart Urinalysis System is divided into three distinct structural modules that manage data collection, serialization, wireless transmission, and machine learning computation.

Module 1: Hardware Data Module

The Hardware Module is the physical core of IoT-Based Smart Urinalysis System. It is built around an Arduino Uno R4 Wi-Fi microcontroller housed within a custom-integrated opaque enclosure with a matte-black interior and controlled white illumination to eliminate ambient light interference. The module integrates a digital pH sensor, an analog TDS sensor for Specific Gravity, and a dual-optical array consisting of a color sensor and a camera module. Upon a user trigger from the keypad controller, the microcontroller acquires the raw pH, SG, and RGB values, packages them into a JSON structure, and transmits them to the mobile application over Bluetooth Low Energy

Figure 4.1

IoT-Based Smart Urinalysis Device



Figure 4.1 presents the IoT-Based Smart Urinalysis System enclosed in a custom-built opaque black enclosure. The enclosure was constructed to serve a dual purpose: housing the internal electronic components and providing a light-controlled environment

for urine sample imaging. As shown in both the front and top views, the exterior of the enclosure features a 4×4 membrane keypad for direct user input, an OLED display module for on-device feedback, and a protruding optical sensor assembly mounted on the top panel, which accommodates the urine sample cup during analysis. The enclosure's matte black interior and exterior minimize ambient light interference during color acquisition, ensuring consistent HSV readings across all test conditions. All wiring is externally routed to allow accessibility during the testing and calibration phase of the study.

Figure 4.2
System Wiring Diagram

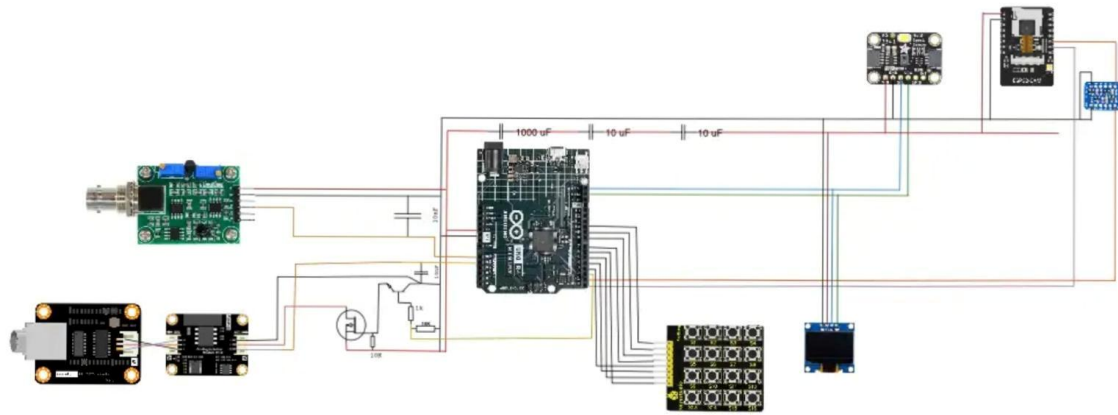


Figure 4.2 illustrates the complete wiring schematic of the IoT-Based Smart Urinalysis System, showing the interconnection of all hardware components centered around the Arduino Uno R3 microcontroller. On the left side, the analog pH sensor module and the TDS/EC sensor (used for Specific Gravity estimation) are wired to the Arduino's analog input pins, with a transistor-based switching circuit and decoupling capacitors (1000 μ F, 10 μ F, 10 μ F) integrated along the power rail to stabilize voltage and suppress electrical noise during multi-sensor operation. The AS7341 11-Channel visible light sensor and the ESP32-CAM module are connected via the I²C bus on the right side of the board, enabling

optical data acquisition and wireless BLE transmission respectively. User interface components are the 4×4 keypad matrix and the OLED display are also connected to the Arduino's digital I/O pins. The overall wiring architecture reflects a multi-modal sensing design where all sensor inputs converge at the microcontroller for synchronized data acquisition and serial transmission to the BLE module for mobile application communication.

Figure 4.3

Sample JSON Payload

```
{
  "device": "URINE-TEST-001",
  "version": "1.0",
  "type": "urinalysis",
  "sensors": {
    "temp_c": 25.4,
    "pH": 6.15,
    "tds_ppm": 450.0,
    "ec_us_cm": 900.0,
    "ec_sample_us_cm": 1800.0,
    "sg": 1.0155,
    "color": {
      "r": 245,
      "g": 210,
      "b": 95,
      "hex": "#F5D25F",
      "lux": 120.5,
      "cct": 3200
    },
    "camera": {
      "r": 240,
      "g": 205,
      "b": 88,
      "hex": "#F0CD58"
    }
  }
}
```

Figure 4.3 presents a sample JSON payload transmitted by the Arduino-based hardware module to the mobile application via Bluetooth Low Energy (BLE). The payload is structured as a single JSON object containing four primary fields. The device and version fields serve as metadata identifiers, designating the hardware unit as "URINE-TEST-001"

running firmware version "1.0", while the type field confirms the packet as a urinalysis transmission.

The `sensors` object encapsulates the raw readings from all onboard sensing modules: `temp_c` records ambient temperature at 25.4°C for reference; `pH` returns a reading of 6.15; `tds_ppm` and `ec_us_cm` provide the raw TDS and electrical conductivity values at 450.0 ppm and 900.0 µS/cm respectively, with `ec_sample_us_cm` representing the blank-corrected EC at 1800.0 µS/cm used for Specific Gravity derivation; and `sg` reports the computed Specific Gravity value of 1.0155. The nested `color` object contains the RGB output of the AS7341 11-Channel Visible Light Sensor (r: 245, g: 210, b: 95), derived from its spectral channel data and converted to a standard RGB representation, along with its hexadecimal equivalent (#F5D25F), illuminance in lux (120.5), and correlated color temperature in Kelvin (3200 K). The AS7341's multi-channel spectral sensing capability enables more precise colorimetric characterization of the urine sample compared to conventional single-chip RGB sensors, as it captures discrete visible light bands across the spectrum before mapping to RGB values.

The `camera` object contains a separate RGB reading derived from the ESP32-CAM image capture (r: 240, g: 205, b: 88, hex: #F0CD58), which serves as the optical input for HSV conversion and color-based hydration classification within the mobile application. Upon receipt, the mobile application parses this JSON structure, converts the camera RGB values to HSV, and feeds all fused parameters into the EF-KNN classification algorithm for hydration level determination and abnormality screening.

Module 2: Mobile Application

The mobile application is the data-processing and visualization hub of the system, developed for both iOS and Android platforms. Wireless data enters through the Bluetooth manager, which routes the transmission to the JSON parsing module. The parsed data flows into the fusion and conversion module, which computes the mean RGB value and converts it to the HSV color space. From there, the data is passed to the EF-KNN algorithm for classification, rendered on the dashboard for the user, and stored in the local database for offline persistence. The application requires user authentication and supports three primary functions: starting a urinalysis test, viewing test results with health recommendations, and viewing historical logs.

Figure 4.4

Login and Registration Screen

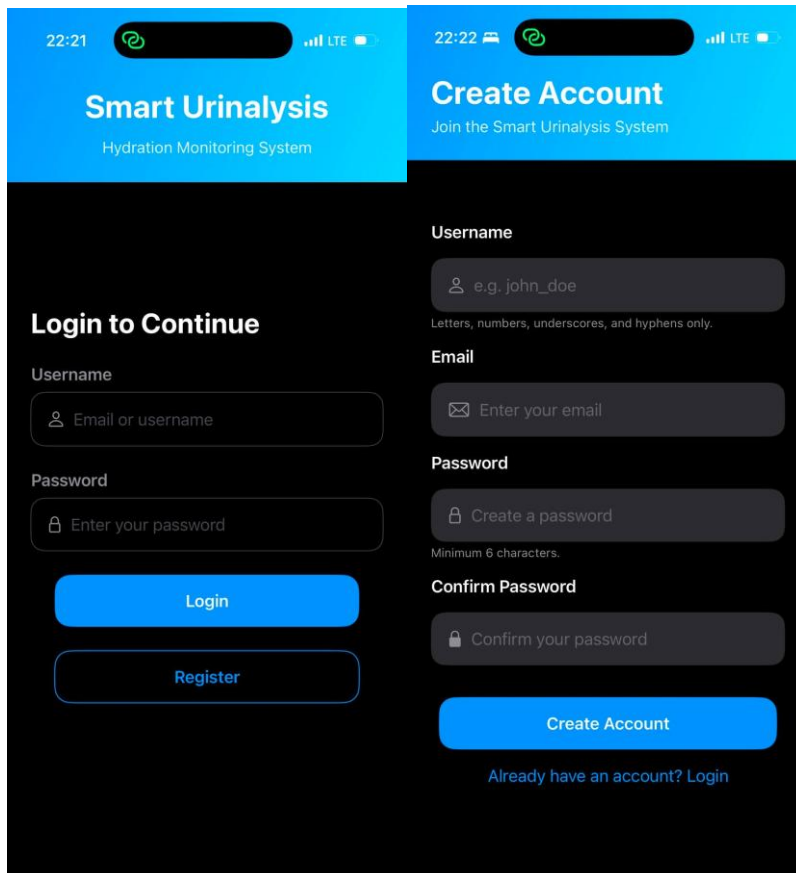


Figure 4.4 presents the authentication interface of the mobile application. The Login screen prompts the user to enter their registered username or email address alongside their password to gain access to the system, with a "Register" option available for new users. The Registration screen allows new users to create an account by providing a unique username (restricted to letters, numbers, underscores, and hyphens), a valid email address, and a password of at least six characters confirmed through a secondary input field. Both screens adopt a dark-themed UI with a gradient blue header bearing the "Smart Urinalysis Hydration Monitoring System" branding, consistent with the application's overall visual design language.

Figure 4.5

Dashboard / Home Screen

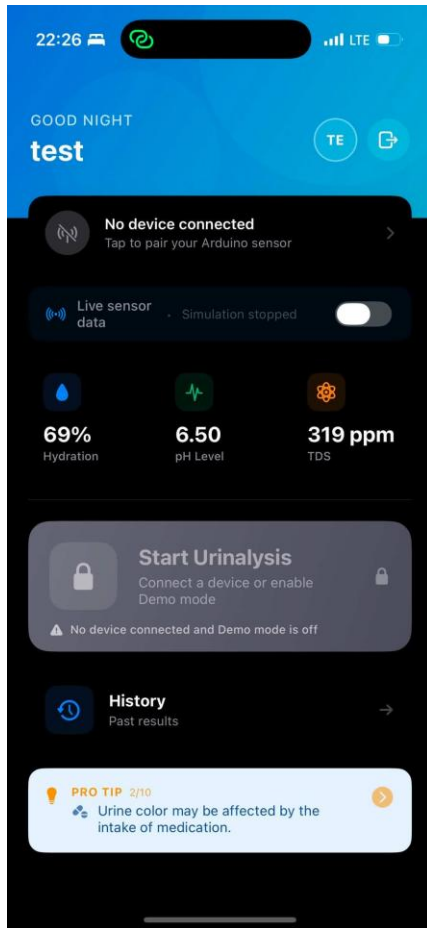


Figure 4.5 presents the main dashboard of the application, which serves as the central navigation hub after a successful login. The screen greets the authenticated user by name and displays a real-time summary of three key indicators which are Hydration percentage, pH Level, and TDS derived from the most recent test or simulation data. A BLE connection status banner at the top alerts the user when no device is paired, prompting them to tap and initiate pairing. A Live Sensor Data toggle enables simulation mode for demonstration purposes. The dashboard also provides quick-access buttons for initiating a new urinalysis test via "Start Urinalysis" and reviewing past results via "History," along with rotating pro tips related to hydration and urinary health at the bottom of the screen.

Figure 4.6

Bluetooth Screen

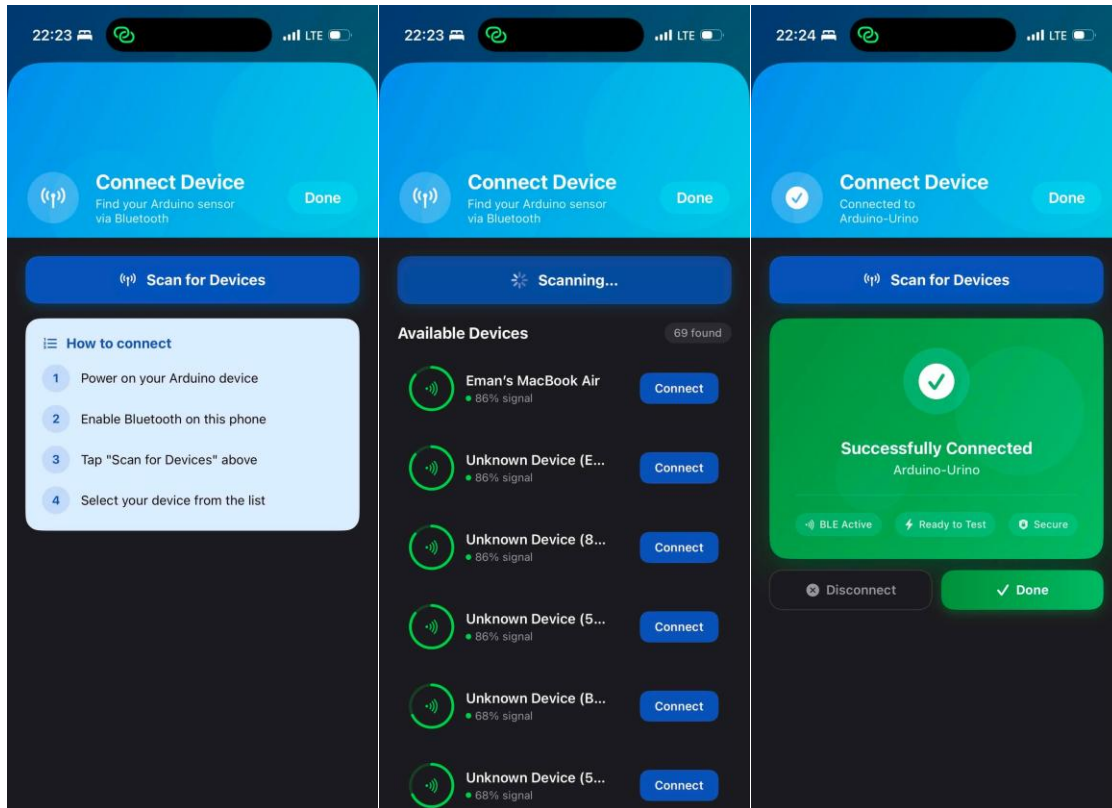


Figure 4.6 illustrates the Bluetooth Low Energy (BLE) device pairing interface. Upon tapping "Scan for Devices", the application initiates a BLE scan and populates the "Available Devices" list with all discovered nearby Bluetooth peripherals, displaying their signal strength alongside each entry. Once the user selects the designated hardware module (identified as "Arduino-Urino"), the application establishes a BLE connection and displays a confirmation screen showing a green "Successfully Connected" status badge accompanied by three status indicators which are BLE Active, Ready to Test, and Secure confirming that the hardware is paired and ready for sample analysis.

Figure 4.7

Waiting / Listening Screen

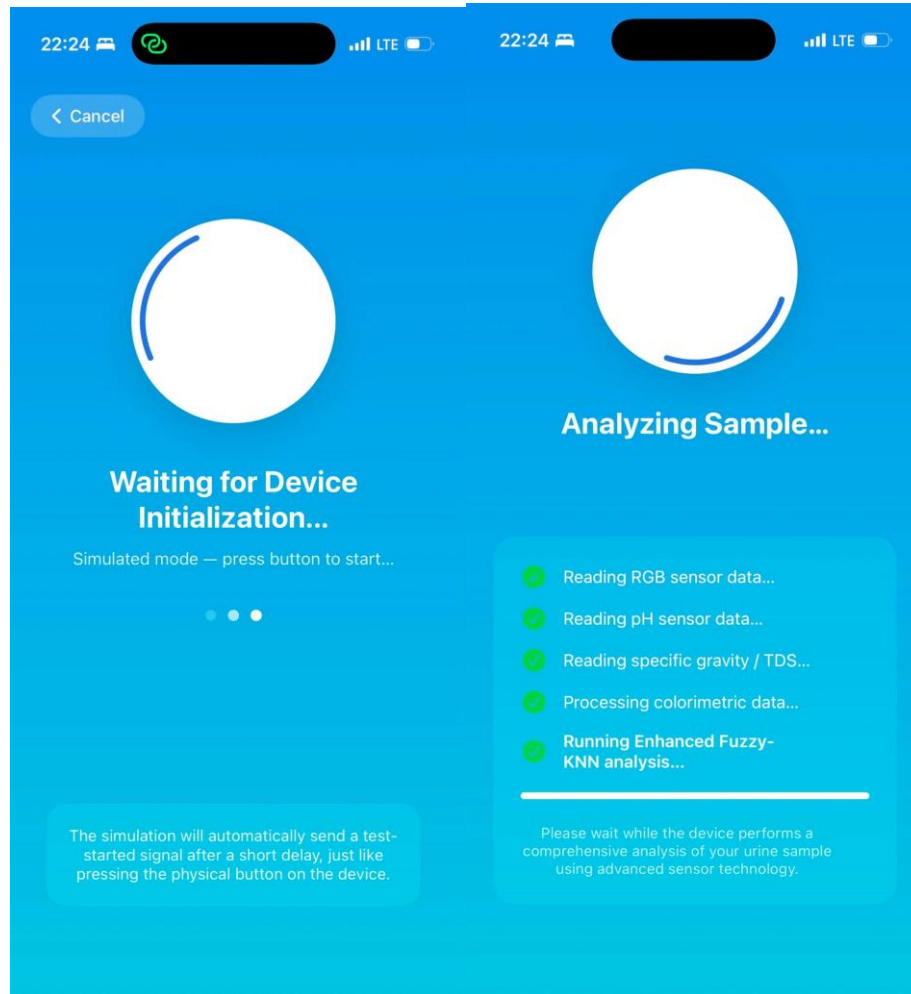


Figure 4.7 presents the waiting screen displayed after a test is initiated. The application enters a listening state, represented by an animated circular progress indicator, while it waits for the hardware module to complete its sensor initialization sequence and transmit the test-start signal over BLE (Image 7). Once the signal is received, the screen transitions to the "Analyzing Sample" state (Image 8), which presents a sequential checklist of processing steps which are reading RGB sensor data, reading pH sensor data, reading Specific Gravity/TDS data, processing colorimetric data, and running the Enhanced Fuzzy-

KNN analysis confirming to the user that each stage of the diagnostic pipeline has been completed before the final results are rendered.

Figure 4.8

Test Result Screen

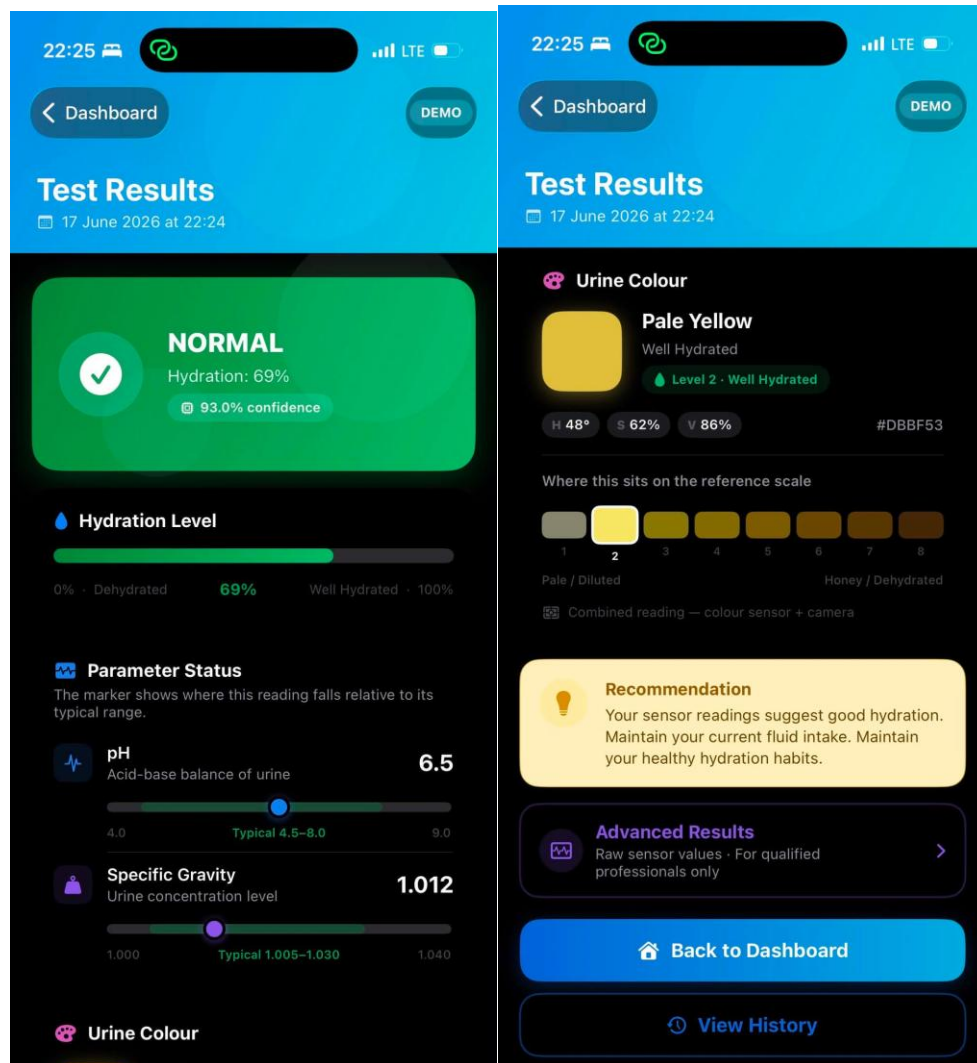


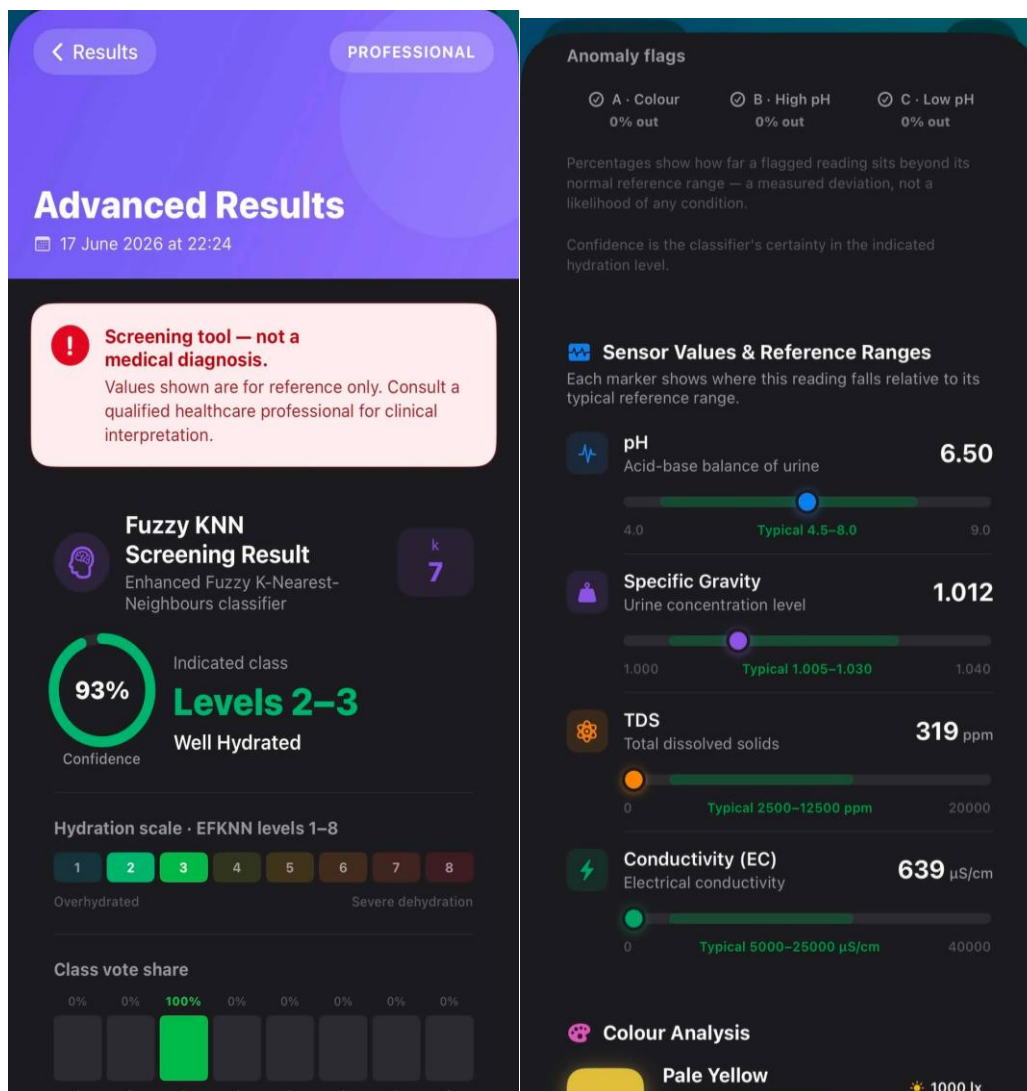
Figure 4.8 presents the primary test results interface displayed to the user upon completion of a urinalysis analysis, composed of two scrollable sections. The upper section features a prominently colored status card rendered in green indicating a classification of NORMAL with a hydration level of 69% and an EF-KNN algorithm confidence score of 93.0%, accompanied by a Hydration Level slider that visualizes the user's position along a

continuous 0%–100% scale from Dehydrated to Well Hydrated, and a Parameter Status section displaying individual reference-range sliders for pH (6.5, within the typical 4.5–8.0 range) and Specific Gravity (1.012, within the typical 1.005–1.030 range). The lower section presents the Urine Color analysis, classifying the sample as "Pale Yellow" at Level 2 (Well Hydrated) with HSV values of H: 48°, S: 62°, V: 86° and hex code #DBBF53, displayed alongside an eight-level visual reference color scale spanning Level 1 (Pale/Dilute) through Level 8 (Honey/Dehydrated), followed by a plain-language Recommendation card advising the user to maintain their current fluid intake, and an Advanced Results entry point restricted to qualified healthcare professionals for access to the raw sensor breakdown and full EF-KNN screening output..

Figure 4.9

Advanced Results Screen





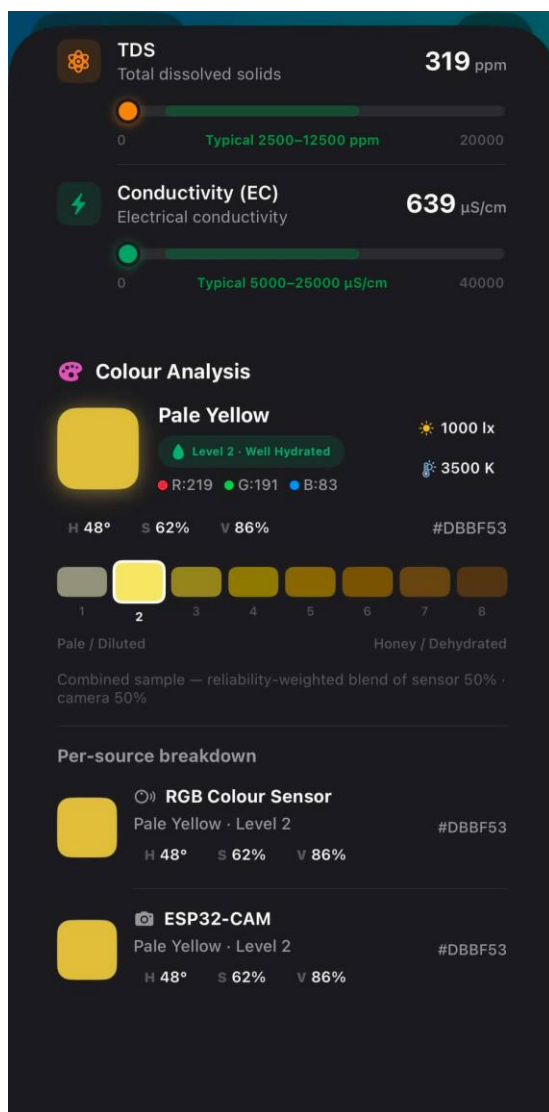


Figure 4.9 presents the Advanced Results interface, which is gated behind a Professional Use Only disclaimer screen that explicitly warns users against self-diagnosis, stating that the raw sensor values are intended exclusively for qualified healthcare professionals such as physicians, nurses, and licensed laboratory technicians, and listing three cautionary advisories that interpreting technical values without clinical training may lead to incorrect conclusions, that abnormal readings do not automatically indicate disease, and that the device functions solely as a preliminary screening tool and does not replace clinical urinalysis or professional medical judgment requiring the user to actively confirm

their professional capacity by tapping "I Am a Qualified Professional I Understand" before proceeding. Upon confirmation, the Advanced Results screen displays the EF-KNN Screening Result indicating hydration class Levels 2–3 (Well Hydrated) at 93% confidence with $K=7$ neighbors, accompanied by a full eight-level hydration scale visualization and a class vote share chart confirming 100% vote concentration at Level 2, followed by an Anomaly Flags section reporting zero out-of-bound deviation across Flag A (Color), Flag B (High pH), and Flag C (Low pH), and a Sensor Values and Reference Ranges section presenting individual reference-range sliders for all raw parameters where pH (6.50, typical 4.5–8.0), Specific Gravity (1.012, typical 1.005–1.030), TDS (319 ppm, typical 2500–12500 ppm), and Conductivity/EC (639 $\mu\text{S}/\text{cm}$, typical 5000–25000 $\mu\text{S}/\text{cm}$) each with interactive markers showing where the reading falls relative to its established clinical reference band..

Figure 4.10

Test History Screen

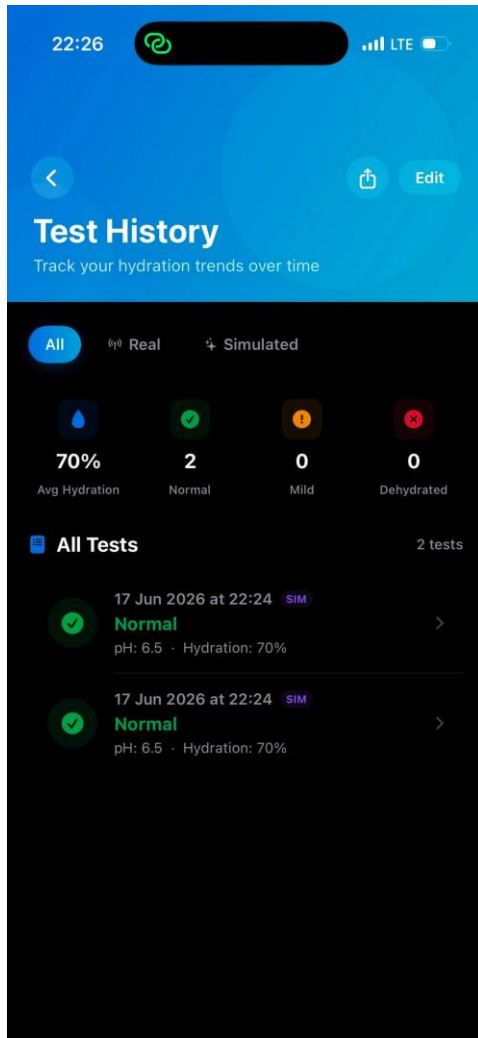


Figure 4.10 presents the Test History interface, which provides the user with a chronological log of all previously completed urinalysis tests stored in the local database, subtitled "Track your hydration trends over time." A summary statistics bar near the top displays aggregate metrics across all recorded sessions average hydration percentage (70%), and the total count of Normal (2), Mild (0), and Dehydrated (0) results giving the user an at-a-glance overview of their hydration trends. Filter tabs allow the user to toggle the list view between All tests, Real (device-acquired) tests, and Simulated (demo mode)

tests. The All Tests section lists each individual test entry with its timestamp, classification label (e.g., Normal in green), pH reading, hydration percentage, and a "SIM" badge distinguishing simulated entries from hardware-acquired results, with a chevron on each row allowing the user to tap through to the full detailed result view for that session.

4.3 Project Capabilities and Limitations

The IoT-Based Smart Urinalysis System, housed in a custom-integrated opaque enclosure, performs real-time, multi-modal sensing of three core urinary parameters. It measures urine pH through a digital pH sensor, estimates Specific Gravity (SG) using a Total Dissolved Solids (TDS) sensor that derives solute concentration from the sample's Electrical Conductivity (EC), and captures urine color through a dual-optical array composed of the AS7341 11-channel spectral sensor and the ESP32 camera module. The color readings from both optical sources are fused into a single mean color value, which is then converted into the Hue-Saturation-Value (HSV) color space. These three fused parameters which are pH, SG, and HSV are processed by an Enhanced Fuzzy k-Nearest Neighbors (EF-KNN) algorithm within an offline mobile application, which automatically classifies the user's hydration level and performs preliminary screening for out-of-bound physiological markers, primarily color anomalies and abnormally low (acidic) or high (alkaline) pH levels. Through this integration, the system provides an objective, portable alternative to subjective visual urinalysis for hydration monitoring in resource-limited settings.

Despite these capabilities, the system operates within several defined limitations. The most significant constraint stems from the electrical conductivity probe used to

estimate Specific Gravity. The prototype relies on a TDS sensor fitted with a $K=1$ conductivity probe, whereas accurately resolving the full range of urinary solute concentrations requires a higher-constant probe, ideally $K=10$. A $K=1$ probe saturates at lower conductivity levels and is therefore only able to reliably measure up to approximately half of the target Specific Gravity range. The appropriate $K=10$ probe was unavailable for purchase through local suppliers throughout the development period, and procurement was not feasible within the project's timeframe and budget. To accommodate this hardware ceiling, the researchers diluted the synthetic urine samples with distilled water at a ratio of 1:1, and 1:2 for the higher-concentration formulations, bringing the resulting conductivity readings within the $K=1$ probe's effective measurement range. This dilution preserved the proportional relationships between samples and allowed valid testing, but it represents a workaround necessitated by hardware availability rather than an ideal measurement configuration.

Second, the study lacked access to a true gold-standard instrument for color validation. The proposed approach of using an iPhone 16 camera to extract HSV values did not yield accurate ground-truth color data, and the available online references only described pre-established standard colors rather than the actual color of urine, leaving the optical channel without a clinically rigorous benchmark. Finally, because the synthetic urine formulations were prepared to fall squarely within expected physiological values, the dataset contained few of the overlapping or ambiguous boundary cases that fuzzy-membership weighting is designed to resolve. Consequently, the EF-KNN algorithm showed no measurable improvement over the traditional k-NN model during evaluation,

as the clean separability of the controlled samples did not exercise the algorithm's primary advantage.

Module 3: EF-KNN Classification Module

The EF-KNN Classification Module is the algorithmic core of the system, implemented entirely within the mobile application to enable offline, real-time hydration classification and preliminary abnormality screening without reliance on external servers or cloud computation. Upon receiving the parsed sensor payload from Module 2, the classification module accepts three fused input features which are pH, Specific Gravity (SG), and the HSV-converted mean color value and processes them through the Enhanced Fuzzy k-Nearest Neighbors algorithm at a deployed neighbor count of $k=3$, determined through stratified cross-validation during training. The predataset underlying the classifier defines twelve discrete classification targets: eight hydration levels and four abnormality flags.

Table 4.1*Hydration Level Classification Targets*

Classification Level	Hydration Status	Target SG	Target pH	Target Color	H (°)	S (%)	V (%)	Hex
Level 1	Diluted	< 1.005	4.5–8.0	Pale / Transparent	5–7	7	98	#FAF9E9
Level 2	Well Hydrated	1.005–1.009	4.5–8.0	Pale Yellow	5–3	17	100	#FFFAD4
Level 3	Minimal Dehydration	1.010–1.014	4.5–8.0	Yellow	5–2	27	98	#F9F1B5
Level 4	Minimal Dehydration	1.015–1.019	4.5–8.0	Darker Yellow	5–8	44	92	#EBE885
Level 5	Significant Dehydration	1.020–1.024	4.5–8.0	Amber	5–4	56	88	#E0D462
Level 6	Significant Dehydration	1.025–1.029	4.5–8.0	Dark Amber / Yellow	5–2	75	85	#D9C336
Level 7	Serious Dehydration	1.030–1.035	4.5–8.0	Ochre / Amber	4–3	70	80	#CBA13E
Level 8	Serious Dehydration	> 1.035	4.5–8.0	Honey / Brownish	4–3	70	64	#A38331

Table 4.1 presents the eight hydration level classification targets embedded in the EF-KNN predataset. The levels span a Specific Gravity range from below 1.005 (Level 1, Diluted) to above 1.035 (Level 8, Serious Dehydration), each associated with a clinically established pH range of 4.5–8.0 and a fixed HSV centroid derived from colorimetric calibration. The color progression reflects the increasing concentration of urinary pigments with dehydration severity, with Saturation rising from 7% at Level 1 to a peak of 75% at Level 6 and Value decreasing from 100% at Level 2 to 64% at Level 8, transitioning visually from pale transparent tones through yellow and amber to deep honey-brownish hues at the most severe dehydration classifications.

Table 4.2

Abnormality Flag Classification Targets

Flag	Targeted Abnormality	Operational SG	pH Trigger	Target Color Profile	Target Hue Range	Hex Precedent
Flag A	Abnormal (Color Flag)	Any Baseline SG	4.5–8.0 (Normal)	Red, Green, or Deep Brown	0°–20° (Red) / 90°–140° (Green)	#C0392B / #27AE60
Flag B	Abnormal (Alkaline)	Any Baseline SG	> 8.0	Standard Baseline Yellow	Matched to hydration layer	Dependent on base layer
Flag C	Abnormal (Acidic)	Any Baseline SG	< 4.5	Standard Baseline Yellow	Matched to hydration layer	Dependent on base layer

Table 4.2 presents the three abnormality flags evaluated in parallel with hydration level classification. Flag A targets color anomalies by detecting Hue values falling outside the expected yellow spectrum specifically 0° – 20° for red-toned samples and 90° – 140° for green-toned samples which may indicate the presence of blood, biliverdin, or other clinically significant urinary pigments. Flag B triggers on a pH reading exceeding 8.0, signaling an alkaline condition potentially associated with urinary tract infection or renal tubular acidosis, while Flag C triggers on a pH reading below 4.5, signaling an acidic condition potentially linked to metabolic acidosis or diabetic ketoacidosis. Both Flag B and Flag C are overlaid on top of the underlying hydration classification, allowing the system to simultaneously report a hydration level and an abnormality flag from a single test. The EF-KNN algorithm assigns fuzzy membership weights to each input feature prior to distance computation, Specific Gravity at 0.40, Hue at 0.30, and pH at 0.15 and derives a confidence score from the aggregated fuzzy membership of the $k=3$ nearest neighbors, providing a more clinically nuanced output than the rigid majority-vote assignment of standard k-NN.

4.4 Project Evaluation

This section presents the structured engineering and usability evaluations conducted to validate the system's performance. The results are organized directly by the study's core research objectives to systematically assess hardware precision, network efficiency, algorithmic reliability, and overall software product quality.

Objective 1 Validation: Hardware Sensor Accuracy Analysis

The raw sensor accuracy of the prototype was evaluated by comparing 150 sensor readings against calibrated reference instruments using RMSE and MAPE, with an acceptance threshold of $\text{MAPE} < 10\%$. An independent samples t-test ($\alpha = 0.05$) was applied to determine whether a statistically significant difference existed between the prototype and the reference tools.

Table 4.3

IoT-Based Smart Urinalysis vs. Reference tools

Sensor	RMSE	MAPE	t	p-value	PASS/FAIL
pH	0.5559 pH units	7.72%	-1.6814	0.0937	PASS
SG	0.001701 SG units	0.13%	+0.1456	0.8843	PASS
HSV	10.4480°	40.03%	+3.1046	0.0021	FAIL

Table 4.3 presents the raw sensor accuracy evaluation of the prototype across three sensing modalities which are pH, Specific Gravity (SG), and HSV color assessed against calibrated reference instruments using RMSE, MAPE, and an independent samples t-test at $\alpha = 0.05$, with a MAPE threshold of less than 10% as the acceptance criterion. The pH sensor achieved a MAPE of 7.72% (RMSE = 0.5559 pH units, $p = 0.0937$), and the Specific Gravity sensor recorded the highest accuracy with a MAPE of 0.13% (RMSE = 0.001701 SG units, $p = 0.8843$) both passing the threshold with no statistically significant difference from their respective reference instruments, failing to reject the null hypothesis. In contrast, the HSV optical channel recorded a MAPE of 40.03% (RMSE = 10.4480°, $p = 0.0021$), far exceeding the acceptance threshold and rejecting the null hypothesis, confirming a

statistically significant deviation from the iPhone 16 Pro ImageJ ground truth. These deviations are attributed to specular LED reflection within the imaging enclosure, which systematically distorts the captured color values particularly at the dilute and concentrated extremes of the urine color spectrum.

Table 4.4

IoT-Based Smart Urinalysis HSV vs. Predataset Reference Vectors

Sensor	RMSE	MAPE	t	p-value	PASS/FAIL
HSV	21.6639°	29.23%	−0.5491	0.5837	PASS

Table 4.4 presents a secondary evaluation comparing the IoT-Based Smart Urinalysis HSV readings directly against the embedded predataset reference vectors used by the EF-KNN classifier as labeled training instances, rather than the external ImageJ reference. The one-sample t-test yielded $t = -0.5491$ and $p = 0.5837$, failing to reject the null hypothesis and confirming no statistically significant difference between the sensor output and the predataset reference vectors returning a PASS result. This is the critical finding: despite failing the raw colorimetric comparison in Table 4.3, the Arduino HSV readings do not significantly deviate from the labeled feature vectors the EF-KNN was trained on, confirming that the sensor and classifier share the same systematic bias and explaining why classification accuracy remains high despite the elevated MAPE against the external reference.

Table 4.5*IoT-Based Smart Urinalysis Readings vs. Laboratory Urinalysis Results*

Sensor	RMSE	MAPE	t	p-value	PASS/FAIL
pH	0.7145 pH units	9.689%	-2.8236	0.0051	PASS
SG	0.0190 SG units	0.94%	+3.1155	0.0020	PASS

Table 4.5 presents the prototype's pH and Specific Gravity accuracy evaluated against the clinical urinalysis results, using the same RMSE, MAPE, and independent samples t-test. The Specific Gravity sensor again performed strongly, recording a MAPE of 0.94% (RMSE = 0.0190 SG units), comfortably within the acceptance threshold; however, the t-test returned $t = +3.1155$, $p = 0.0020$, indicating a small but statistically significant positive offset of approximately 0.005 SG units relative to the lab. This significance is a consequence of the Urinalysis lab results reporting SG in coarse 0.005 increments while the prototype resolves continuously, so a minor consistent bias becomes statistically detectable despite the negligible practical error. The pH sensor recorded a MAPE of 9.689% (RMSE = 0.7145 pH units, $t = -2.8236$, $p = 0.0051$), exceeding the 10% threshold and rejecting the null hypothesis. This deviation is attributed to the limited resolution of the clinical reference, which reports pH only in 0.5-unit steps (5.0, 6.0, 6.5, 7.5, 9.0) and saturates at the extremes of its measurement range most notably for the strongly acidic specimen, where the clinical urinalysis results read 5.0. While the prototype's pH channel passes against the calibrated reference instrument (Table 4.3, MAPE = 7.72%), the elevated error in Table 4.5 reflects the coarseness and saturation of the standard Urinalysis rather than a deficiency in the sensor itself.

Table 4.6*IoT-Based Smart Urinalysis Color Readings vs. Laboratory Urinalysis Results*

Comparison	N	Abnormal excluded	Exact match	Within ± 1	Weighted κ	Wilcoxon p	PASS/ FAIL
Color (1-8 scale)	102	48	100%	100%	1.000	1.000	PASS

Table 4.6 presents the color agreement between the IoT-Based Smart Urinalysis prototype and the clinical urinalysis lab results on the ordinal 1–8 hydration color scale. Of the 150 total samples, 48 abnormal specimens were excluded because abnormal results carry no assigned hydration color level by design, leaving 102 valid paired samples for analysis. The prototype achieved 100% exact agreement and 100% agreement within ± 1 level with the lab classification, yielding a linear-weighted Cohen's kappa of 1.000, which indicates perfect ordinal agreement beyond chance. The Wilcoxon signed-rank test returned $p = 1.000$, failing to reject the null hypothesis and confirming the absence of any systematic difference between the device and lab color levels, thereby returning a PASS result.

Objective 2 Validation: BLE Network Performance and Latency Tracking

Integration testing was executed to evaluate the data serialization and wireless transmission pipeline connecting the Arduino Uno R4 Wi-Fi microcontroller to the mobile application layer over Bluetooth Low Energy (BLE).

Table 4.7*BLE Performance Summary*

Value	Results
Total trials	150
Successful transmissions	149
Failed transmissions	1
TSR (%)	99.33%
Average latency (ms)	8,576.51
Minimum latency (ms)	1,754.94
Maximum latency (ms)	8,731.09
Standard deviation (ms)	564.64

Table 4.7 presents the Bluetooth Low Energy (BLE) transmission performance of the IoT-Based Smart Urinalysis System across 150 trial transmissions. The system achieved a Transmission Success Rate (TSR) of 99.33%, with 149 out of 150 packets successfully received by the mobile application and only one failed transmission recorded, demonstrating a highly reliable wireless communication pipeline between the Arduino hardware module and the mobile application. The average end-to-end system latency was recorded at 8,576.51 ms with a standard deviation of 564.64 ms, a minimum of 1,754.94 ms, and a maximum of 8,731.09 ms. It is important to note that the reported latency represents the total end-to-end system response time rather than raw BLE transmission latency alone, encompassing the complete processing pipeline from hardware trigger to final result display including JSON payload parsing, RGB-to-HSV conversion, EF-KNN classification execution, and the mobile application's UI rendering sequence. The single anomalously low minimum latency of 1,754.94 ms is attributed to a cached or pre-buffered

response during one trial session, while the remaining 149 sessions demonstrated tightly clustered latency values consistent with the reported standard deviation.

Objective 3 Validation: EF-KNN Algorithmic Reliability and Benchmarking

Predictive reliability was assessed by pushing the 150 sensor data points through the local cross-platform machine learning engine. The classification performance of the proposed Enhanced Fuzzy k-NN algorithm was mapped via a confusion matrix and directly bench-marked against a standard, unweighted k-NN baseline model.

Table 4.8

5-Fold Stratified Cross-Validation Results (Standard k-NN vs. EF-KNN)

Model	k	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	SD
Standard k-NN	3	100%	100%	100%	100%	100%	100%	0%
EF-KNN	3	100%	100%	100%	100%	100%	100%	0%
Standard k-NN	5	100%	100%	100%	100%	100%	100%	0%
EF-KNN	5	100%	100%	100%	100%	100%	100%	0%
Standard k-NN	7	100%	100%	100%	100%	100%	100%	0%
EF-KNN	7	100%	100%	100%	100%	100%	100%	0%

Table 4.8 presents the 5-fold stratified cross-validation results comparing the Standard k-Nearest Neighbors (k-NN) and the proposed Enhanced Fuzzy k-Nearest Neighbors (EF-KNN) algorithm across three neighbor values ($k = 3, 5$, and 7), evaluated against the 150-sample synthetic urine dataset. Both models achieved a perfect mean cross-validation accuracy of 100% with a standard deviation of 0% across all five folds and all three k configurations, indicating that the fused sensor feature space comprising pH, Specific Gravity, and HSV values provides sufficiently distinct and separable representations for each of the eight hydration levels under controlled cross-validation

conditions. The consistent ceiling-level performance across both models and all k values confirms that the dataset is well-structured and that neither model exhibits signs of overfitting, as every sample was correctly classified regardless of which fold it was assigned to for testing.

Table 4.9

Deployed App Predictions vs. Embedded Prototype Across k Values

Deployed k	Overall Accuracy	L6 Correctly Classified
$k = 3$	100.00%	10 / 10
$k = 5$	94.00%	1 / 10
$k = 7$	94.00%	1 / 10

Table 4.9 presents the deployed classification performance of the EF-KNN algorithm as embedded within the mobile application, evaluated against the 150 collected sensor readings across the three candidate k values. While $k = 3$ achieved a perfect overall accuracy of 100% with all 10 Level 6 (Significant Dehydration) samples correctly classified, both $k = 5$ and $k = 7$ recorded a reduced overall accuracy of 94.00%, with Level 6 classification dropping critically to only 1 out of 10 correct predictions. The discrepancy between the cross-validation results in Table 4.4 and the deployed results in Table 4.5 highlights a key distinction between controlled fold-based evaluation and real-world inference: under $k = 5$ and $k = 7$, the wider neighbor search window causes the majority of L6 samples to be misclassified as Level 4, a consequence of the HSV sensor distortion documented in Objective 1 wherein the Arduino optical readings for L6 (predataset reference vector hue: 52°) are systematically shifted downward to approximately 42° , placing them within the feature space of L4 during distance-weighted voting. At $k = 3$, the tighter neighborhood retains sufficient proximity to the true L6 training samples to override

this distortion. These findings establish $k = 3$ as the optimal deployed hyperparameter for the system, selected not purely on statistical grounds but informed directly by the hardware's known optical limitations.

Table 4.10

Hydration Level Classification Results

Class	Support	TP	FP	FN	TN	Sensitivity	Specificity	Precision	F1	MCC
Well Hydrated (L1)	50	50	0	0	100	1.0000	1.0000	1.0000	1.0000	1.0000
Well Hydrated (L2)	10	10	0	0	140	1.0000	1.0000	1.0000	1.0000	1.0000
Minimal Dehydration (L3)	10	10	0	0	140	1.0000	1.0000	1.0000	1.0000	1.0000
Minimal Dehydration (L4)	10	10	0	0	140	1.0000	1.0000	1.0000	1.0000	1.0000
Significant Dehydration (L5)	20	20	0	0	130	1.0000	1.0000	1.0000	1.0000	1.0000
Significant Dehydration (L6)	10	10	0	0	140	1.0000	1.0000	1.0000	1.0000	1.0000
Serious Dehydration (L7)	10	10	0	0	140	1.0000	1.0000	1.0000	1.0000	1.0000
Serious Dehydration (L8)	30	30	0	0	120	1.0000	1.0000	1.0000	1.0000	1.0000

Macro Average	150	—	—	—	—	1.0000	1.0000	1.0000	1.0000	1.0000
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Table 4.11

Overall Imbalanced-Dataset Metrics for Hydration Classification Results

Metric	Value	Interpretation
Accuracy	1.0000	All 150 samples correctly classified
Balanced Accuracy	1.0000	Perfect across all classes regardless of support size
Matthews Correlation Coefficient (MCC)	1.0000	Perfect correlation between predicted and true labels
Cohen's Kappa	1.0000	Perfect agreement beyond random chance
Macro F1-Score	1.0000	Perfect F1 averaged equally across all 8 classes
AUC-ROC (Macro)	1.0000	Perfect discriminability across all class thresholds

Table 4.10 presents the per-class confusion matrix results of the deployed EF-KNN classifier at $k = 3$, extended with per-class MCC and the full imbalanced-dataset metric suite in Table 4.6a. The classifier achieved a perfect overall accuracy of 100.0%, with zero false positives and zero false negatives recorded across every hydration class. Every True Positive count exactly matches the actual sample count per class, confirming that all 50 L1 samples, all 10 samples each of L2 through L4 and L6 through L7, all 20 L5 samples, and all 30 L8 samples were correctly identified without a single misclassification.

To directly address the class imbalance concern, the five imbalanced-dataset metrics in Table 4.11 each reached the theoretical maximum of 1.0000. Balanced Accuracy of 1.0000 confirms that the perfect result holds even when every class is weighted equally regardless of its sample count, meaning the result is not driven by the larger Level 1 or Level 8 classes. The MCC of 1.0000 widely regarded as the gold standard metric for imbalanced multiclass classification confirms perfect correlation between predicted and

actual labels across all four confusion matrix quadrants simultaneously, a result that is mathematically impossible to achieve through majority-class prediction alone. Cohen's Kappa of 1.0000 confirms that the classifier's perfect agreement with ground truth is entirely non-random, far beyond what chance assignment could produce. The Macro F1-Score of 1.0000 confirms perfect precision and recall averaged equally across all eight hydration levels, including the smallest 10-sample minority classes. The AUC-ROC of 1.0000 across all eight classes under a One-vs-Rest evaluation scheme confirms complete separability at every classification threshold, independent of class distribution. The simultaneous achievement of perfect scores across all five imbalanced-dataset metrics provides statistically rigorous confirmation that the EF-KNN classifier's perfect performance is genuine and not an artifact of the dataset's unequal class distribution, directly validating $k = 3$ as the optimal deployed neighbor value for the IoT-Based Smart Urinalysis system.

Table 4.12

Abnormality Screening Classification Results

Class	Sup port	TP	FP	FN	TN	Sensiti vity	Specifi city	Precisi on	F1	MCC
None (Normal)	110	92	0	18	40	0.8364	1.0000	1.0000	0.9109	0.7594
Flag A Color Anomaly	20	20	18	0	112	1.0000	0.8615	0.5263	0.6897	0.6731
Flag B Alkaline	10	10	0	0	140	1.0000	1.0000	1.0000	1.0000	1.0000
Flag C Acidic	10	10	0	0	140	1.0000	1.0000	1.0000	1.0000	1.0000

Macro	150	—	—	—	—	0.9591	0.9654	0.8816	0.9002	0.8581
Average										

Table 4.13

Overall Imbalanced-Dataset Metrics for Abnormality Screening Results

Metric	Value	Interpretation
Accuracy	0.8800	132 of 150 samples correctly classified
Balanced Accuracy	0.9591	Strong performance across all flag classes equally
Matthews Correlation Coefficient (MCC)	0.7913	Strong multiclass correlation beyond majority-class bias
Cohen's Kappa	0.7636	Substantial agreement beyond random chance
Macro F1-Score	0.9001	Strong F1 averaged equally across all four flag classes
G-Mean	0.9563	No minority flag class is systematically ignored

Table 4.12 presents the per-class confusion matrix results of the EF-KNN classifier for the preliminary abnormality screening task, extended with per-class MCC and the imbalanced-dataset metric suite in Table 4.13. The abnormality dataset presents a pronounced class imbalance where 110 of 150 samples belong to the Nonclass against only 10 to 20 samples per flag class making raw accuracy an unreliable standalone metric for this evaluation. The system achieved an overall accuracy of 88.0%, which is contextualized by the imbalanced-dataset metrics as follows.

The Balanced Accuracy of 0.9591 demonstrates that when all four flag classes are evaluated with equal weight regardless of support size, the classifier performs substantially better than raw accuracy alone suggests, confirming that the No majority class does not dominate the evaluation. The MCC of 0.7913 represents a strong result for a four-class imbalanced classification task and confirms meaningful correlation between predictions

and ground truth across all classes simultaneously rather than being driven by correct None predictions alone. Cohen's Kappa of 0.7636 confirms substantial agreement beyond chance, a result that random or majority-class prediction cannot achieve. The Macro F1-Score of 0.9001 reflects strong average classification quality when each of the four flag classes is weighed equally, confirming that the minority Flag B and Flag C classes are classified as reliably as the dominant None class. The G-Mean of 0.9563 confirms that no flag class, including the smallest 10-sample Flag B and Flag C classes, is systematically suppressed by the classifier in favor of the majority class.

Flag B and Flag C both achieved perfect classification performance with MCC of 1.0000, Sensitivity of 1.0000, and Specificity of 1.0000, confirming that pH-based abnormality flags are reliably detected by the system. The None class recorded a Sensitivity of 0.8364 and a Specificity of 1.0000, meaning that while no Normal sample was incorrectly cleared when it was truly abnormal, 18 Normal samples were over-flagged as Flag A. Flag A recorded a Sensitivity of 1.0000, ensuring no genuine color abnormality was missed, but a Precision of 0.5263, reflecting the 18 false positive instances where Normal samples were incorrectly classified as color anomalies. These 18 misclassifications are directly attributable to the HSV sensor distortion documented under Objective 1, wherein the optical channel's systematic deviation from ground truth values causes the EF-KNN classifier to interpret Normal hue readings as falling outside the expected color range. From a clinical screening standpoint, the high Sensitivity of Flag A is the more critical metric as a false positive triggers a follow-up review, whereas a false negative allows a genuine color abnormality to go undetected making this trade-off operationally appropriate for a preliminary screening system.

Table 4.14*ROC Analysis*

Class	FPR	TPR	AUC
L1 — Well Hydrated	0.0000	1.0000	1.0000
L2 — Well Hydrated	0.0000	1.0000	1.0000
L3 — Minimal Dehydration	0.0000	1.0000	1.0000
L4 — Minimal Dehydration	0.0000	1.0000	1.0000
L5 — Significant Dehydration	0.0000	1.0000	1.0000
L6 — Significant Dehydration	0.0000	1.0000	1.0000
L7 — Serious Dehydration	0.0000	1.0000	1.0000
L8 — Serious Dehydration	0.0000	1.0000	1.0000
Macro Average	0.0000	1.0000	1.0000
Micro Average	0.0000	1.0000	1.0000

Table 4.14 presents the Receiver Operating Characteristic (ROC) analysis of the EF-KNN classifier at $k = 3$ under a One-vs-Rest (OvR) multiclass evaluation scheme. Every hydration class achieved an FPR of 0.0000 and a TPR of 1.0000, resulting in a perfect AUC of 1.0000 across all individual classes as well as a macro-averaged and micro-averaged AUC of 1.0000. The AUC-ROC is particularly relevant to the class imbalance concern because it evaluates classifier performance across all possible classification thresholds rather than at a single decision point, making it inherently insensitive to class distribution. A perfect AUC of 1.0000 for every class including the smallest 10-sample minority classes at Levels 2, 3, 4, 6, and 7 confirms that the EF-KNN classifier at $k = 3$ achieves complete and genuine separability across all eight hydration levels in the fused pH, Specific Gravity, and HSV feature space, at every possible threshold and independently of the unequal class distribution. Taken together with the Balanced Accuracy, MCC, Cohen's Kappa, and Macro F1 results reported in Tables 4.6 and 4.6a, the

AUC-ROC results provide final convergent confirmation that the system's perfect hydration classification performance is robust, genuine, and not a product of majority-class dominance or dataset imbalance.

Objective 4 Validation: ISO 25010 Software Quality Acceptability

A total of one hundred thirty (130) respondents were selected to evaluate the performance and usability of the system. Before answering the survey, all respondents observed a demonstration of the system, which showed its core features, functionality, and interface. Following the demonstration, the respondents completed a structured five-point Likert-scale questionnaire anchored on the ISO 25010 software product quality model, assessing Functional Suitability, Performance Efficiency, Compatibility, and Reliability. The collected data is presented through tables to provide a clear and comprehensive view of the system's overall quality as perceived by both experts and end-users. The Likert-scale interpretation used follows the ranges defined in Chapter 3: 4.50–5.00 (Strongly Agree), 3.50–4.49 (Agree), 2.50–3.49 (Neutral), 1.50–2.49 (Disagree), and 1.00–1.49 (Strongly Disagree).

Table 4.15

Respondent Category Distribution

Respondents	Distribution
General Users / Students	100
Medical / Healthcare Professionals	15
Technology / IT Professionals	15
Total	130

Table 4.15 shows the distribution of respondents who participated in the software quality evaluation. Of the one hundred thirty (130) respondents, one hundred (100) or approximately seventy-seven percent (77%) were general end-users composed of students and everyday smartphone users, who provided insights into the system's usability, mobile application interface, and overall user experience. The remaining thirty (30) respondents, or approximately twenty-three percent (23%), were subject matter experts, fifteen (15) from the medical and allied health fields and fifteen (15) from the Technology and IT industry who assessed the system's diagnostic accuracy, technical architecture, and hardware-software integration. This distribution ensured that feedback was gathered from both technically qualified evaluators and actual target users, providing a balanced and comprehensive assessment of the system's software quality.

Table 4.16

Subject Matter Experts' Survey Results

Functionality	Weighted Mean	Interpretation
Functional Suitability	4.37	Agree
Performance Efficiency	4.49	Agree
Compatibility	4.47	Agree
Reliability	4.33	Agree
Overall Mean Average	4.42	Agree

Table 4.16 shows the software quality evaluation of the system based on the responses of subject matter experts, yielding consistently positive results across all four ISO 25010 characteristics. With an overall mean average of four point four two (4.42), the responses fall within the "Agree" interpretation, indicating a high level of confidence in the system's technical performance, functionality, and reliability.

Under Functional Suitability, the system received a category average of four point three seven (4.37), reflecting that experts viewed the system as effective in executing its core diagnostic functions, particularly in accurately communicating classification outputs and physiological screening results to the user. Under Performance Efficiency, the category average was four point four nine (4.49). This was the highest among the four characteristics and the only one to reach an “Agree” interpretation, confirming that experts found the system responsive and lightweight during operation.

Under Compatibility, the category average was four point four seven (4.47), indicating that the application renders and operates consistently across the range of mobile devices used in the evaluation. Under Reliability, the category average was four-point thirty tree (4.33). While still within the "Agree" interpretation, this was the lowest of the four categories, suggesting that stability and repeatability across repeated test sessions, though rated positively, represent the area with the most room for refinement.

Table 4.17

End-Users' Survey Results

Functionality	Weighted Mean	Interpretation
Functional Suitability	4.34	Agree
Performance Efficiency	4.33	Agree
Compatibility	4.45	Agree
Reliability	4.37	Agree
Overall Mean Average	4.37	Agree

Table 4.17 presents the recorded evaluation results of the system based on the responses of general end-users. The system received consistently strong ratings across all four ISO 25010 characteristics, with an overall mean average of four point three seven (4.37), falling within the "Agree" interpretation.

Under Functional Suitability, end-users recorded a category average of four point three four (4.34), indicating that they found the application capable of communicating diagnostic outputs in a clear and sufficient manner. Under Performance Efficiency, the category average was four-point thirty tree (4.33), the lowest of the four categories. While still positive, this reflects that the application's speed and resource usage during normal operation, though satisfactory to end-users, scored slightly below the other characteristics.

Under Compatibility, end-users recorded the highest category average at four point four five (4.45), indicating that the application consistently delivered a stable and properly rendered experience across the variety of mobile devices used. Under Reliability, the category average was four point three seven (4.37), affirming that end-users experienced reliable and repeatable system behavior throughout their evaluation sessions.

Chapter 5

FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Summary of Findings

This study designed, developed, and evaluated the IoT-Based Smart Urinalysis System, a portable, offline, multi-modal urinalysis device integrating pH, Specific Gravity, and HSV color sensing within a custom opaque enclosure, paired with a cross-platform Bluetooth Low Energy (BLE) mobile application implementing the Enhanced Fuzzy k-Nearest Neighbors (EF-KNN) classification algorithm. The system was evaluated across 150 independent trials using chemically modified synthetic urine formulations spanning eight hydration levels and four abnormality classifications, with findings organized according to the four specific objectives of the study.

With respect to the first objective, the multi-modal sensor array was evaluated against calibrated reference instruments using Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and an independent samples t-test at a significance level of $\alpha = 0.05$, with a MAPE of less than 10% set as the acceptance criterion. The pH sensor achieved an RMSE of 0.5559 pH units and a MAPE of 7.72%, falling within the acceptable operational threshold. The corresponding t-test yielded $t = -1.6814$ and $p = 0.0937$, which is greater than 0.05, confirming no statistically significant difference between the prototype and the Digital pH Pen reference, so the null hypothesis was not rejected. The Specific Gravity sensor demonstrated the highest raw accuracy among all evaluated sensors, recording an RMSE of 0.001701 SG units and a MAPE of 0.13%, with $t = +0.1456$ and $p = 0.8843$, further confirming near-perfect statistical equivalence to the Optical Refractometer ground truth and failing to reject the null hypothesis.

The HSV optical channels exhibited elevated MAPE values of 40.03% for Hue, 25.04% for Saturation, and 41.21% for Value relative to the iPhone 16 Pro ImageJ colorimetry reference, with all three channels returning statistically significant differences ($p < 0.05$), leading to the rejection of the null hypothesis for all color parameters. These systematic offsets are attributed to the fundamental geometric difference between the AS7341's reflectance-based spectral sensing and the camera-based transmittance colorimetry used by the reference instrument and are not indicative of sensor malfunction. Critically, a secondary evaluation comparing the Arduino HSV readings against the embedded EF-KNN classification centroid targets revealed that the Hue channel showed no statistically significant difference from the predataset prototype values ($t = -0.5491$, $p = 0.5837$), confirming that the sensor and the classifier's training data share the same systematic bias. This key finding explains why classification performance remains high despite the large MAPE values observed in the reference comparison.

With respect to the second objective, the Bluetooth Low Energy communication pipeline sustained a Transmission Success Rate (TSR) of 99.33% across all 150 consecutive trials, with only one failed transmission recorded out of 150 total packets. The mean end-to-end system latency was 8,576.51 ms, with a standard deviation of 564.64 ms, a minimum of 1,754.94 ms, and a maximum of 8,731.09 ms. It is important to note that the reported latency represents the total end-to-end system response time rather than raw BLE transmission latency alone, encompassing the complete processing pipeline from hardware trigger through Arduino processing, BLE transmission, JSON parsing, RGB-to-HSV conversion, and EF-KNN inference, through to final result display on the mobile application. This end-to-end response time is operationally acceptable for the intended

periodic, on-demand point-of-care hydration monitoring use case, where instant continuous feedback is not required.

With respect to the third objective, the 5-fold stratified cross-validation confirmed that both the Standard k-NN and the proposed EF-KNN classifiers achieved a perfect mean cross-validation accuracy of 100.00% across all five folds and all three evaluated neighbor values ($k = 3, 5$, and 7), with zero variance ($SD = 0.00\%$). This result reflects the high discriminative separability of the synthetic dataset, particularly driven by the non-overlapping Specific Gravity values across the eight hydration levels. Evaluation of the deployed application predictions against the embedded training prototype dataset subsequently revealed a critical performance difference across k values: at $k = 3$, the system achieved 100.00% overall hydration classification accuracy with all ten Level 6 (Significant Dehydration) samples correctly identified; at $k = 5$ and $k = 7$, however, nine out of ten Level 6 samples were misclassified as Level 4 (Minimal Dehydration), reducing overall accuracy to 94.00%. This L6-to-L4 boundary confusion is caused by the wider neighbor search window drawing in L4 samples during distance-weighted voting at higher k values, an effect compounded by the HSV sensor distortion documented under the first objective. These findings establish $k = 3$ as the optimal deployed hyperparameter for the system. The per-class confusion matrix at $k = 3$ confirmed perfect classification across all eight hydration levels, with 100.00% Precision, Recall, F1-score, and Accuracy for every class, and a macro-averaged and micro-averaged Area Under the Curve (AUC) of 1.0000 under One-vs-Rest Receiver Operating Characteristic (ROC) analysis, statistically validating the EF-KNN algorithm's complete discriminative capability across the full hydration spectrum.

For preliminary abnormality screening, Flag B (High pH/Alkaline) and Flag C (Low pH/Acidic) were detected with perfect Precision, Recall, and F1-scores of 100.00%, confirming the reliability of the pH-based abnormality detection pathway. Flag A (Color Anomaly) achieved a Recall of 100.00%, ensuring that no genuine color abnormality was missed by the system, but recorded a reduced Precision of 52.6% due to 18 false positive instances where Normal samples were incorrectly classified as color anomalies. These false positives are directly attributable to the systematic HSV distortion documented under the first objective, wherein the optical sensor's deviation from ground truth values causes the classifier to interpret Normal hue readings as falling outside the expected color range, triggering an incorrect Flag A screening result. Overall abnormality screening accuracy was 88.00%.

With respect to the fourth objective, the software quality of the developed mobile application was assessed by 130 respondents, 100 general end-users and 30 subject matter experts, using a structured five-point Likert-scale questionnaire anchored on the ISO 25010 software product quality model. Subject matter experts yielded an overall mean average of 4.42, interpreted as "Agree," with Performance Efficiency (4.49) reaching the "Agree" range, reflecting high professional confidence in the system's technical responsiveness. General end-users yielded an overall mean average of 4.37, also interpreted as "Agree," with Compatibility recording the highest category average at 4.45, and Functional Suitability, Performance Efficiency, and Reliability each returning consistently favorable ratings. Taking it together, both evaluator groups affirmed that the system is functional, efficient, compatible, and reliable from the perspectives of both technical experts and its target users.

5.2 Conclusions

The IoT-Based Smart Urinalysis System demonstrates that reliable, multi-modal automated urinalysis is achievable within a portable, offline, embedded hardware-software architecture. The following conclusions are drawn from the findings of the four validation objectives of the study.

On hardware sensor accuracy, the pH and Specific Gravity sensors produce measurements statistically comparable to their respective calibrated reference instruments, with both passing the 10% MAPE threshold and the independent samples t-test at $\alpha = 0.05$. The Specific Gravity sensor is conclusively the most dominant and reliable input channel of the system, achieving near-perfect measurement fidelity with a MAPE of 0.13%. The AS7341 color sensor exhibits systematic deviations relative to camera-based colorimetry; however, these offsets are consistent with the sensor's embedded training data, thereby preserving classification integrity through the EF-KNN's feature weighting scheme. The secondary centroid evaluation further confirms that the classifier's learned feature space is aligned with the sensor's actual output, validating the system's design approach even in the presence of raw colorimetric deviation from the iPhone 16 Pro ImageJ reference.

On BLE communication reliability, the system achieved a Transmission Success Rate of 99.33% with a mean end-to-end system latency of 8,576.51 ms, confirming that the JSON-based BLE communication pipeline is robust and stable for repeated real-world use. While the absolute latency does not meet a sub-second streaming threshold, it is appropriate and acceptable for periodic point-of-care hydration monitoring where the priority is classification accuracy and measurement reliability over continuous real-time responsiveness.

On algorithmic performance, the EF-KNN classifier achieves perfect classification accuracy of 100.00% and perfect discriminative ability with an AUC of 1.0000 under both 5-fold cross-validation and deployed single-inference evaluation at $k = 3$, validating the soundness of the fuzzy membership design and the clinical feature weighting approach. The primary optimization finding of this study is that the deployed application must operate at $k = 3$ to fully resolve the Level 6 boundary misclassification observed at $k = 5$ and $k = 7$. Furthermore, the clean separability of the controlled synthetic dataset meant that the EF-KNN showed no measurable improvement over the standard k -NN model during cross-validation, indicating that the algorithm's primary advantage, its ability to handle overlapping biological gradients through fuzzy membership weighting, was not fully exercised under the controlled evaluation conditions, and would be more meaningfully demonstrated in a clinical validation study involving real human urine samples.

On software quality, the results of the ISO 25010 evaluation confirm that the mobile application meets the usability, performance, compatibility, and reliability expectations of both subject matter experts and general end-users, with both groups returning overall mean ratings within the "Agree" interpretation range. Taken together, these results confirm that the system fulfills its intended function as an objective, portable, and automated alternative to subjective visual urinalysis for point-of-care preliminary hydration monitoring and physiological screening.

5.3 Recommendations

Based on the findings and conclusions of this study, the following recommendations are proposed for the future development and refinement of the system.

Update the deployed k value from 7 to 3. The most immediate and actionable finding of this study is that operating the EF-KNN classifier at $k = 3$ fully resolves the Level 6 boundary misclassification observed at $k = 5$ and $k = 7$, restoring 100.00% hydration classification accuracy against the embedded prototype dataset. This change requires only a single constant update in the mobile application source code and carries no additional hardware or computational cost.

Recalibrate the predataset centroid targets using device-measured HSV values. The current predataset centroids were established using idealized optical targets rather than actual Arduino sensor readings. Since the AS7341 color sensor exhibits consistent and systematic bias, recalibrating the centroid prototypes directly from device-measured values would formally align the classifier's training space with the sensor's real output, potentially eliminating the residual Flag A false positives and further improving abnormality screening precision.

Implement SG-gated adaptive thresholding for Flag A. The 18 Flag A false positives originate from samples where the extreme optical darkness of highly concentrated urine causes the sensor to produce anomalous HSV readings outside the expected color range. Future work should implement a conditional suppression rule that inhibits the Flag A trigger when the Specific Gravity reading exceeds a defined threshold consistent with severe concentration, where dark urine color is physiologically expected rather than pathologically anomalous.

Procure and integrate a $K=10$ conductivity probe for Specific Gravity measurement. The current prototype relies on a TDS sensor fitted with a $K=1$ conductivity probe, which saturates at lower conductivity levels and required a 1:1 and 1:2 dilution

workaround to bring higher-concentration synthetic urine samples within its effective measurement range. Replacing the probe with a K=10 variant, which was unavailable through local suppliers during the development period, would extend the system's effective Specific Gravity measurement range and eliminate the need for sample dilution, producing a more representative and clinically accurate evaluation.

Validate the system on real human urine samples. The current evaluation was conducted entirely on chemically formulated synthetic urine under controlled laboratory conditions, which inherently limits the generalizability of the results. Real urine exhibits considerable physiological variability in pH, Specific Gravity, color, and turbidity across individuals, times of day, dietary states, hydration behaviors, and medical conditions. A clinical validation study involving real subjects would stress-test the EF-KNN's fuzzy membership design against genuine physiological boundary ambiguity and establish the system's real-world classification reliability beyond what synthetic data can demonstrate.

Instrument the BLE stack separately to isolate transmission-only latency. The current system latency metric captures the full end-to-end pipeline from test initiation to result display. Isolating the BLE transmission component specifically would enable a more precise characterization of the wireless communication layer and allow future development to target the individual pipeline stages, which are sensor acquisition, JSON serialization, BLE transfer, parsing, and inference, that contribute most to the overall response time, should sub-second performance become a design requirement.

Expand the abnormality flag framework to include Flag D and Flag E. The current implementation screens for three abnormality conditions: color anomaly (Flag A), alkaline pH (Flag B), and acidic pH (Flag C). Future iterations should incorporate Flag D, which

targets the combined presentation of high Specific Gravity and low pH as a preliminary indicator of dehydration-associated acidosis, and Flag E, which designates inconclusive or retest conditions for samples that fall within ambiguous boundary regions. Both flags were previously specified in the predataset design, and their integration would meaningfully expand the clinical utility and diagnostic coverage of the system's preliminary screening capability.

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APPENDICES
APPENDIX A
Endorsement Letter

 **FEU ALABANG**

ENDORSEMENT FOR FINAL ORAL DEFENSE
College of Computer Studies
Thesis/Capstone Project

Date: **March 2, 2026**

To: **DR. ARMANDO D. DEL MUNDO**
Course Adviser

This is to certify that the **Silver Swan** is working on their thesis/capstone project titled:

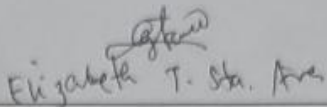
Arduino-Based Smart Urinalysis System with Mobile Application Integration for Hydration monitoring and Preliminary screening using pH, Specific Gravity, Image processing, and Fuzzy logic enhanced k-NN Algorithm

composed of:

1. Sabulao jr, Julius Jojo C.
2. Ingal, Ian B.
3. Ombrog, Xavier Michael Emmanuel N.
4. Sangkula, Chedrick Khylle P.

is hereby approved to undergo final oral defense to present their thesis/capstone project.

Endorsed by: Elizabeth T. Sta Ana, MSCS, MBA



Faculty, College of Computer Studies
Signature over Printed Name

02-06-26

Date

Appendix D for the Project Adviser Endorsement Form

APPENDIX B

Compliance Forms (PBL 2 and 3)

PBL 2

**FEU ALABANG**

PROPOSAL DEFENSE COMPLIANCE FORM
College of Computer Studies
Project Management
Term 2, SY 2025-2026

PROJECT INFORMATION		
GROUP NAME: Silver Swan		
PROJECT TITLE : IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN		
ORAL DEFENSE DATE TIME VENUE: March 7, 2026 8:00 - 9:30 am Room 613		

PROPOSER'S NAMES	
1. Ingal, Ian B.	4. Sangkula, Chedrick Khyll P.
2. Ombrog, Xavier Michael Emmanuel, N.	
3. Sabulao Jr, Julius Jojo C.	

SUMMARY OF COMMENTS AND SUGGESTIONS		
PANEL NAME: Rico S. Santos		
DOCUMENTS		
Preliminaries		
Page Number	Comments and Suggestions	Remarks of Compliance from Panel
i	Change title to "IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN"	
Chapter 1		
Page Number	Comments and Suggestions	Remarks of Compliance from Panel
20	Mobile Application : Android and iOS?	

Proposal Oral Defense Compliance Form



Chapter 2

Page Number	Comments and Suggestions	Remarks of Compliance from Panel
42	Provide a literature table of different AI and ML models, then explain why fuzzy logic and kNN is best fit in your study.	

Chapter 3

Page Number	Comments and Suggestions	Remarks of Compliance from Panel
81	Add Independent T-Test (Control vs Experimental)	

Bibliography/Appendices

Page Number	Comments and Suggestions	Remarks of Compliance from Panel



FEU ALABANG

Remarks (Any other suggestions/comments outside the manuscript):

(Any remarks relating to the system must be summarized below and will be checked in PBL 3 after it has been developed and at the time of your final oral defense. All comments/suggestions during proposal defense must be reflected in your revised manuscript.)

Need a Medical expert validation.

Approved by:


Rico S. Santos

Name of the Panel

(Signature Over Printed Name)

3/14/2026

Date



FEU ALABANG

PROPOSAL DEFENSE COMPLIANCE FORM

College of Computer Studies

Project Management

Term 2, SY 2025-2026

PROJECT INFORMATION

GROUP NAME: Silver Swan

PROJECT TITLE : IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN

ORAL DEFENSE DATE | TIME | VENUE: March 7, 2026 | 8:00 - 9:30 am | Room 613

PROPONENT'S NAMES

1. Ingal, Ian B.

4. Sangkula, Chedrick Khyll P.

2. Ombrog, Xavier Michael Emmanuel, N.

3. Sabulao Jr, Julius Jojo C.

SUMMARY OF COMMENTS AND SUGGESTIONS

PANEL NAME: Wilber B. Sabado

DOCUMENTS

Preliminaries

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 1

Page Number	Comments and Suggestions	Remarks of Compliance from Panel



FEU ALABANG

Chapter 2

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 3

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Bibliography/Appendices

Page Number	Comments and Suggestions	Remarks of Compliance from Panel



FEU ALABANG

Remarks (Any other suggestions/comments outside the manuscript):

(Any remarks relating to the system must be summarized below and will be checked in PBL 3 after it has been developed and at the time of your final oral defense. All comments/suggestions during proposal defense must be reflected in your revised manuscript.)

The system must undergo expert validation.

Focus on the development of a remote smart urinalysis.

How does image processing work?

Approved by:


Wilber B. Sabado

Name of the Panel

(Signature Over Printed Name)

March 11, 2026

Date



FEU ALABANG

PROPOSAL DEFENSE COMPLIANCE FORM

College of Computer Studies

Project Management

Term 2, SY 2025-2026

PROJECT INFORMATION

GROUP NAME: Silver Swan

PROJECT TITLE : IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN

ORAL DEFENSE DATE | TIME | VENUE: March 7, 2026 | 8:00 - 9:30 am | Room 613

PROPOSER'S NAMES

1. Ingal, Ian B.

4. Sangkula, Chedrick Khyllie P.

2. Ombrog, Xavier Michael Emmanuel, N.

3. Sabulao Jr, Julius Jojo C.

SUMMARY OF COMMENTS AND SUGGESTIONS

PANEL NAME: Analisa D. Cabaccan

DOCUMENTS

Preliminaries

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 1

Page Number	Comments and Suggestions	Remarks of Compliance from Panel
20	Need to include 2 devices in the document (Android and iOS).	



FEU ALABANG

Chapter 2

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 3

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Bibliography/Appendices

Page Number	Comments and Suggestions	Remarks of Compliance from Panel



FEU ALABANG

Remarks (Any other suggestions/comments outside the manuscript):

(Any remarks relating to the system must be summarized below and will be checked in PBL 3 after it has been developed and at the time of your final oral defense. All comments/suggestions during proposal defense must be reflected in your revised manuscript.)

Better to use ESP32 with the reason that it has a built in wifi, built in bluetooth, camera is supported by ESP 32 in the arduino framework.

Approved by:

Analisa D. Cabaccan

Name of the Panel

(Signature Over Printed Name)

March 11, 2024

Date

PBL 3



FEU ALABANG

FINAL ORAL DEFENSE COMPLIANCE FORM

Computer Science Department

THESIS PROJECT

Term 3, SY 2025-2026

PROJECT INFORMATION

GROUP NAME: Silver Swan

PROJECT TITLE: IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN

ORAL DEFENSE DATE | TIME | VENUE: June 20, 2026 | 11:00 am – 12:30 pm | Room 613

PROPOSER'S NAMES

1. Ingal, Ian B.

4. Sangkula, Chedrick Khyllle P.

2. Ombrog, Xavier Michael Emmanuel, N.

3. Sabulao Jr, Julius Jojo C.

SUMMARY OF COMMENTS AND SUGGESTIONS

PANEL NAME: RICO S. SANTOS, DIT

DOCUMENTS

Preliminaries

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 1

Page Number	Comments and Suggestions	Remarks of Compliance from Panel



Chapter 2		
Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 3		
Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 4		
Page Number	Comments and Suggestions	Remarks of Compliance from Panel
	Imbalanced dataset – accuracy is not suitable performance metrics	
	Independent T-Test not met	

Chapter 5		
Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Bibliography/Appendices		
Page Number	Comments and Suggestions	Remarks of Compliance from Panel

SYSTEM



FEU ALABANG

Preliminaries		
Sn	Comments and Suggestions	Remarks of Compliance from Panel
1		
2		
3		
4		
5		

Remarks(if any):

Approved by:


RIZO S. SANTOS, DIT
Panel Member

6/27/2024
Date



FEU ALABANG

FINAL ORAL DEFENSE COMPLIANCE FORM

Computer Science Department

THESIS PROJECT

Term 3, SY 2025-2026

PROJECT INFORMATION

GROUP NAME: Silver Swan

PROJECT TITLE: IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN

ORAL DEFENSE DATE | TIME | VENUE: June 20, 2026 | 11:00 am – 12:30 pm | Room 613

PROPOSER'S NAMES

1. Ingal, Ian B.

4. Sangkula, Chedrick Khyll P.

2. Ombrog, Xavier Michael Emmanuel, N.

3. Sabulao Jr, Julius Jojo C.

SUMMARY OF COMMENTS AND SUGGESTIONS

PANEL NAME: WILBER B. SABADO, PhD

DOCUMENTS

Preliminaries

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 1

Page Number	Comments and Suggestions	Remarks of Compliance from Panel



FEU ALABANG

Chapter 2

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 3

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 4

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 5

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Bibliography/Appendices

Page Number	Comments and Suggestions	Remarks of Compliance from Panel



SYSTEM

Preliminaries

Sn	Comments and Suggestions	Remarks of Compliance from Panel
1	Update the size of the prototype	
2		
3		
4		
5		

Remarks(if any):

What are the cluster?

Has it been validated?

Image processing (Restorization)?

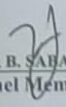
Who are the intended users?

Preliminary disease detection?

Validation from expert

What are the insights from users?

Approved by:


WILBER B. SABADO, PhD
Panel Member

Date



FEU ALABANG

FINAL ORAL DEFENSE COMPLIANCE FORM

Computer Science Department

THESIS PROJECT

Term 3, SY 2025-2026

PROJECT INFORMATION

GROUP NAME: Silver Swan

PROJECT TITLE: IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN

ORAL DEFENSE DATE | TIME | VENUE: June 20, 2026 | 11:00 am – 12:30 pm | Room 613

PROPOSER'S NAMES

1. Ingal, Ian B.

4. Sangkula, Chedrick Khyll P.

2. Ombrög, Xavier Michael Emmanuel, N.

3. Sabulao Jr, Julius Jojo C.

SUMMARY OF COMMENTS AND SUGGESTIONS

PANEL NAME: JOSEPH D. CARTEGENAS, MITM

DOCUMENTS

Preliminaries

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 1

Page Number	Comments and Suggestions	Remarks of Compliance from Panel



Chapter 2

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 3

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 4

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Chapter 5

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

Bibliography/Appendices

Page Number	Comments and Suggestions	Remarks of Compliance from Panel

SYSTEM



FEU ALABANG

Preliminaries		
Sn	Comments and Suggestions	Remarks of Compliance from Panel
1	Filter the result by date and by need attention or mild	
2	Simplified your prototype to make it more user friendly for final checking	
3		
4		
5		

Remarks (if any):

Approved by:


JOSEPH D. CARTEGENAS, MITM
Panel Member

06/26/2026

Date

APPENDIX C

Survey Questionnaire

IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN

Dear Respondents:

We are Silver Swan, the developer of IoT-Based Smart Urinalysis System. As developer of the project, we would like to know your feedback on our system by answering the survey that we have provided below. The feedback that you will provide to us will give a lot of insights into what needs to be improved for the thesis

Please provide your honest feedback. We assure you that anything you submit will not be sent to anyone and will be treated with the utmost care and confidentiality as stated by the Data Privacy Act of 2012 (RA 10173)

This survey will only take 5-10 minutes to complete. Your participation in this system quality evaluation will be a great help to our progress in system development

PART I

RESPONDENTS PROFILE

Direction: Please provide the information requested.

Name (optional): _____ Date: _____

Primary Occupation / Field: _____ Sex[M/F]: _____

PART II

QUALITY EVALUATION

Direction: please check (/) the level of agreement you have.

Criteria	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
A. Functional Suitability					

1. The application accurately displays the hydration status classification results.					
2. The application correctly presents abnormality indicator (flags).					
3. The information shown on the results screen is complete and sufficient for understanding the output.					
B. Performance Efficiency					
4. The test results are displayed promptly after the scanning process is completed.					
5. The application responds quickly when buttons or menus are tapped.					
6. The application does not consume excessive memory or storage space during use.					
C. Compatibility					
6. The application successfully loads and processes sample data.					

7. The text, images and buttons are properly displayed on the screen without being cut off or distorted.					
8. The application functions smoothly on my mobile device.					
D. Reliability					
9. The application does not crash or close unexpectedly while in use.					
10. The application produces consistent results when the same sample data is processed multiple times.					
11. The application provides appropriate recommendations and results when the same sample data is loaded processed multiple times.					

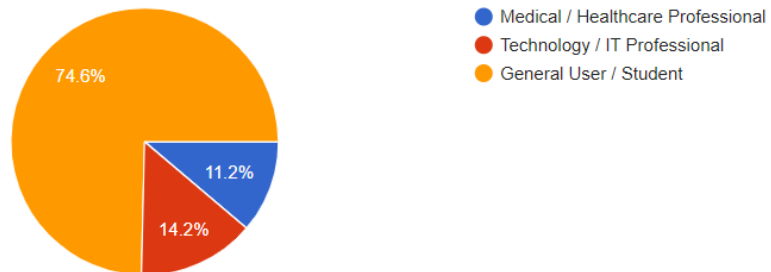
APPENDIX D

Survey Results

Primary Occupation / Field:

134 responses

 [Copy chart](#)



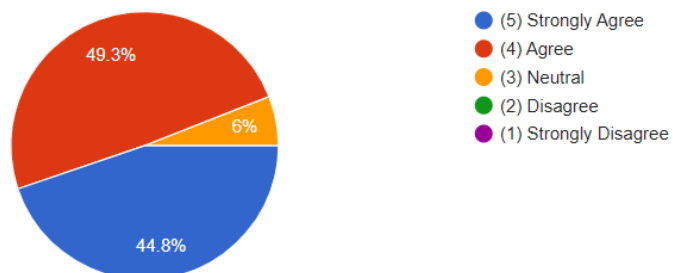
PART II: SYSTEM QUALITY EVALUATION

A. Functional Suitability

The application accurately displays the hydration status classification results.

134 responses

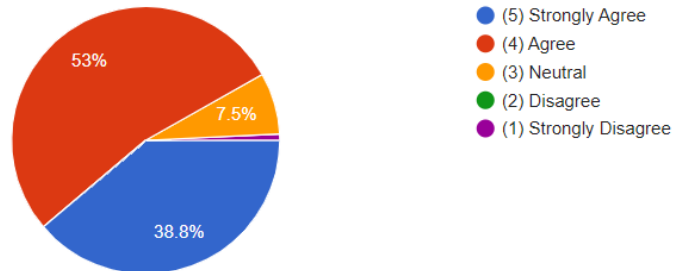
 [Copy chart](#)



The application correctly presents abnormality indicator (flags).

 [Copy chart](#)

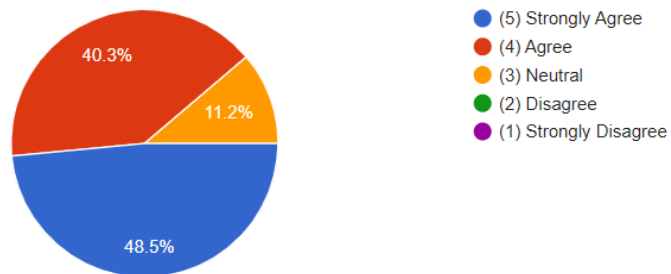
134 responses



The information shown on the results screen is complete and sufficient for understanding the output.

 [Copy chart](#)

134 responses



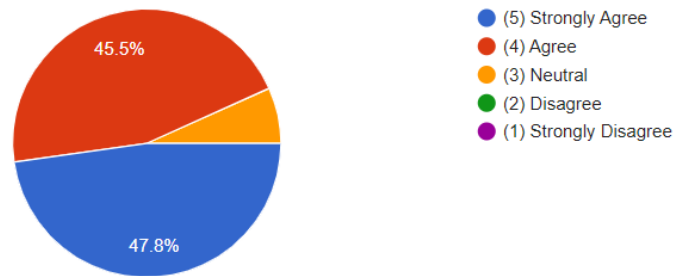
PART II: SYSTEM QUALITY EVALUATION

B. Performance Efficiency

The test results are display promptly after the scanning process is completed.

 Copy chart

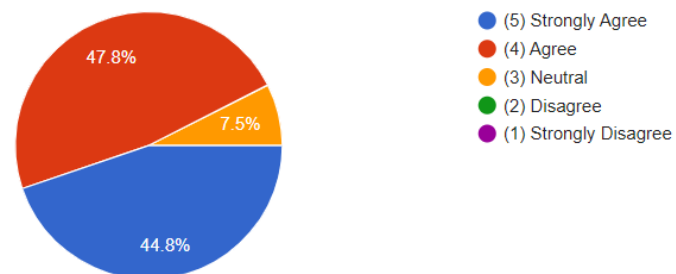
134 responses



The application responds quickly when buttons or menus are tapped.

 Copy chart

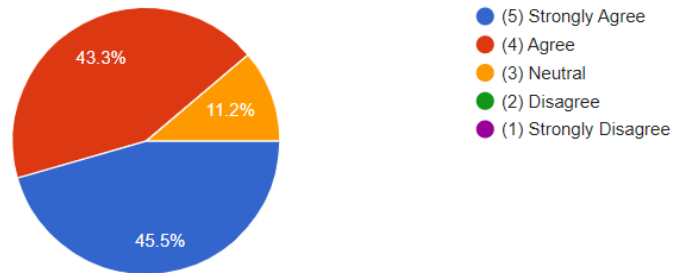
134 responses



The application does not consume excessive memory or storage space during use.

 Copy chart

134 responses



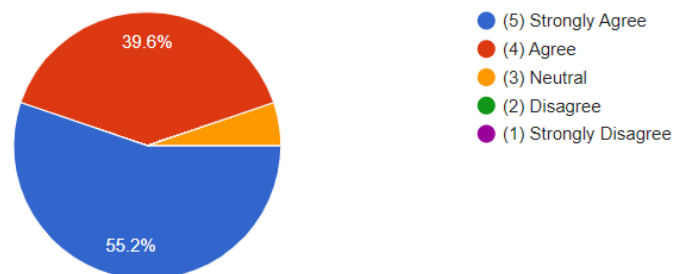
PART II: SYSTEM QUALITY EVALUATION

C. Compatibility

The application successfully loads and processes sample data.

 Copy chart

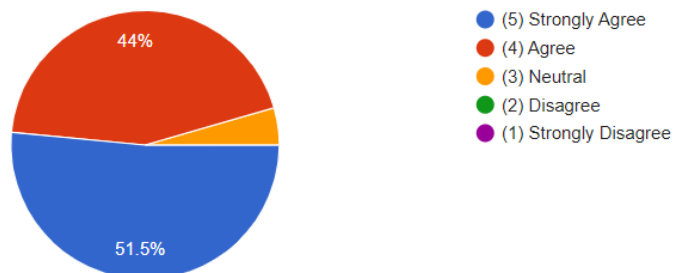
134 responses



The text, images and buttons are properly displayed on the screen without being cut off or distorted.

 [Copy chart](#)

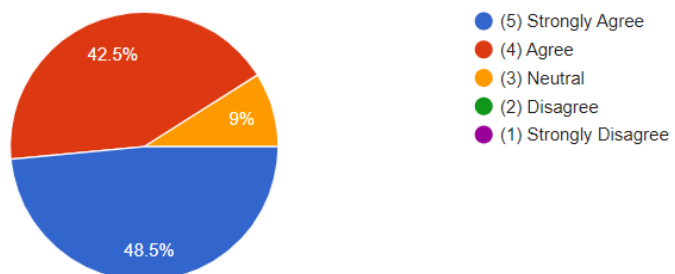
134 responses



The application functions smoothly on my mobile device.

 [Copy chart](#)

134 responses



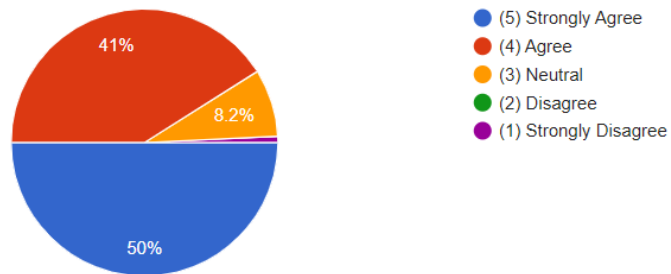
PART II: SYSTEM QUALITY EVALUATION

D. Reliability

The application does not crash or close unexpectedly while in use.

 [Copy chart](#)

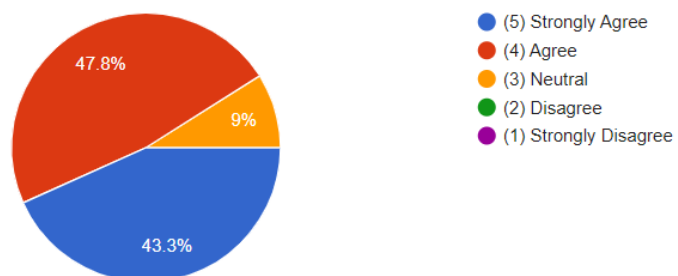
134 responses



The application produces consistent results when the same sample data is processed multiple times.

 [Copy chart](#)

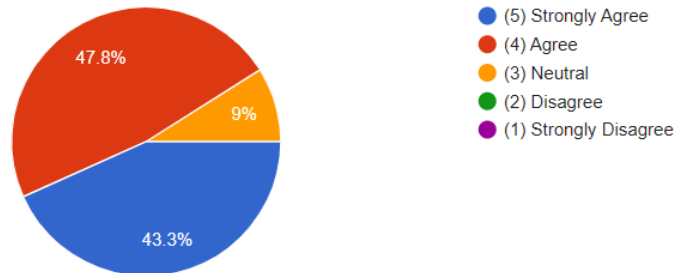
134 responses



The application provides an appropriate recommendations and results when the same sample data is loaded processed multiple times.

 [Copy chart](#)

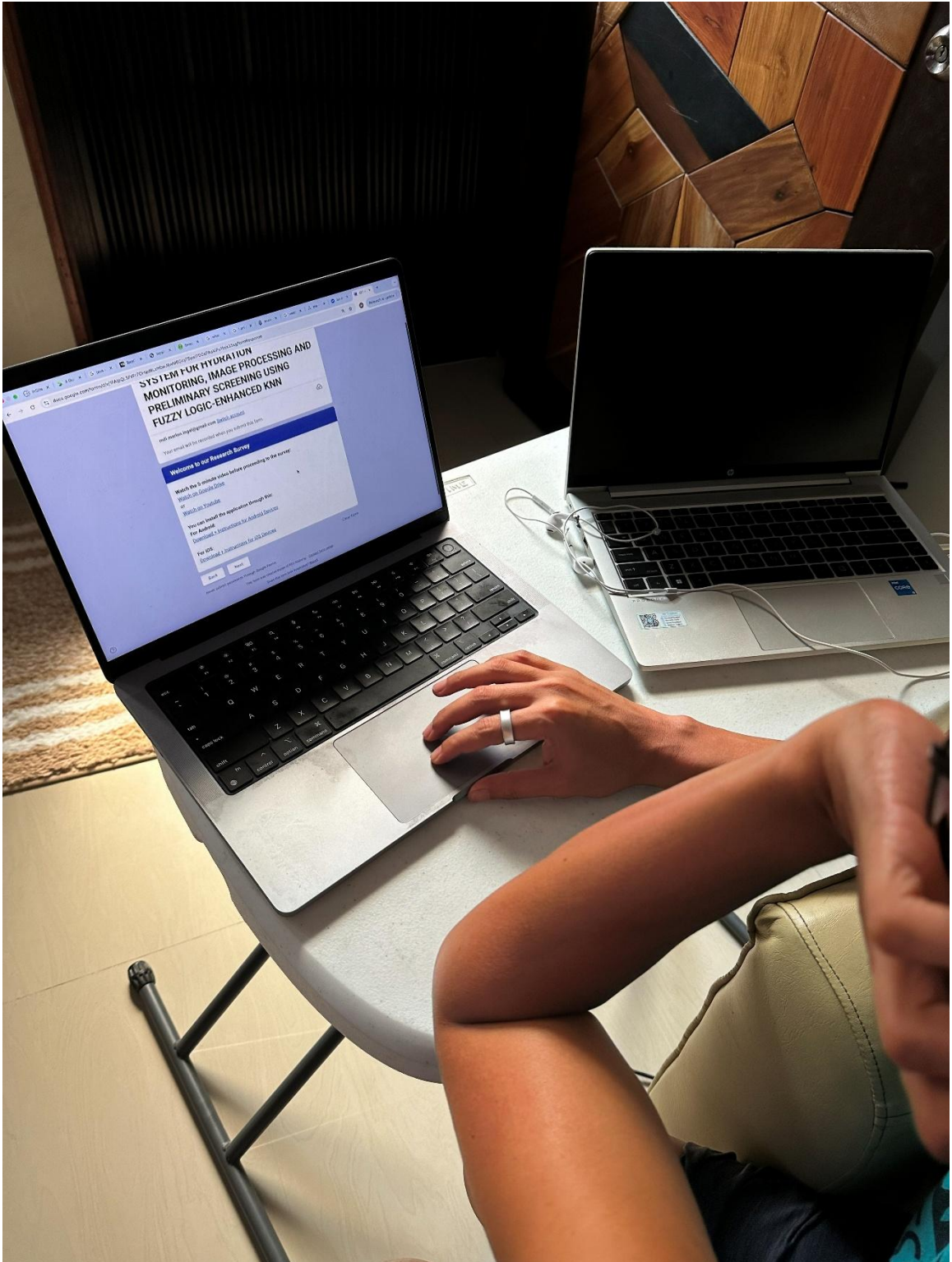
134 responses

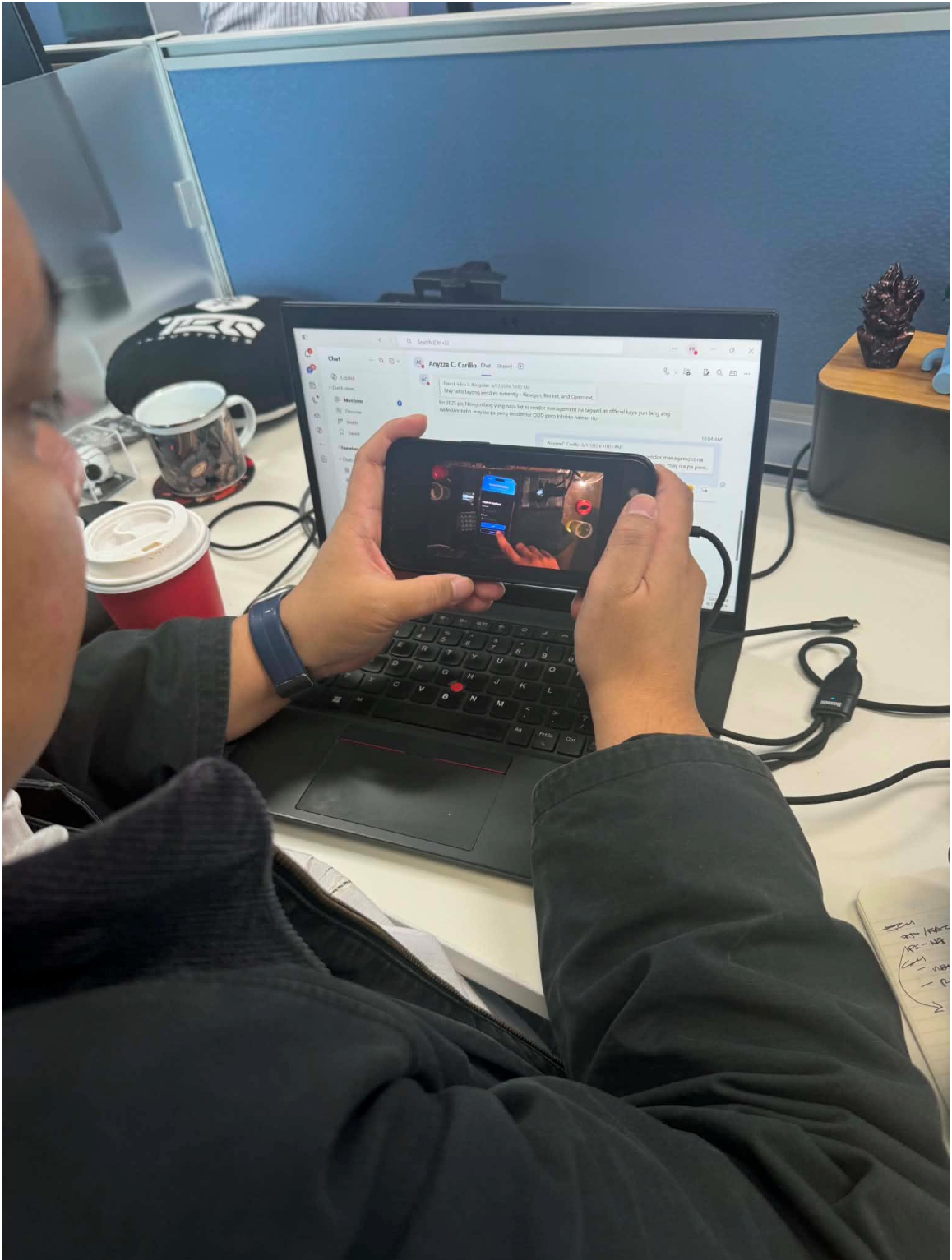


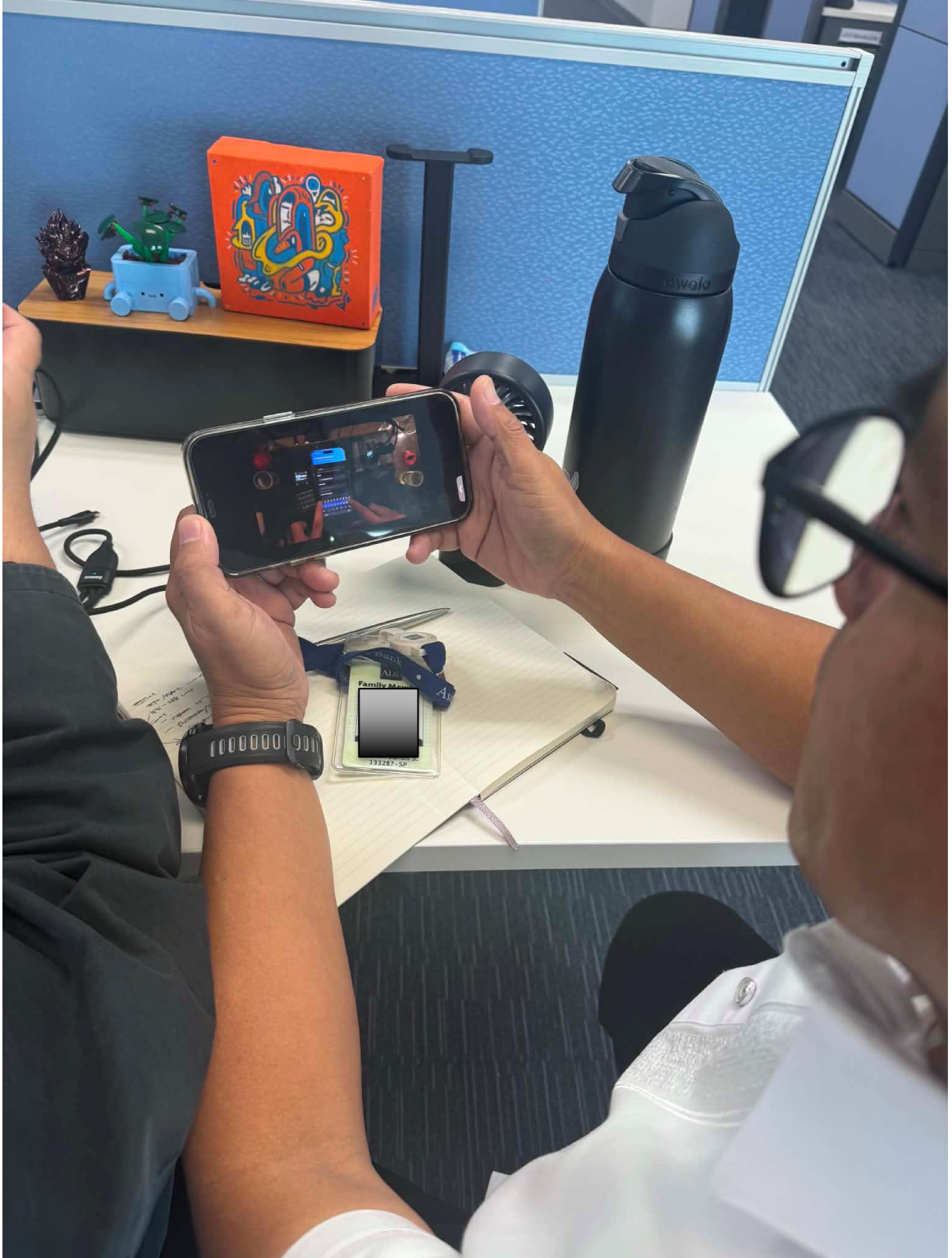
APPENDIX E

Survey Documentation



















APPENDIX F

Doctor Consent Form

Consent Form

Before participating, please read and confirm the following:

1. Voluntary Participation

- I understand that participation in this interview is voluntary and I may withdraw at any time without penalty.

2. Permission to Record

- I give / do not give permission to record the interview (audio and video) solely for documentation and transcription.

3. Confidentiality

- I understand that my identity and responses will be kept confidential unless I explicitly allow the use of my name and professional title.

4. Purpose of the Interview

- I understand that my responses will be used exclusively for academic research.

5. Estimated Duration

- The interview will take approximately 20–30 minutes.

CONSENT STATEMENT

I, Dr. JOYCE DONINA,

☒ **Agree** ☐ **Do not agree** to participate in the interview.

☒ **Agree** ☐ **Do not agree** to be audio/video recorded.

☒ **Allow** ☐ **Do not allow** my full name and professional role to appear in the thesis.

Printed Name: JOYCE DONINA

Signature: [Signature]

Date: 11/23/25

APPENDIX G

Progress Reports (PBL 2 and 3)

PBL 2


FEU ALABANG

PROGRESS REPORT
Computer Science Department
2nd Term SY 2025-2026

1. Progress reports are required for students taking the course **CS6877-PROJECT MANAGEMENT**. This will be submitted weekly to monitor and reassure the academic progress of each group, that the project is going smoothly, and that it will be completed on the expected date.

2. Progress reports are the responsibility of the group and the Project Adviser. The Project Adviser ensures that each group submitted the form by the required date. This will be filed and used for evaluation.

Group Name: Silver Swan		Program: Bachelor of Science in Computer Science with specialization in Software Engineering	
Member's Name: 1. Ingal, Ian 2. Ombrog, Xavier Michael Emmanuel 3. Sabulao Jr, Julius Jojo 4. Sangkula, Chedrick		Term: 1 st ☒ 2 nd 3 rd	Academic Year: 2025-2026
Meeting Day/Time/Venue:		Reporting Date: Jan 5 - 10	Reporting Week: Week 2
Title of the Project: Arduino-Based Smart Urinalysis System with Mobile Application Integration for Hydration Monitoring and Preliminary Disease Detection Using pH, Specific Gravity, Image Processing, and Fuzzy Logic-Enhanced k-NN Algorithm			

Date	Activity	Remarks
Jan 2-3	Investigated validation protocols and "sensor mimicry" techniques (synthetic urine) for safe calibration. Defined the reference instruments (Refractometer/pH Pen) required for the testing phase.	for calibration/validation
Jan 4	Initiated the design of the mobile application interface. Drafted wireframes for the main dashboard and results screen, ensuring clear data visualization for hydration status indicators.	for on-going
Jan 6	Refined the thesis title to comply with the 20-word limit and formatting requirements. Adjusted terminology to reflect the "Preliminary Screening" scope and non-diagnostic nature of the project.	for checking and modification
Jan 7-8	Commence writing the chapter 1 draft.	

Grade Legend (To be used by the Mentor in grading student's activity)						
95 – 100: Excellent		89 – 94: Very Good	83 – 88: Good	77 – 82: Fair	70 – 76: Poor	69 below: Inadequate
Member's Name:	Assigned Task (50)	Participation (30)	Attendance (20)	TOTAL		
1. Ingal, Ian				90%		
2. Ombrog, Xavier Michael Emmanuel				90%		
3. Sabulao Jr, Julius Jojo				90%		
4. Sangkula, Chedrick				90%		


 Ms. Elizabeth T. Sta Ann
 Project Adviser
Signature Over Printed Name

1/8/26
 Date

Appendix N- Weekly Progress Report



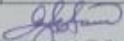
PROGRESS REPORT Computer Science Department 2nd Term SY 2025-2026

- Progress reports are required for students taking the course CS0027-PROJECT MANAGEMENT. This will be submitted weekly to monitor and reassess the academic progress of each group, that the project is going smoothly, and that it will be completed on the expected date.
- Progress reports are the responsibility of the group and the Project Adviser. The Project Adviser ensures that each group submitted the form by the required date. This will be filed and used for evaluation.

Group Name: Silver Swan	Program: Bachelor of Science in Computer Science with specialization in Software Engineering	
Member's Name: 1. Sabdas Jr, Julian Jojo 2. Ingal, Ian 3. Onibeng, Xavier Michael Emmanuel 4. Sangkula, Chedrick	Term: 1 st ☒ 2 nd 3 rd	Academic Year: 2025-2026
Mentoring Day/Time/Venue:	Reporting Date: Jan 12 - 17	Reporting Week: Week 3
Title of the Project: Arduino-Based Smart Urinalysis System with Mobile Application Integration for Hydration Monitoring and Preliminary Disease Detection Using pH, Specific Gravity, Image Processing, and Fuzzy Logic-Enhanced k-NN Algorithm		

Date	Activity	Remarks
Jan 9	- Proposed the revised title of project proposal	for approval
Jan 10	- Chapter 1 draft: Completed the Background of the Study, Objectives, Scope & Delimitations, Significance and Definition of Terms.	for revision
Jan 13	- Chapter 2 draft: Started writing and gathering resources for Local literature/studies and foreign literature/studies	for checking
Jan 14	- Purchased essential components: Onhand: - Arduino R4, pH Sensor, TDS Sensor, Camera Module To be delivered: - RGB Sensor (Jan 16)	for checking of resources

Grade Legend (To be used by the Mentor in grading student's activity)				
95 – 100: Excellent		89 – 94: Very Good	83 – 88: Good	77 – 82: Fair
		70 – 76: Poor	69 below: Inadequate	
Member's Name:	Assigned Task (50)	Participation (30)	Attendance (20)	TOTAL
1. Sabdas Jr, Julian Jojo				95.2
2. Onibeng, Xavier Michael Emmanuel				95.2
3. Ingal, Ian				95.2
4. Sangkula, Chedrick				95.2


Ms. Elizabeth T. Sta Ana
Project Adviser
Signature Over Printed Name

Date _____



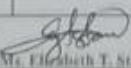
PROGRESS REPORT Computer Science Department 2nd Term SY 2025-2026

- Progress reports are required for students taking the course **CS0027:PROJECT MANAGEMENT**. This will be submitted weekly to monitor and reassess the academic progress of each group, that the project is going smoothly, and that it will be completed on the expected date.
- Progress reports are the responsibility of the group and the Project Adviser. The Project Adviser ensures that each group submitted the form by the required date. This will be filed and used for evaluation.

Group Name: Silver Swan	Program: Bachelor of Science in Computer Science with specialization in Software Engineering		
Member's Name: 1. Sabdan Jr, Julius Jojo 2. Ingal, Ian 3. Onibing, Xavier Michael Emmanuel 4. Sangkula, Chedrick	Term: 1 st ☒ 2 nd 3 rd	Academic Year: 2025-2026	
Mentoring Day/Time/Venue:	Reporting Date: Jan 19 - 24	Reporting Week: Week 4	
Title of the Project: Arduino-Based Smart Urinalysis System with Mobile Application Integration for Hydration Monitoring and Preliminary Disease Detection Using pH, Specific Gravity, Image Processing, and Fuzzy Logic-Enhanced k-NN Algorithm			

Date	Activity	Remarks
Jan 17	Developed the initial Gantt chart draft covering activities from the 2nd to the 3rd term.	} for checking
Jan 18	Completed the draft of Chapter 2.	
Jan 21	Began development of the project prototype.	

Grade Legend (To be used by the Mentor in grading student's activity)				
95 - 100: Excellent	89 - 94: Very Good	83 - 88: Good	77 - 82: Fair	70 - 76: Poor
69 below: Inadequate				
Member's Name:	Assigned Task (50)	Participation (30)	Attendance (20)	TOTAL
1. Sabdan Jr, Julius Jojo				78.75
2. Onibing, Xavier Michael Emmanuel				78.75
3. Ingal, Ian				78.75
4. Sangkula, Chedrick				78.75


Ms. Elizabeth T. Sta Ana
Project Adviser
Signature Over Printed Name

1/22/2026
Date



FEU ALABANG

PROGRESS REPORT Computer Science Department 2nd Term SY 2025-2026

1. Progress reports are required for students taking the course **CS8927-PROJECT MANAGEMENT**. This will be submitted weekly to monitor and reassess the academic progress of each group, that the project is going smoothly, and that it will be completed on the expected date.
2. Progress reports are the responsibility of the group and the Project Adviser. The Project Adviser ensures that each group submitted the form by the required date. This will be filed and used for evaluation.

Group Name: Silver Swan	Program: Bachelor of Science in Computer Science with specialization in Software Engineering	
Member's Name: 1. Sabulao Jr, Julius Jojo 2. Ingal, Ian 3. Ombrog, Xavier Michael Emmanuel 4. Sangkula, Chedrick	Term: 1 st ☒ 2 nd 3 rd	Academic Year: 2025-2026
Mentoring Day/Time/Venue:	Reporting Date: Jan 26 - 31	Reporting Week: Week 5
Title of the Project: Arduino-Based Smart Urinalysis System with Mobile Application Integration for Hydration Monitoring and Preliminary Disease Detection Using pH, Specific Gravity, Image Processing, and Fuzzy Logic-Enhanced k-NN Algorithm		

Date	Activity	Remarks
Jan 26	Started writing chapter 3 draft	
Jan 28	(Hardware) Check if the sensors, arduino, and wires, display screen are working (Hardware) Identified lacking materials which are m to f and m to m arduino wire	
Jan 31	- Acquired the lacking components 40 pins of f to m and m to m arduino wires	
Feb 1	- Establish the connection of the display screen and the arduino - Finished writing the chapter 3 draft	

Grade Legend (To be used by the Mentor in grading student's activity)				
95 – 100: Excellent	89 – 94: Very Good	83 – 88: Good	77 – 82: Fair	70 – 76: Poor
69 below: Inadequate				
Member's Name:	Assigned Task (50)	Participation (30)	Attendance (20)	TOTAL
1. Sabulao Jr, Julius Jojo				100%
2. Ombrog, Xavier Michael Emmanuel				100%
3. Ingal, Ian				100%
4. Sangkula, Chedrick				100%


Ms. Elizabeth T. Sta Ana
Project Adviser

Signature Over Printed Name

Date



PROGRESS REPORT

Computer Science Department

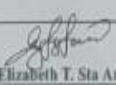
2nd Term SY 2025-2026

1. Progress reports are required for students taking the course **CS0027-PROJECT MANAGEMENT**. This will be submitted weekly to monitor and reassess the academic progress of each group, that the project is going smoothly, and that it will be completed on the expected date.
2. Progress reports are the responsibility of the group and the Project Adviser. The Project Adviser ensures that each group submitted the form by the required date. This will be filed and used for evaluation.

Group Name: Silver Swan	Program: Bachelor of Science in Computer Science with specialization in Software Engineering	
Member's Name: 1. Sabulan Jr, Julius Jojo 2. Ingal, Ian 3. Ombrog, Xavier Michael Emmanuel 4. Sangkula, Chedrick	Term: 1 st ☒ 2 nd 3 rd	Academic Year: 2025-2026
Mentoring Day/Time/Venue:	Reporting Date: Feb 2 - 7	Reporting Week: Week 6
Title of the Project: Arduino-Based Smart Urinalysis System with Mobile Application Integration for Hydration Monitoring and Preliminary Disease Detection Using pH, Specific Gravity, Image Processing, and Fuzzy Logic-Enhanced k-NN Algorithm		

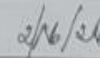
Date	Activity	Remarks
Feb 2	Finalizing the draft manuscript of chapter 1, 2, and 3.	
Feb 4	- Identified the component, Camera module without FIFO is not applicable (Wrong order). - Ordered camera module with FIFO again.	
Feb 7 - 8	Trial and error of connecting the sensors.	

Grade Legend (To be used by the Mentor in grading student's activity)				
95 - 100: Excellent 89 - 94: Very Good 83 - 88: Good 77 - 82: Fair 70 - 76: Poor 69 below: Inadequate				
Member's Name:	Assigned Task (50)	Participation (30)	Attendance (20)	TOTAL
1. Sabulan Jr, Julius Jojo				100 4
2. Ombrog, Xavier Michael Emmanuel				100 2
3. Ingal, Ian				100 2
4. Sangkula, Chedrick				100 2


Ms. Elizabeth T. Sta Ana

Project Adviser

Signature Over Printed Name


Date

Date

PBL 3


FEU ALABANG

PROGRESS REPORT

Computer Science Department/Information Technology Department
3rd Term SY 2025-2026

Progress reports are required for students taking the course CS0072/IT0019-PROJECT MANAGEMENT. This will be submitted weekly to monitor and measure the academic progress of each group, that the project is going smoothly, and that it will be completed on the expected date. Progress reports are the responsibility of the group and the Project Adviser. The Project Adviser ensures that each group submitted the form by the required date. This will be filed and used for evaluation.

Group Name: Silver Swan	Program: Bachelor of Science in Computer Science with specialization in Software Engineering		
Member's Name: 1. Sabulao Jr, Julius Jojo C. 2. Ingat, Ian B. 3. Sangkula, Chadrick Khyllie P. 4. Onibing, Xavier Michael Emmanuel N. 5.	Term: 1 st 2 nd ☒ 3 rd	Academic Year: 2025-2026	
Thursday/13:00-14:00/Venue: Library room	Reporting Date: April 29, 2026 - May 6, 2026		Reporting Week: 1

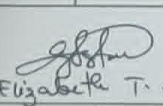
Title of the Project: IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN

Date	Activity	Remarks
29/04/26	-Ordered male-to-male Arduino wire (10 cm, 20 cm) -Ordered male to female Arduino wire (10 cm) -Fixed the wiring of the Arduino. -Planned the layout and materials for the prototype housing	
04/05/26	-Code adjustments for the sensors. -Calibrated the sensors. -Testing of the camera, TDS sensor, PH meter. -Tested the prototype in synthetic urine.	

Grade Legend (To be used by the Mentor in grading student's activity)

95 – 100: Excellent 89 – 94: Very Good 83 – 88: Good 77 – 82: Fair 70 – 76: Poor 69 below: Inadequate

Member's Name:	Assigned Task (50)	Participation (30)	Attendance (20)	TOTAL
1. Sabulao Jr, Julius Jojo C.				} 180
2. Ingat, Ian B.				
3. Sangkula, Chadrick Khyllie P.				
4. Onibing, Emmanuel N.				
5.				


Elizabeth T. Sta. Ana, MScS, MBA
 Project Adviser
Signature Over Printed Name

Date _____

Appendix N- Weekly Progress Report



FEU ALABANG

PROGRESS REPORT

Computer Science Department/Information Technology Department
3rd Term SY 2025-2026

1. Progress reports are required for students taking the course CS6047/IT6049-PROJECT MANAGEMENT. This will be submitted weekly to monitor and measure the academic progress of each group, that the project is going smoothly, and that it will be completed on the expected date.
2. Progress reports are the responsibility of the group and the Project Adviser. The Project Adviser ensures that each group submitted the form by the required date. This will be filed and used for evaluation.

Group Name: Silver Swan		Program: Bachelor of Science in Computer Science with specialization in Software Engineering	
Member's Name: 1. Sabulao Jr, Julius Jojo C. 2. Ingol, Ian R. 3. Sangkula, Chedrick Khyllie P. 4. Ombrog, Xavier Michael Emmanuel N.		Term: 1 st 2 nd ☒ 3 rd	Academic Year: 2025-2026
Thursday/11:00-14:00/Venue: Library room		Reporting Date: May 9, 2026 - May 13, 2026	Reporting Week: 2
Title of the Project: IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN			

Date	Activity	Remarks
09/05/26	-Resolder the keyboard pins ✓ -Fixed the wirings of ESP32 camera. -Rearranged the wirings of the arduino to sensors ✓	- ok - done w/ the box - in a box w/ proper connection
11/05/26	-BLE of android connection fixed ✓ -Capacitors order arrived. -Re configure the sensors.	- done - re-configuring

Grade Legend (To be used by the Mentor in grading student's activity)				
95 – 100: Excellent 89 – 94: Very Good 83 – 88: Good 77 – 82: Fair 70 – 76: Poor 69 below: Inadequate				
Member's Name:	Assigned Task (50)	Participation (30)	Attendance (20)	TOTAL
1. Sabulao Jr, Julius Jojo C.				} 98%
2. Ingol, Ian R.				
3. Sangkula, Chedrick Khyllie P.				
4. Ombrog, Xavier Michael Emmanuel N.				
5.				

Ms. Elizabeth T. Sta Ana, MSCS, MBA

Project Adviser

Signature Over Printed Name

5-14-2026

Date



FEU ALABANG

PROGRESS REPORT

Computer Science Department/Information Technology Department
3rd Term SY 2025-2026

- Progress reports are required for students taking the course CS0077/IT0019-PROJECT MANAGEMENT. This will be submitted weekly to monitor and reassess the academic progress of each group, that the project is going smoothly, and that it will be completed on the expected date.
- Progress reports are the responsibility of the group and the Project Adviser. The Project Adviser ensures that each group submitted the form by the required date. This will be filed and used for evaluation.

Group Name: Silver Swan	Program: Bachelor of Science in Computer Science with specialization in Software Engineering	
Member's Name: 1. Sabulao Jr, Julius Jojo C. 2. Ingal, Ian B. 3. Sangkula, Chedrick Khyllie P. 4. Onibrog, Xavier Michael Emmanuel N.	Term: [] 1 st [] 2 nd [] ☒ 3 rd	Academic Year: 2025-2026
Thursday/13:00-14:00/Venue: Library room	Reporting Date: May 16, 2026 - May 20, 2026	Reporting Week: 3
Title of the Project: IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN		

Date	Activity	Remarks
16/05/26	-Removed the lights on the sensor. -Create a draft box of the Arduino casing.	
18/05/26	-Assembled the box case of Arduino. -Assembled the wiring inside the box.	
18/05/26	-Testing of urinalysis with box casing. -Planning for better improvements on the casing.	

Grade Legend (To be used by the Mentor in grading student's activity)				
95 – 100: Excellent 89 – 94: Very Good 83 – 88: Good 77 – 82: Fair 70 – 76: Poor 69 below: Inadequate				
Member's Name:	Assigned Task (50)	Participation (30)	Attendance (20)	TOTAL
1. Sabulao Jr, Julius Jojo C.				98.2
2. Ingal, Ian B.				
3. Sangkula, Chedrick Khyllie P.				
4. Onibrog, Xavier Michael Emmanuel N.				
5.				

Ms. Elizabeth T. Sta Ana, MSCS, MBA

Project Adviser

Signature Over Printed Name

Date

Appendix N- Weekly Progress Report



FEU ALABANG

PROGRESS REPORT

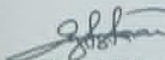
Computer Science Department/Information Technology Department
3rd Term SY 2025-2026

1. Progress reports are required for students taking the course **CS0027/IT0039-PROJECT MANAGEMENT**. This will be submitted weekly to monitor and reassess the academic progress of each group, that the project is going smoothly, and that it will be completed on the expected date.
2. Progress reports are the responsibility of the group and the Project Adviser. The Project Adviser ensures that each group submitted the form by the required date. This will be filed and used for evaluation.

Group Name: Silver Swan	Program: Bachelor of Science in Computer Science with specialization in Software Engineering	
Member's Name: 1. Sabulao Jr, Julius Jojo C. 2. Ingal, Ian D. 3. Sangkula, Chedrick Khyllie P. 4. Ombrog, Xavier Michael Emmanuel N.	Term: 1 st 2 nd ☒ 3 rd	Academic Year: 2025-2026
Thursday: 13:00-14:00/Venue: Library room	Reporting Date: May 23, 2026 - May 27, 2026	Reporting Week: 4
Title of the Project: IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN		

Date	Activity	Remarks
23/05/26	-Tds sensor re calibrated. -Android app and ios application fixed.	
25/05/26	-Wiring fixtures of the arduino. -Fixed the camera sensor problem, to acquire more accurate results.	
27/05/26	-Updated the wirings of the arduino (new wiring diagram).. -Installed capacitors for better result acquisition.	

Grade Legend (To be used by the Mentor in grading student's activity)				
95 – 100: Excellent		89 – 94: Very Good	83 – 88: Good	77 – 82: Fair
		70 – 76: Poor	69 below: Inadequate	
Member's Name:	Assigned Task (50)	Participation (30)	Attendance (20)	TOTAL
1. Sabulao Jr, Julius Jojo C.				100%
2. Ingal, Ian D.				
3. Sangkula, Chedrick Khyllie P.				
4. Ombrog, Xavier Michael Emmanuel N.				
5.				


Ms. Elizabeth T. Sta Ana, MScS, MBA
Project Adviser
Signature Over Printed Name

Date

Appendix N- Weekly Progress Report



FEU ALABANG

PROGRESS REPORT

Computer Science Department/Information Technology Department
3rd Term SY 2025-2026

- Progress reports are required for students taking the course **CS0027/IT0039-PROJECT MANAGEMENT**. This will be submitted weekly to monitor and reinforce the academic progress of each group; that the project is going smoothly; and that it will be completed on the expected date.
- Progress reports are the responsibility of the group and the Project Adviser. The Project Adviser ensures that each group submitted the form by the required date. This will be filed and used for evaluation.

Group Name: Silver Swan	Program: Bachelor of Science in Computer Science with specialization in Software Engineering	
Member's Name: 1. Sabutan Jr, Julius Jojo C. 2. Ingal, Ian D. 3. Sangkula, Chadrick Khylin P. 4. Ombong, Xavier Michael Emmanuel N.	Term: 1 st 2 nd ☒ 3 rd	Academic Year: 2025-2026
Thursday/13:00-14:00/Venue: Library room	Reporting Date: May 29, 2026 - June 3, 2026	Reporting Week: 5
Title of the Project: IOT-BASED SMART URINALYSIS SYSTEM FOR HYDRATION MONITORING, IMAGE PROCESSING AND PRELIMINARY SCREENING USING FUZZY LOGIC-ENHANCED KNN		

Date	Activity	Remarks
29/05/26	-Testing for specific gravity using refractometer. -Tested the sensors. -Sensors re-calibrated.	
30/05/26	-Fixed the wirings of the arduino. -Fixed the housing of the arduino urinalysis. -Sensors adjustments.	
01/06/26	-Bought materials for testing (Food coloring, baking soda, salt, lemon, Vinegar). -Ph testing using the Ph meter. -Initial test for different colors of the urine. -Identified the rgb sensor problem.	
03/06/26	-Fixing the rgb sensor. -Ordered D/Robot Gravity: AS3741 (1-Channel visible light sensor.	

Grade Legend (To be used by the Mentor in grading student's activity)				
95 - 100: Excellent	89 - 94: Very Good	83 - 88: Good	77 - 82: Fair	70 - 76: Poor
				69 below: Inadequate
Member's Name:	Assigned Task (50)	Participation (30)	Attendance (20)	TOTAL
1. Sabutan Jr, Julius Jojo C.				100%
2. Ingal, Ian D.				
3. Sangkula, Chadrick Khylin P.				
4. Ombong, Xavier Michael Emmanuel N.				

Ms. Elizabeth T. Sta Ana, MSCS, MBA
Project Adviser
Signature Over Printed Name

Date

Appendix N- Weekly Progress Report

APPENDIX H

Adviser Forms

 **FEU ALABANG**

Date: December 18, 2025

Faculty, College of Computer Studies
FEU Alabang
South Corporate Avenues, Filinvest City, Alabang, Muntinlupa

Dear Ms. Elizabeth T. Sta Ana,

Good day!

The undersigned students are currently enrolled in **CS0027 - CS Project Management**, pursuing a Bachelor of Science in Computer Science with specializations in Software Engineering.

We humbly request your knowledge, expertise, and help as project adviser for our thesis/capstone project entitled **Arduino-Based Smart Urinalysis System with Mobile Application Integration for Hydration Monitoring and Preliminary Disease Detection Using pH, Specific Gravity, Image Processing, and Fuzzy Logic-Enhanced k-NN Algorithm**. We believe that your qualifications as a project adviser will be beneficial and will significantly improve our research.

Thank you very much and we are hoping for your kind approval with this matter.

Respectfully yours,

Silver Swan

Sabulao Jr, Julius Jojo


Ingal, Ian Batu


Ombrog, Xavier Michael Emmanuel Novio


Sankula, Hedrick Khylle Piang



FEU ALABANG

APPROVAL SHEET

(For Project Adviser only)

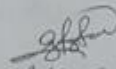
Please Check



APPROVED



DISAPPROVED


Elizabeth T. Sta Ana

Faculty, College of Computer Studies
Signature over Printed Name

16-18-2025

Date

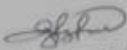


FEU ALABANG

PROJECT ADVISER ROLES AND RESPONSIBILITIES

1. The project adviser shall monitor the project, requiring the students to report the progress of their project weekly. The project adviser shall meet the group at least thirty (30) minutes in a week.
2. The project adviser shall ensure that the project paper is written in the prescribed format.
3. The project adviser shall give suggestions and comments on how the project could be strengthened. In doing so, the project adviser shall guide the development of the project, and check the content of the paper towards the betterment of its output, and not just a defense-oriented one.
4. The project adviser shall rate each student according to the student's participation/performance in the project as determined from the regular consultation meeting. This shall constitute ten percent (10%) of the student's final course grade.
5. The project adviser shall examine the project output and certify that the project is ready and complete for ORAL defense by conducting a mock defense.
6. The project adviser shall ensure that four (4) copies of the complete project documents are submitted on time and that each one is readable and presentable. (Three copies for the head panel and members)
7. During the ORAL DEFENSE:
 - a. The project adviser may be present during the project defense to give moral support to the mentees.
 - b. The project adviser may provide final statements in the absence of the students after the oral presentation and before the deliberation of the final verdict by the panel committee.
 - c. The project adviser shall leave the room during the deliberation of the final verdict and may be present upon the announcement of the final verdict of the defense.

I am stating that I have taken time to read and understand the guidelines and roles of a Project Adviser. By signing this document, I fully acknowledge and agree to the contents of this document and I indicate that I have the capacity of doing my job as a Project Adviser of Silver Swan group.


Elizabeth T. Sta. Ana
Faculty, College of Computer Studies
Signature Over Printed Name

12-18-2021
Date



FEU ALABANG

MENTORING COMMITMENT FORM
Computer Science and Information Technology Departments
2nd Term and School Year 2025-2026

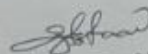
Date: December 18, 2025

I, Elizabeth T. Sta Ana, one of the faculty members of the College of Computer Studies under the CS Department, commit to the group/team as their Project Adviser, at least thirty (30) minutes of my time outside my residency hours starting on _____

Below are the details of my mentoring:

Course Code and Name:	CS0027 - CS Project Management
Section:	AN33
Course Adviser:	Armando D. Del Mundo
Term and School Year:	3 rd year 2025-2026
Group/Team Name:	Silvery Swan
Project Title:	Admission and Vignette system with mobile application integration
Group Members	Sabino Julius, Ensal Jan, Omphrey Xavier, Sangita Chiriele
Client (Company Name):	
Mentoring Day:	
Mentoring Time:	
Mentoring Venue:	

Note for FEU Tech Associates: By signing in this form, the mentor acknowledges that he will voluntarily perform the mentoring duty outside of his regular working hours. Furthermore, it is understood that any amount to be paid by the College in the performance of this duty shall not be considered a demandable part of the mentor's salary or an established College practice or precedent but merely an act of gratuity on the part of the College which may be discontinued or revised at any time at the College's sole discretion.


Elizabeth T. Sta Ana

Faculty, College of Computer Studies
Signature Over Printed Name

12-18-2025

Date



PROJECT TEAM ASSIGNMENT FORM

Bachelor of Science in Computer Science with Specialization in Software Engineering
2nd Term, SY 2025-2026

This form is for organizing a project team, and a Roles and Responsibilities table for documenting all the members of a cross-functional project team, including their roles and responsibilities.

PROJECT INFORMATION

GROUP NAME: Silver Swan

PROJECT TITLE: Arduino-Based Smart Urinalysis System with Mobile Application Integration for Hydration Monitoring and Preliminary Disease Detection Using pH, Specific Gravity, Image Processing, and Fuzzy Logic-Enhanced k-NN Algorithm

DATE : December 18, 2025

Roles and Responsibilities

Researcher's Name	Role/s	Responsibility
1. Sabulao Jr, Julius Jojo	Project Manager / Mobile Application Developer / System Tester	• Oversees overall project planning, scheduling, and coordination • Ensures milestones and deliverables are completed on time • Assists in mobile application development and feature integration • Tests mobile application functions and system workflows • Verifies that outputs are clearly labeled as non-diagnostic • Coordinates system integration testing between hardware and software • Assists in evaluation, validation, and final system presentation
2. Ingol, Ian Batu	Documentation Head /System Tester / Primary Communicator	• Serves as the main communicator between the research group and: ◦ Thesis adviser ◦ Panel members ◦ External professionals (e.g., pediatrician interview) • Handles official emails, messages, and coordination related to the thesis • Leads thesis documentation and ensures compliance with APA 7th edition • Prepares and consolidates the Background of the Study, RRL, and Methodology • Analyzes related studies and research findings to support system design • Documents expert interviews, ethical considerations, and regulatory compliance • Conducts system testing from a research and usability perspective



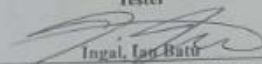
FEU ALABANG

3. Ombrog, Xavier Michael Emmanuel Novio	Lead Developer / Hardware Engineer / System Tester	• Records test results, observations, and limitations for analysis and reporting • Designs and develops the Arduino-based hardware module • Integrates pH, specific gravity, color sensors, and camera • Performs sensor calibration and hardware-level validation • Conducts hardware functionality and accuracy testing • Troubleshoots sensor and microcontroller issues • Ensures stable data transmission to the mobile application • Assists in system optimization and reliability testing
4. Sankula, Chedrick Khylle Piang	Mobile Application Developer / System Tester / Quality Assurance Lead	• Leads mobile application development and user interface design • Implements image processing and classification algorithms • Develops test cases and testing protocols for the system • Leads system-wide testing, validation, and error analysis • Ensures consistency, reliability, and usability of system outputs • Verifies compliance with safety, ethical, and legal requirements • Coordinates testing documentation and quality reporting

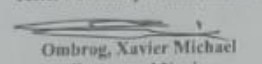
Approved by:


Sabunio Jr, Julius Jojo

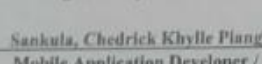
Project Manager / Mobile
Application Developer / System
Tester


Ingal, Ian Bath

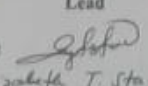
Documentation Head / System
Tester / Primary Communicator


Ombrog, Xavier Michael
Emmanuel Novio

Lead Developer / Hardware
Engineer / System Tester


Sankula, Chedrick Khylle Piang
Mobile Application Developer /
System Tester / Quality Assurance
Lead

Noted by:


Elizabeth T. Sta. Ana

Project Adviser

Signature Over Printed Name

12-18-2025

Date

APPENDIX I

Certificate of Medical Validation

CERTIFICATE OF MEDICAL VALIDATION

I, Dr. Joyce Domina, hereby certify my voluntary participation in an interview with the research group of Ingal, Ian B., Ombrog, Xavier Michael Emmanuel N., Sabulao Jr., Julius Jojo C., and Sangkula, Chedrick Khyllle P., undergraduate researchers at FEU Alabang.

The group is developing the "IoT-Based Smart Urinalysis System for Hydration Monitoring, Image Processing and Preliminary Screening Using Fuzzy Logic-Enhanced KNN." During the interview, they presented a clinical reference pre-dataset containing eight (8) hydration levels and three (3) abnormality flags based on Urine Specific Gravity, pH, and HSV Color values. Upon my clinical review, I confirm that the classification boundaries and associated physiological conditions therein are medically consistent with accepted urinalysis standards.

I acknowledge this pre-dataset is aggregated from peer-reviewed clinical literature not from primary human subject data and serves solely as an algorithmic reference guide. I hereby grant approval for its use in the thesis manuscript and prototype, provided that appropriate citations are maintained and the system is not represented as a substitute for clinical laboratory diagnosis.

This certification is issued for academic and documentary purposes only.

Sincerely,

Dr. Joyce Domina
Lic. #095313
Signature 
Date Signed: 6/17/26

APPENDIX J

Certificate of Technical Validation

CERTIFICATE OF TECHNICAL VALIDATION

I, MARLOU INCAL, hereby certify my voluntary participation in a demonstration and technical review with the research group of Ingal, Ian B., Ombeog, Xavier Michael Emmanuel N., Sabulao Jr., Julius Jojo C., and Sangkula, Chedrick Khyllie P., undergraduate researchers at FEU Alabang.

The group is developing the "IoT-Based Smart Urinalysis System for Hydration Monitoring, Image Processing and Preliminary Screening Using Fuzzy Logic-Enhanced KNN." The researchers presented the system's companion application and demonstrated its connection to the Arduino-based device. They showed the app establishing communication with the hardware, receiving the Urine Specific Gravity, pH, and Color values transmitted from the device, and displaying the classification results produced by the Fuzzy Logic-Enhanced K-Nearest Neighbors (EF-KNN) algorithm.

Upon my technical review, I confirm that the application receives its measurements from the connected Arduino device in real time and that the values shown in the app correspond to the actual readings transmitted by the hardware. To the extent observable during this review, the data displayed is not hardcoded, pre-loaded, or manually entered, and the application behaves consistently with the architecture and data flow described by the researchers.

I acknowledge that this validation pertains to the technical operation and integrity of the application's connection to the device and its handling of the transmitted data, and not to the system's clinical or diagnostic accuracy. I hereby grant approval for the use of this assessment in the thesis manuscript and prototype, provided that the system is not represented as a certified medical or laboratory-grade diagnostic device.

This certification is issued for academic and documentary purposes only.

Sincerely,

Name MARION J. WAL
Company BDO ULI BANK / SENIOR MANAGER / ARCHITECT
Signature [Signature]
Date Signed: JUNE 28, 2024

APPENDIX K

Urinalysis Result of Samples

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SSMC Health, Inc.

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3F, Sycamore Centre, Buencamino St., Cupang, Muntinlupa City
Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmciphil.com

NAME: F1 DATE: June 24, 2026
AGE: SEX: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE:

URINALYSIS

Color: Straw Transparency: Reaction: Acidic - 6.0 Specific Gravity: 1.000 Sugar: Albumin: Pregnancy test: Method:	Epithelial cells: Pus cells: /hpf RBC: /hpf Amorphous Substances Mucus threads: Bacteria: Cast: Yeast cells: others:
--	--

Analyzed by: Akhila Elisse B. Capuno, RMT Verified by: Raphael G. Dalope, RMT Noted by: Iftizar N. Haron, MD, DPSP
 Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
 Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

SSMC Health - Rosario, 2nd Floor, Andico Bldg. Gen. Trias, Drive, Rosario Cavite *SSMC Health - Sta. Rosa, Old National Highway (in front of Mercury Drug) Bulbago, Sta. Rosa, Laguna *SSMC Health - Calamba, 2nd J. Alcaid Business Center Crossing, Calamba, Laguna *SSMC Health - Muntinlupa, 3F, Sycamore Centre (in front of Honda Cars - Alabang) Buencamino St. Cupang, Muntinlupa City

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Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmciphil.com

NAME: F 1.2 DATE: June 24, 2026
AGE: SEX: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE:

URINALYSIS

Color: Straw Transparency: Reaction: Acidic - 6.0 Specific Gravity: 1.000 Sugar: Albumin: Pregnancy test: Method:	Epithelial cells: Pus cells: /hpf RBC: /hpf Amorphous Substances Mucus threads: Bacteria: Cast: Yeast cells: others:
--	--

Analyzed by: Akhila Elisse B. Capuno, RMT Verified by: Raphael G. Dalope, RMT Noted by: Iftizar N. Haron, MD, DPSP
 Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
 Medical Technologist Chief Medical Technologist Pathologist

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 3F, Sycamore Centre, Buencamino St., Cupang, Muntinlupa City
 Tel. Nos. : 8875437
 Email: info@ssmchealth.com Website: www.ssmciphil.com

NAME: F2 SEX: DATE: June 24, 2026
 AGE: COMPANY: FAR EASTERN UNIVERSITY ALABANG
 BIRTHDATE :

URINALYSIS

Color: Light Yellow	Epithelial cells: /hpf
Transparency: /hpf	Pus cells: /hpf
Reaction: Acidic - 6.5	RBC:
Specific Gravity: 1.010	Amorphous Substances
Sugar:	Mucus threads:
Albumin:	Bacteria:
Pregnancy test:	Cast:
Method:	Yeast cells:
	others:

Analyzed by: *[Signature]* Verified by: *[Signature]* Noted by: *[Signature]*
 AKHILA ELISSE B. CAPUNO, RMT RAPHAEL G. DALOPE, RMT IFTIZAR N. HIRON, MD, DPSP
 Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
 Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

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 Tel. Nos. : 8875437
 Email: info@ssmchealth.com Website: www.ssmciphil.com

NAME: F3 SEX: DATE: June 24, 2026
 AGE: COMPANY: FAR EASTERN UNIVERSITY ALABANG
 BIRTHDATE :

URINALYSIS

Color: Yellow	Epithelial cells: /hpf
Transparency: /hpf	Pus cells: /hpf
Reaction: Acidic - 6.5	RBC:
Specific Gravity: 1.010	Amorphous Substances
Sugar:	Mucus threads:
Albumin:	Bacteria:
Pregnancy test:	Cast:
Method:	Yeast cells:
	others:

Analyzed by: *[Signature]* Verified by: *[Signature]* Noted by: *[Signature]*
 AKHILA ELISSE B. CAPUNO, RMT RAPHAEL G. DALOPE, RMT IFTIZAR N. HIRON, MD, DPSP
 Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
 Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

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Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmcphil.com

NAME: F4 DATE: June 24, 2026
AGE: SEX: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE:

URINALYSIS

Color: Amber	Epithelial cells: /hpf
Transparency: /hpf	Pus cells: /hpf
Reaction: Acidic - 6.5	RBC:
Specific Gravity: 1.015	Amorphous Substances
Sugar:	Mucus threads:
Albumin:	Bacteria:
Pregnancy test:	Cast:
Method:	Yeast cells:
	others:

Analyzed by: *Akhila* Verified by: *Raphael G. Dalope* Noted by: *Iftizar H. Haron*
AKHILA ELISSE B. CAPUNO, RMT **RAPHAEL G. DALOPE, RMT** **IFTIZAR H. HARON, MD, DPSP**
 Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
 Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

SSMC Health - Rosario, 2nd Floor, Andico Bldg. Gen. Trias, Drive, Rosario Cavite *SSMC Health - Sta. Rosa, Old National Highway (in front of Mercury Drug) Balibago, Sta. Rosa, Laguna *SSMC Health - Calamba, 2nd J. Alcañal Business Ctr. Crossing, Calamba, Laguna *SSMC Health - Muntinlupa, 3F, Sycamore Centre (in front of Honda Cars - Alabang) Buencamino St. Cupang, Muntinlupa City

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Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmcphil.com

NAME: F5 DATE: June 24, 2026
AGE: SEX: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE:

URINALYSIS

Color: Amber	Epithelial cells: /hpf
Transparency: /hpf	Pus cells: /hpf
Reaction: Acidic - 6.0	RBC:
Specific Gravity: 1.020	Amorphous Substances
Sugar:	Mucus threads:
Albumin:	Bacteria:
Pregnancy test:	Cast:
Method:	Yeast cells:
	others:

Analyzed by: *Akhila* Verified by: *Raphael G. Dalope* Noted by: *Iftizar H. Haron*
AKHILA ELISSE B. CAPUNO, RMT **RAPHAEL G. DALOPE, RMT** **IFTIZAR H. HARON, MD, DPSP**
 Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
 Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

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Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmcphil.com

NAME: F 5.2 DATE: June 24, 2026
AGE: SEX: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE :

URINALYSIS

Color: Amber	Epithelial cells: /hpf
Transparency: Acidic - 6.5	Pus cells: /hpf
Reaction: 1.020	RBC:
Specific Gravity:	Amorphous Substances
Sugar:	Mucus threads:
Albumin:	Bacteria:
Pregnancy test:	Cast:
Method:	Yeast cells:
	others:

Analyzed by: *Akhila* Verified by: *Raphael G. Dalope* Noted by: *IFTIZAR N. HARON*
AKHILA ELISSE B. CAPUNO, RMT RAPHAEL G. DALOPE, RMT IFTIZAR N. HARON, MD, DPSP
Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

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Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmcphil.com

NAME: F6 DATE: June 24, 2026
AGE: SEX: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE :

URINALYSIS

Color: Dark Yellow	Epithelial cells: /hpf
Transparency: Acidic - 5.0	Pus cells: /hpf
Reaction: 1.025	RBC:
Specific Gravity:	Amorphous Substances
Sugar:	Mucus threads:
Albumin:	Bacteria:
Pregnancy test:	Cast:
Method:	Yeast cells:
	others:

Analyzed by: *Akhila* Verified by: *Raphael G. Dalope* Noted by: *IFTIZAR N. HARON*
AKHILA ELISSE B. CAPUNO, RMT RAPHAEL G. DALOPE, RMT IFTIZAR N. HARON, MD, DPSP
Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

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Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmciphil.com

NAME: F7 DATE: June 24, 2026
AGE: SEX: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE :

URINALYSIS

Color: Dark Yellow	Epithelial cells: /hpf
Transparency: /hpf	Pus cells: /hpf
Reaction: Acidic - 5.0	RBC:
Specific Gravity: 1.030	Amorphous Substances
Sugar:	Mucus threads:
Albumin:	Bacteria:
Pregnancy test:	Cast:
Method:	Yeast cells:
	others:

Analyzed by:  **AKHILA ELISSE B. CAPUNO, RMT**
Lic. No. 0129911
Medical Technologist

Verified by:  **RAPHAEL G. DALOPE, RMT**
Lic. No. 0117594
Chief Medical Technologist

Noted by:  **IFTIZAR N. HARON, MD, DPSP**
Lic. No. 0119158
Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

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Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmciphil.com

NAME: F8 DATE: June 24, 2026
AGE: SEX: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE :

URINALYSIS

Color: Dark Yellow	Epithelial cells: /hpf
Transparency: /hpf	Pus cells: /hpf
Reaction: Alkaline - 7.5	RBC:
Specific Gravity: 1.020	Amorphous Substances
Sugar:	Mucus threads:
Albumin:	Bacteria:
Pregnancy test:	Cast:
Method:	Yeast cells:
	others:

Analyzed by:  **AKHILA ELISSE B. CAPUNO, RMT**
Lic. No. 0129911
Medical Technologist

Verified by:  **RAPHAEL G. DALOPE, RMT**
Lic. No. 0117594
Chief Medical Technologist

Noted by:  **IFTIZAR N. HARON, MD, DPSP**
Lic. No. 0119158
Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

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Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmciphil.com

NAME: F 8.2 SEX: DATE: June 24, 2026
AGE: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE:

URINALYSIS

Color:	Dark Yellow	Epithelial cells:	
Transparency:		Pus cells:	/hpf
Reaction:	Acidic - 6.0	RBC:	/hpf
Specific Gravity:	1.025	Amorphous Substances	
Sugar:		Mucus threads:	
Albumin:		Bacteria:	
Pregnancy test:		Cast:	
Method:		Yeast cells:	
		others:	

Analyzed by: *Akhila* Verified by: *Raphael G. Dalope* Noted by: *Iftizar H. Haron*
 AKHILA ELISSE B. CAPUNO, RMT RAPHAEL G. DALOPE, RMT IFTIZAR H. HARON, MD, DPSP
 Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
 Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

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3F, Sycamore Centre, Buencamino St., Cupang, Muntinlupa City
Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmciphil.com

NAME: FA (RED) SEX: DATE: June 24, 2026
AGE: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE:

URINALYSIS

Color:	Red	Epithelial cells:	
Transparency:		Pus cells:	/hpf
Reaction:	Acidic - 6.0	RBC:	/hpf
Specific Gravity:	1.000	Amorphous Substances	
Sugar:		Mucus threads:	
Albumin:		Bacteria:	
Pregnancy test:		Cast:	
Method:		Yeast cells:	
		others:	

Analyzed by: *Akhila* Verified by: *Raphael G. Dalope* Noted by: *Iftizar H. Haron*
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 Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
 Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

SSMC Health - Rosario, 2nd Floor, Andico Bldg. Gen. Trias, Drive, Rosario Cavite *SSMC Health - Sta. Rosa, Old National Highway (in front of Mercury Drug) Balibago, Sta. Rosa, Laguna *SSMC Health - Calamba, 2nd J. Alcaid Business Ctr. Crossing, Calamba, Laguna *SSMC Health - Muntinlupa, 3F, Sycamore Centre (in front of Honda Cars - Alabang) Buencamino St. Cupang, Muntinlupa City

SSMC-F-LAB-D006-R.0

SSMC Health, Inc.
Muntinlupa Branch
3F, Sycamore Centre, Buencamino St., Cupang, Muntinlupa City
Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmcphil.com

NAME: FA (GREEN) DATE: June 24, 2026
AGE: SEX: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE:

URINALYSIS

Color: Green	Epithelial cells: /hpf
Transparency: /hpf	Pus cells: /hpf
Reaction: Acidic - 6.0	RBC: /hpf
Specific Gravity: 1.000	Amorphous Substances
Sugar: /hpf	Mucus threads:
Albumin: /hpf	Bacteria:
Pregnancy test: /hpf	Cast: /hpf
Method: /hpf	Yeast cells: /hpf
	others: /hpf

Analyzed by: *[Signature]* Verified by: *[Signature]* Noted by: *[Signature]*
AKHILA ELISSE B. CAPUNO, RMT RAPHAEL G. DALOPE, RMT IFTIZAR H. HARON, MD, DPSP
Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

SSMCHealth-Rosario, 2nd Floor, Andico Bldg, Gen. Trias, Drive, Rosario Cavite *SSMCHealth-Sta. Rosa, Old National Highway (in front of Mercury Drug) Bulibago, Sta. Rosa, Laguna *SSMCHealth-Calamba, 2nd J. Alcadid Business Ctr. Crossing Calamba, Laguna *SSMCHealth-Muntinlupa, 3F, Sycamore Centre (in front of Honda Cars-Alabang) Buencamino St., Cupang, Muntinlupa City

SSMC-F-LAB-D006-R.0

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3F, Sycamore Centre, Buencamino St., Cupang, Muntinlupa City
Tel. Nos. : 8875437
Email: info@ssmchealth.com Website: www.ssmcphil.com

NAME: FB DATE: June 24, 2026
AGE: SEX: COMPANY: FAR EASTERN UNIVERSITY ALABANG
BIRTHDATE:

URINALYSIS

Color: Light Yellow	Epithelial cells: /hpf
Transparency: /hpf	Pus cells: /hpf
Reaction: Alkaline - 9.0	RBC: /hpf
Specific Gravity: 1.000	Amorphous Substances
Sugar: /hpf	Mucus threads:
Albumin: /hpf	Bacteria:
Pregnancy test: /hpf	Cast: /hpf
Method: /hpf	Yeast cells: /hpf
	others: /hpf

Analyzed by: *[Signature]* Verified by: *[Signature]* Noted by: *[Signature]*
AKHILA ELISSE B. CAPUNO, RMT RAPHAEL G. DALOPE, RMT IFTIZAR H. HARON, MD, DPSP
Lic. No. 0129911 Lic. No. 0117594 Lic. No. 0119158
Medical Technologist Chief Medical Technologist Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK

SSMCHealth-Rosario, 2nd Floor, Andico Bldg, Gen. Trias, Drive, Rosario Cavite *SSMCHealth-Sta. Rosa, Old National Highway (in front of Mercury Drug) Bulibago, Sta. Rosa, Laguna *SSMCHealth-Calamba, 2nd J. Alcadid Business Ctr. Crossing Calamba, Laguna *SSMCHealth-Muntinlupa, 3F, Sycamore Centre (in front of Honda Cars-Alabang) Buencamino St., Cupang, Muntinlupa City

SSMC-F-LAB-0006-R.0



SSMC Health, Inc.
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 3F, Sycamore Centre, Buencamino st., Cupang, Muntinlupa City
 Tel. Nos. : 8875437
 Email: info@ssmchealth.com Website: www.ssmciphil.com

NAME: FC
 AGE:
 BIRTHDATE:

SEX:

DATE: **June 24, 2026**
 COMPANY: **FAR EASTERN UNIVERSITY ALABANG**

URINALYSIS

Color: Yellow Transparency: Reaction: Acidic - 5.0 Specific Gravity: 1.030 Sugar: Albumin: Pregnancy test: Method:	Epithelial cells: Pus cells: /hpf RBC: /hpf Amorphous Substances Mucus threads: Bacteria: Cast: Yeast cells: others:
--	--

Analyzed by: 
AKHILA ELISSE B. CAPUNO, RMT
Lic. No. 0129911
 Medical Technologist

Verified by: 
RAPHAEL G. DALOPE, RMT
Lic. No. 0117394
 Chief Medical Technologist

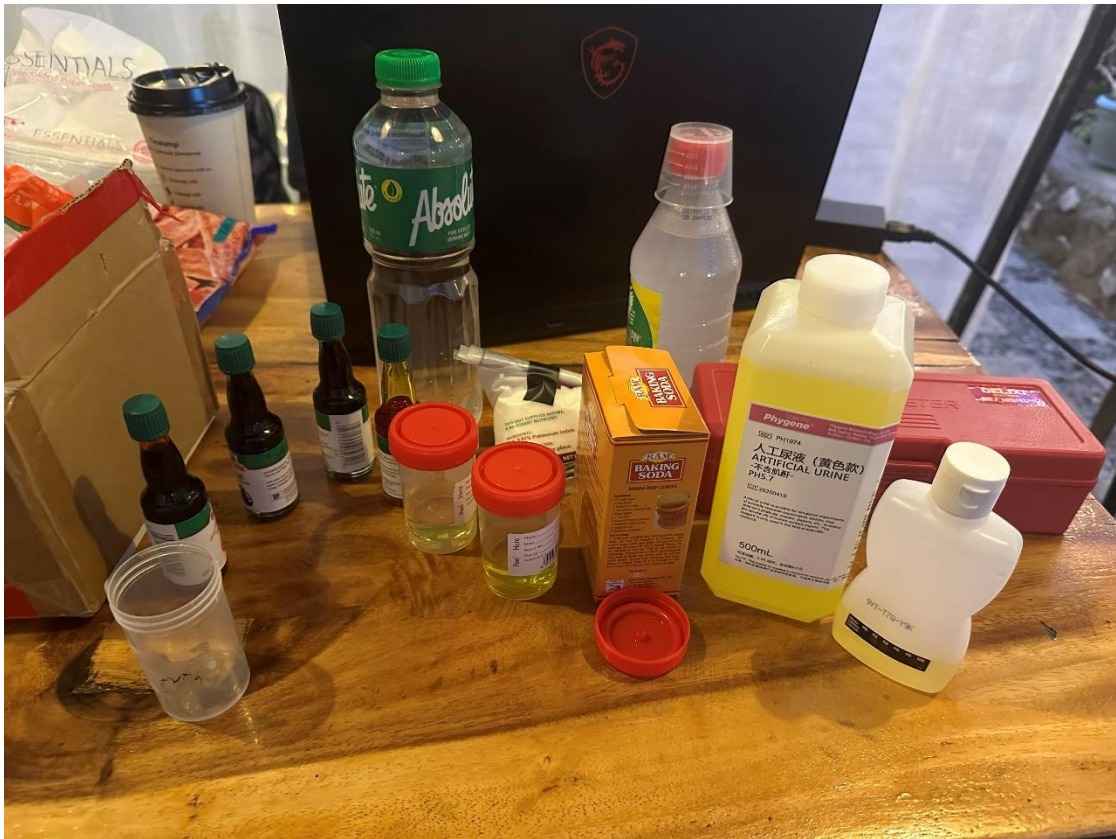
Noted by: 
IFTIZAR H. HARON, MD, DPSP
Lic. No. 0119158
 Pathologist

LOCATIONS OF OTHER SSMC HEALTH SYSTEM NETWORK
 SSMC Health - Rosario, 2nd Floor, Andico Bldg. Gen. Trias Drive, Rosario Cavite *SSMC Health - Sta. Rosa, Old National Highway (in front of Mercury Drug) Balibago, Sta. Rosa, Laguna *SSMC Health - Calamba 2nd J. Alcaid Business Ctr. Crossing Calamba, Laguna *SSMC Health - Muntinlupa, 3F, Sycamore Centre (in front of Honda Cars-Alabang) Buencamino St. Cupang, Muntinlupa City

APPENDIX L

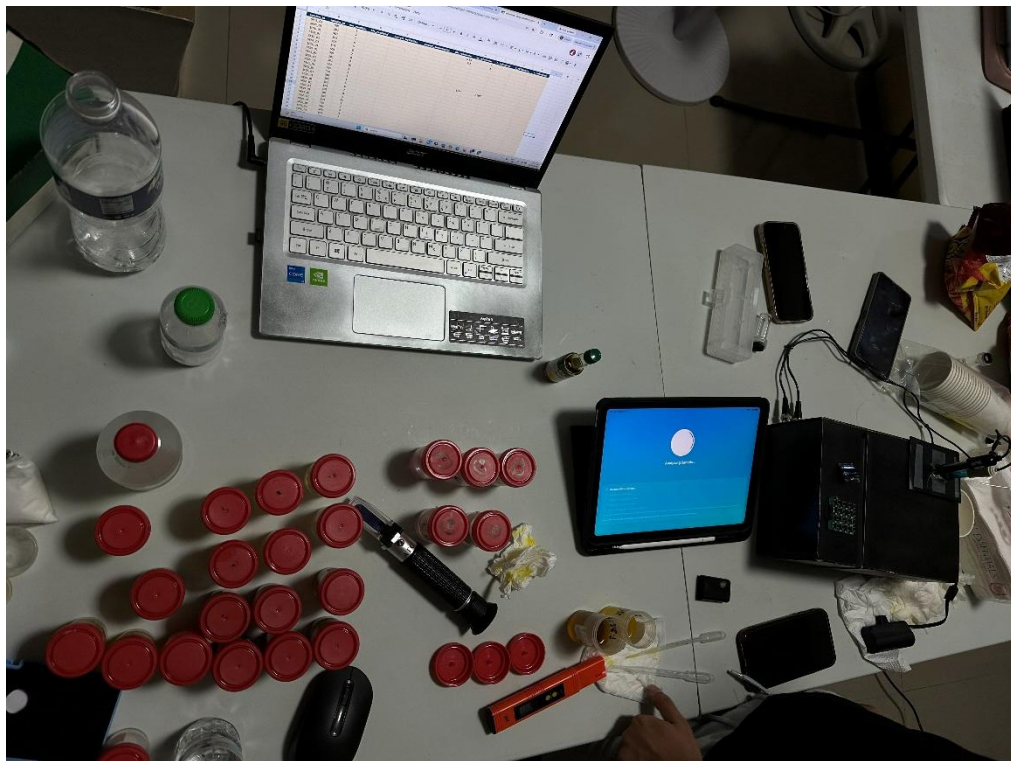
Samples Documentation





APPENDIX M

Testing Documentation





APPENDIX N

Request for Amendments

REQUEST FOR AMENDMENTS IN HARDWARE COMPONENTS AND CALIBRATION METHODOLOGY

Dear Members of the Panel,

We, the Silver Swan, are currently developing our thesis project entitled "IoT-Based Smart Urinalysis System for Hydration Monitoring, Image Processing and Preliminary Screening Using Fuzzy Logic-Enhanced KNN." Following the constructive feedback and recommendations provided during our recent proposal defense, alongside subsequent prototyping and testing, we would like to formally request approval to implement the following adjustments to our hardware specifications and calibration methodology:

Component / Methodology	From	To
Camera Module	OV7670 Camera Module	ESP32-CAM Microcontroller Module <i>Note: This change was recommended by Analisa D. Cabaccan, MSEM, Panelist 2 during our proposal defense due to its advanced processing and multi-functional capabilities.</i>
Color Sensor	TCS34725 RGB Sensor	AS7341 11-Channel Visible Light Sensor <i>Note: Testing revealed that the TCS34725 did not meet our performance expectations and was unsuitable for the specific constraints of our</i>

		<i>fluid testing setup.</i>
Calibration Methodology	Physical Calibration Card	Calibration using liquids <i>Note: The original approach of using a solid reference card to calibrate a liquid medium introduces inaccuracies, as solids and liquids possess fundamentally different optical and refractive properties.</i>

Since our Fuzzy Logic-Enhanced KNN algorithm relies heavily on processing precise HSV (Hue, Saturation, Value) data from urine samples, minimizing environmental interference is vital. Upgrading to the AS7341 11-Channel Visible Light Sensor paired with the ESP32-CAM vastly increases our optical data resolution and leverages advanced processing capabilities. Furthermore, transition from a physical calibration card to liquids allows the system to establish an absolute optical baseline directly inside the fluid testing chamber. This effectively isolates our color readings from external ambient room lighting, solving a major environmental variable and matching the actual physical properties of the fluid medium.

We would like to emphasize that these structural optimizations do not alter our core objectives, title, or algorithmic framework. They are implemented solely to guarantee the scientific precision and reliability of our system.

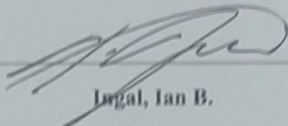
Respectfully yours,
Silver Swan



Sabulao Jr, Julius Jojo C.
Project Manager



Ombrog, Xavier Michael Emmanuel, N.
Lead Developer



Logal, Ian B.
Lead Researcher



Sangkya, Chedrick Khyllle P.
Researcher And Developer

Approved And Acknowledged By:

Elizabeth T. Sta Ana, MSCS, MBA
Thesis Advisor



Rico S. Santos, DIT
Head Panelist

Maricar D. Asuncion, MSCS
Course Advisor



Wilber B. Sabado, PhD
Panelist 1

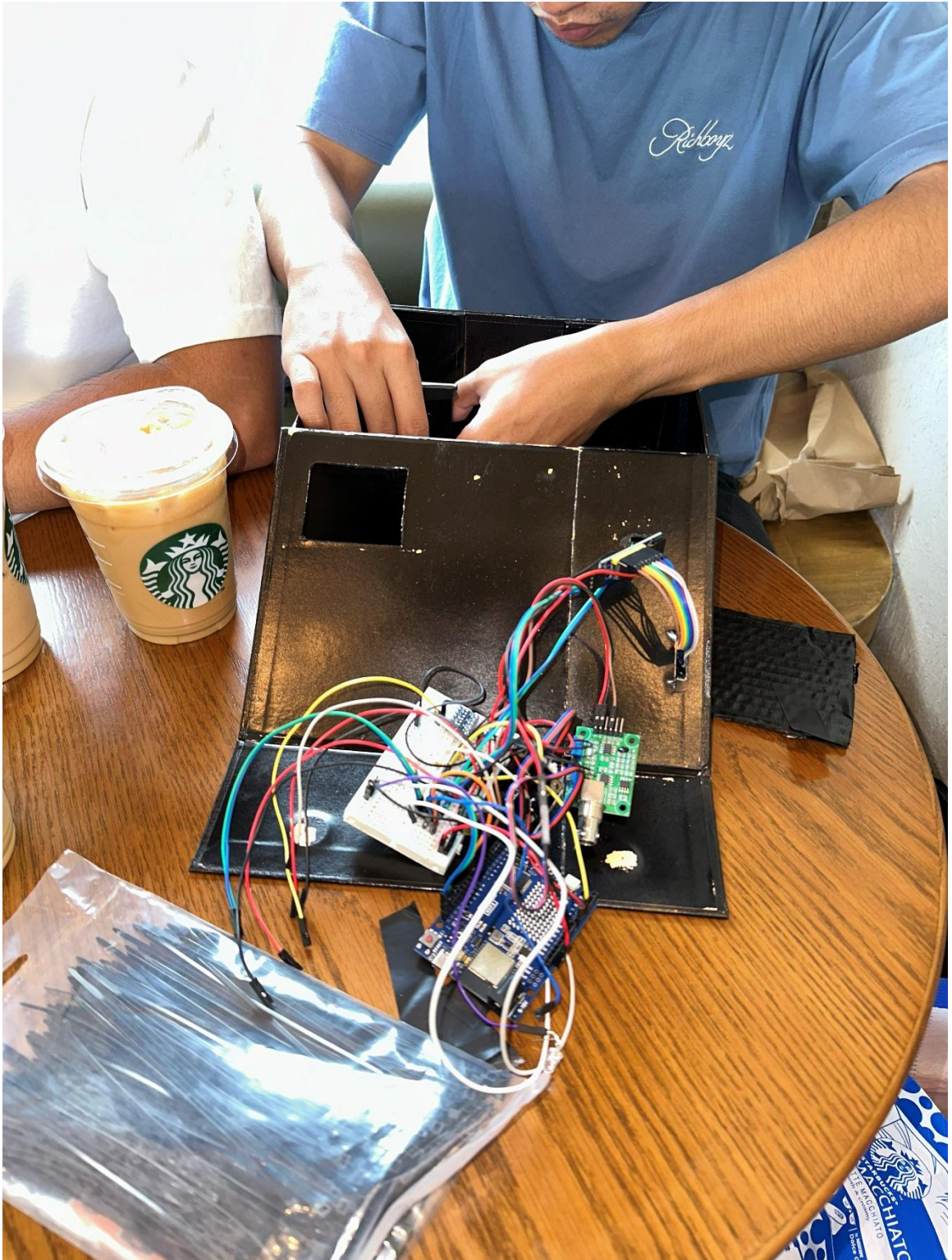


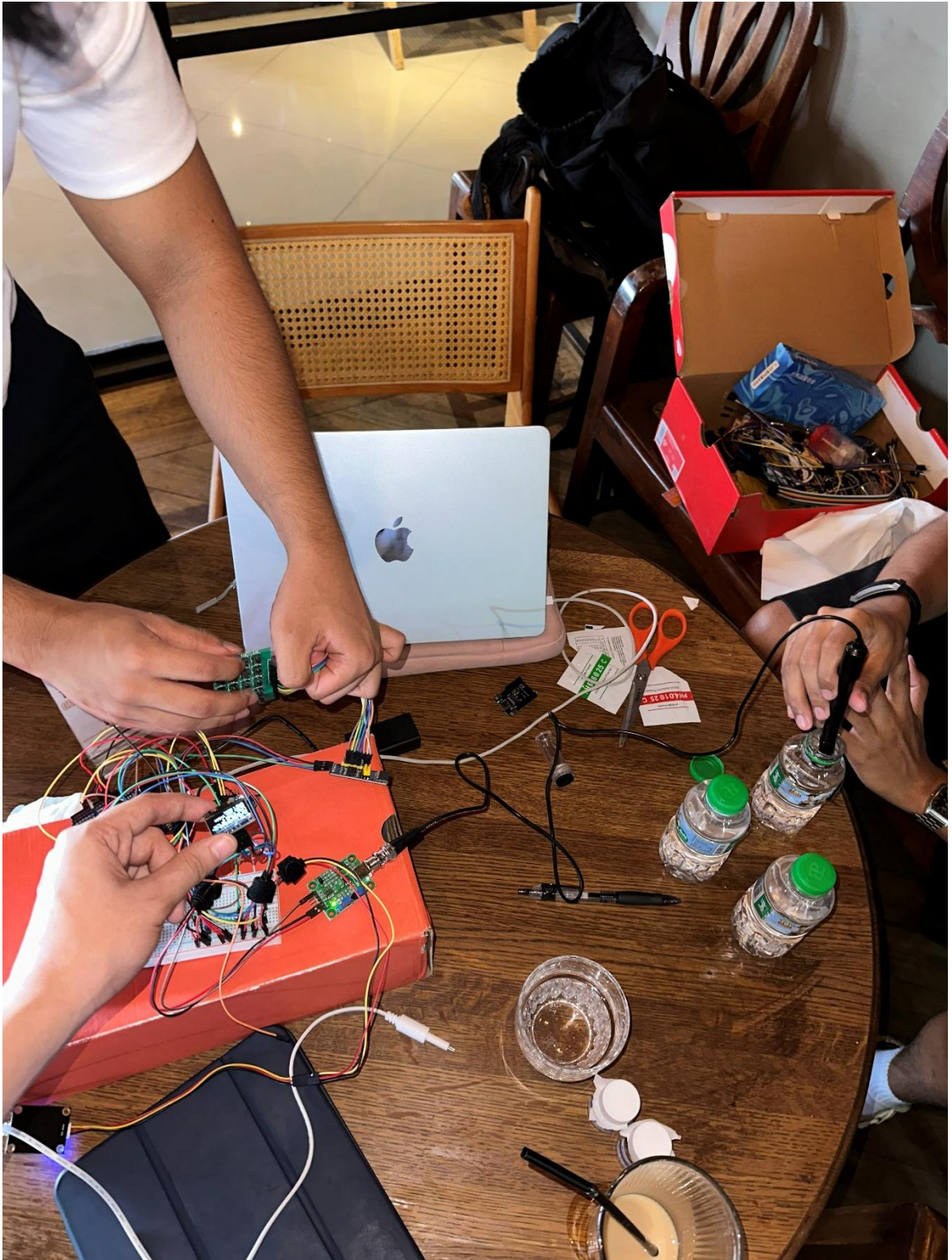
Analisa D. Cabaccan, MSEM
Panelist 2

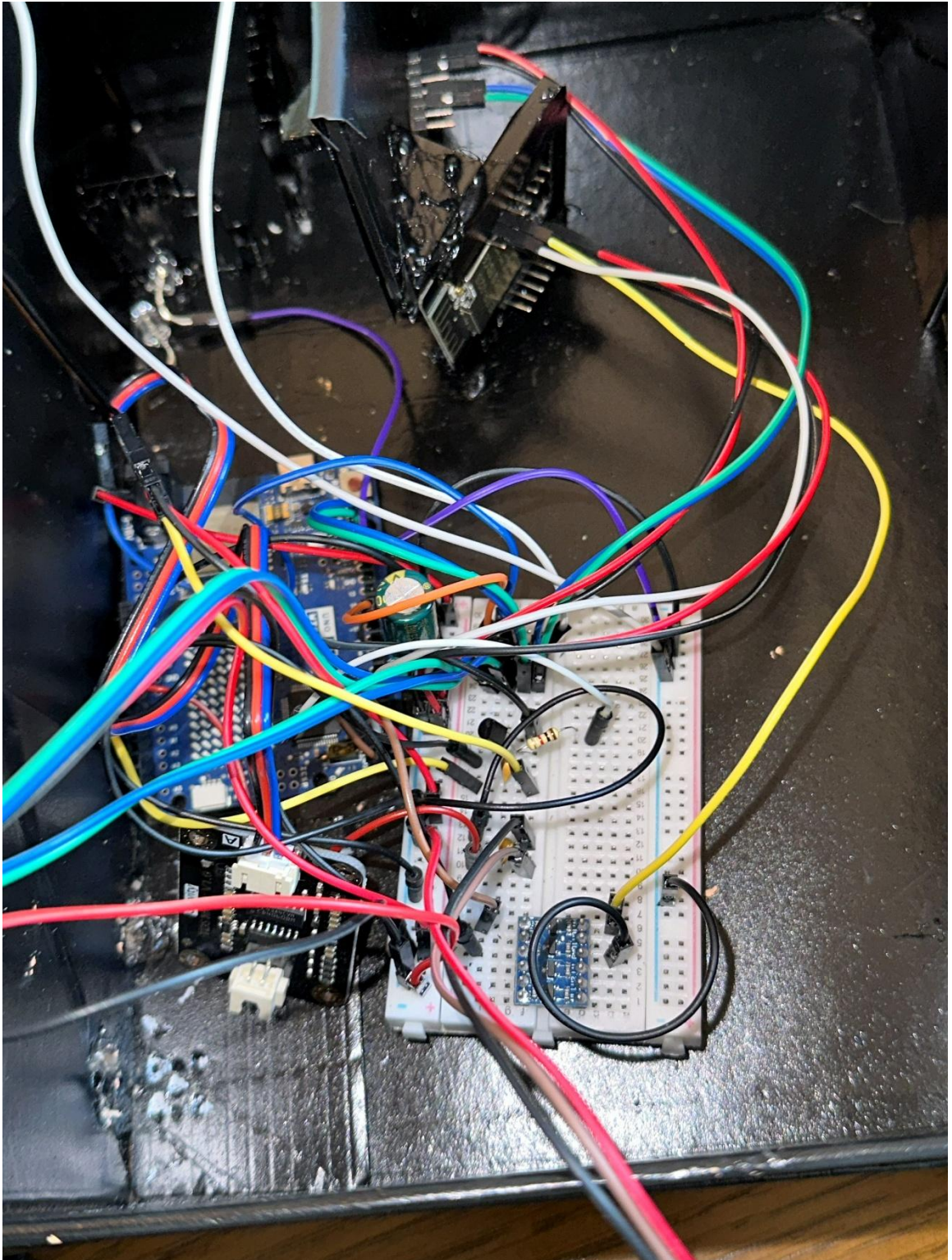
APPENDIX O

Development Documentation









APPENDIX P

Researchers Profile



IAN B. INGAL

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🌐 www.linkedin.com/in/ian-ingal-036030291
🔗 <https://github.com/lan-Ingal>
📍 Blk 5 Lot 17/18 Phase 3 Country Homes
Putatan Muntinlupa City

SUMMARY

Motivated and detail-oriented Computer Science undergraduate at FEU Alabang with strong foundations in software development, data analysis, and problem-solving. Currently in the 3rd year of study, developing practical experience in programming, algorithm design, and system development through academic projects and independent learning.

ACADEMIC BACKGROUND

SOUTHERNSIDE MONTESSORI SCHOOL (2021-2023)

Science, Technology, Engineering, and Mathematics

FAR EASTERN UNIVERSITY - ALABANG (CURRENT)

Bachelor of Science in Computer Science with specialization in Software Engineering

SKILLS

TECHNICAL SKILLS

- **Programming languages:** Basic knowledge of C++, Java, Python
- **Database Management:** Basic knowledge of SQL
- **Software Development Tools:** Visual Studio Code, Android Studio

SOFT SKILLS

- Analytical Thinking
- Cross-functional Team Collaboration
- Communication Skills
- Quality Focus
- Eager to Learn
- Performance Driven
- Attention to Detail
- Effective Time Management

CERTIFICATES

INFORMATION TECHNOLOGY SPECIALIST IN JAVA

Certificate upon request

Oct 9, 2024

INFORMATION TECHNOLOGY SPECIALIST IN PYTHON

Certificate upon request

June, 21 2025



XAVIER MICHAEL EMMANUEL OMBROG

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- ombrogxavieremmanuel@gmail.com
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- <https://github.com/KeMeNoToRiKo>
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SUMMARY

Computer Science student with a solid foundation in software development and technology. Motivated to learn, contribute to innovative projects, and gain hands-on experience in the tech industry.

ACADEMIC BACKGROUND

SAN BEDA COLLEGE ALABANG (2021-2023)

Humanities and Social Sciences

FAR EASTERN UNIVERSITY - ALABANG (CURRENT)

Bachelor of Science in Computer Science with specialization in Software Engineering

SKILLS

TECHNICAL SKILLS

- **Programming languages:** Basic knowledge of C++, Java, Python, Javascript, Swift, Kotlin, NASM Assembly
- **Database Management:** Basic knowledge of SQL
- **Software Development Tools:** Visual Studio Code, Android Studio, XCode
- **Operating System:** Windows, macOS

SOFT SKILLS

- Analytical Thinking
- Problem Solving
- Team Collaboration
- Communication Skills
- Adaptability
- Eager to Learn
- Collaboration Tools Proficiency
- Attention to Detail
- Time Management

CERTIFICATES

INFORMATION TECHNOLOGY SPECIALIST IN JAVA

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Oct 9, 2024

INFORMATION TECHNOLOGY SPECIALIST IN PYTHON

Certificate upon request

June, 21 2025



JULIUS JOJO C. SABULAO JR

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🌐 [linkedin.com/in/julius-jojo-sabulao-jr](https://www.linkedin.com/in/julius-jojo-sabulao-jr)
🔗 <https://github.com/clstljr>
📍 Lot 1 Block 6 Purok 10 South Daang Hari,
Taguig City, Philippines

SUMMARY

Dedicated Computer Science student with a growing foundation in technology and software development. Eager to learn and contribute to innovative projects while gaining hands-on experience in the tech industry.

ACADEMIC BACKGROUND

APEC SCHOOLS - STA. RITA (2021-2023)

Science, Technology, Engineering, and Mathematics

FAR EASTERN UNIVERSITY - ALABANG (CURRENT)

Bachelor of Science in Computer Science with specialization in Software Engineering

SKILLS

TECHNICAL SKILLS

- **Programming languages:** Basic knowledge of C++, Java, Python
- **Database Management:** Basic knowledge of SQL
- **Software Development Tools:** Visual Studio Code, Android Studio
- **Operating System:** Windows
- **Proficient in Microsoft:** Word, Excel, PowerPoint

SOFT SKILLS

- Analytical Thinking
- Problem Solving
- Team Collaboration
- Communication Skills
- Adaptability
- Eager to Learn
- Collaboration Tools Proficiency
- Attention to Detail
- Time Management

CERTIFICATES

INFORMATION TECHNOLOGY SPECIALIST IN JAVA

Certificate upon request

Oct 9, 2024

INFORMATION TECHNOLOGY SPECIALIST IN PYTHON

Certificate upon request

June, 21 2025



Chedrick Khylle P. Sangkula

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🌐 [linkedin.com/in/chedrick-khyll-sangkula](https://www.linkedin.com/in/chedrick-khyll-sangkula)
📄 <https://github.com/Chedlz>
📍 34 Gladiola, Pilar, Las Pinas

SUMMARY

Computer Science student driven by curiosity and innovation with a determination on learning complex problems and a keen eye for detail. Open in learning new things everyday.

ACADEMIC BACKGROUND

St. Andrew's School (2021-2023)

Science, Technology, Engineering, and Mathematics

FAR EASTERN UNIVERSITY - ALABANG (2023- PRESENT)

Bachelor of Science in Computer Science with specialization in Software Engineering

SKILLS

TECHNICAL SKILLS

- **Programming languages:** C++, Java, Python, PHP, XML, Kotlin
- **Database Management:** MySQL, PHP
- **Development Tools:** Visual Studio Code, Android Studio, GitHub
- **Operating System:** Windows
- **Productivity Tools:** Microsoft Office (Word, Excel, PowerPoint)

SOFT SKILLS

- Analytical & Critical Thinking
- Problem-Solving Mindset
- Team Collaboration & Communication
- Adaptability in Fast-Paced Environments
- Time Management & Task Prioritization
- Continuous Learner with Growth Mindset

CERTIFICATES

INFORMATION TECHNOLOGY SPECIALIST IN JAVA

Certificate upon request

Oct 9, 2024

INFORMATION TECHNOLOGY SPECIALIST IN PYTHON

Certificate upon request

June, 21 2025